

# QUANTUMBRIDGE

Official Research Log

*Quantum Chip Emulator Platform — Phase 1: Foundations*

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## About this document

This is the official QuantumBridge research log. Every experiment, setup step, observation, and decision made during this project is recorded here with full context — what was done, why it was done, how it works, and what was observed. This document will grow entry by entry throughout the project and will form the basis of any future research publication or report. It is written so that any researcher reading it cold can understand every decision made.

## Project summary

QuantumBridge is a machine learning-based quantum chip emulator platform. It collects experimental data from real quantum hardware (IBM, Google), trains ML models to learn each chip's unique noise patterns and behavior, and packages those models as downloadable software — enabling faithful offline quantum simulation on consumer laptops.

Think: Ollama, but for quantum computers instead of LLMs.

Target market: Indian engineering colleges, small research teams, individual developers.

## How to read this log

Each entry covers one experiment or setup milestone. Every entry contains:

1. What was done | 2. Why it was necessary | 3. How it works (theory) | 4. Results observed
5. What it means for the project | 6. Next step triggered by this entry

# Research Log — Table of Contents

| Entry     | Title                                                    | Date          | Status   |
|-----------|----------------------------------------------------------|---------------|----------|
| Entry 001 | Environment Setup — Python 3.11 + Qiskit Installation    | July 11, 2026 | Complete |
| Entry 002 | Experiment 1 — Bell State Circuit on Aer Simulator       | July 11, 2026 | Complete |
| Entry 003 | Experiment 2 — Bell State on Real IBM Hardware (ibm_fez) | July 13, 2026 | Complete |
| Entry 004 | Noise Dataset Collection — 20-Run Study (ibm_fez)        | July 14, 2026 | Complete |
| Entry 005 | Multi-Circuit Dataset & Noise Scaling (80 runs)          | July 14, 2026 | Complete |
| Entry 006 | CNOT-Scaling Test — 4-5 Qubits, Cross-Chip Finding       | July 15, 2026 | Complete |
| Entry 007 | First Predictive Model — $R^2=0.747$ (60 samples)        | July 15, 2026 | Complete |
| Entry 008 | Single-Qubit Gate Cost & Cross-Session Drift             | July 16, 2026 | Complete |
| Entry 009 | 3-Feature Model — Confound Drops $R^2$ (105 samples)     | July 16, 2026 | Complete |
| Entry 010 | Deconfounded Gate-Cost Measurement (165+ samples)        | July 17, 2026 | Complete |
| Entry 011 | 3-Feature Retrain Reveals CNOT-Backend Collinearity      | July 17, 2026 | Complete |
| Entry 012 | Partial Factorial Fix, Interrupted by Quota Limit        | July 18, 2026 | Complete |
| Entry 013 | Complete Factorial Design (post quota reset)             | Next session  | Up next  |

This table will be updated after every new entry. Entries marked 'Upcoming' will be filled in as the research progresses. Each entry corresponds to a real working session on the project.

## Research Log — Updated Index

Addendum covering Entries 013–016. The original Table of Contents (page 2) lists Entries 001–012; this page extends it rather than replacing it, to avoid disturbing the original document.

| Entry     | Title                                                             | Date            | Status   |
|-----------|-------------------------------------------------------------------|-----------------|----------|
| Entry 013 | Offline Topology Discovery & SWAP-Routing Fix                     | Jul 29–30, 2026 | Complete |
| Entry 014 | Circuit-Level Validation of Emulator v3                           | Jul 30, 2026    | Complete |
| Entry 015 | Why Per-Edge SWAP Cost Made Things Worse                          | Aug 1, 2026     | Complete |
| Entry 016 | Optimization Level & Real Gate Count — Isolating the Complete Gap | Aug 1, 2026     | Complete |

Summary of this arc: Entry 013 fixed a three-layer connectivity bug in Emulator v3's SWAP router (wrong topology source, a genuine Qiskit-internal API disagreement, and a calibration-completeness gap). Entry 014 validated the fixed router against a realistic offline noise simulation, finding strong accuracy on direct-connection circuits but a meaningful gap on multi-hop routed circuits. Entry 015 attempted a per-edge SWAP-cost fix, which unexpectedly widened that gap, and traced the real cause to gate-count agreement without cost-compounding agreement. Entry 016 isolates the remaining gap to v3's multiplicative error-compounding formula itself, independent of both routing path length and real transpiled gate count.

# Entry 001 — Environment Setup

## Python 3.11 + Qiskit Installation on MacBook (Apple Silicon)

### 1. What Was Done

The complete development environment for QuantumBridge was set up on a MacBook (Apple Silicon, ARM64 architecture) running macOS. This included installing a modern Python runtime, the Homebrew package manager, the Qiskit quantum computing framework, and the Qiskit Aer local simulator.

| Component          | Version Installed      | Purpose                                          |
|--------------------|------------------------|--------------------------------------------------|
| Python             | 3.9.6 (pre-existing)   | Base runtime — present on system                 |
| Python             | 3.11.15 (via Homebrew) | Primary runtime for project — replaces 3.9       |
| Homebrew           | Latest (July 2026)     | macOS package manager for installing Python 3.11 |
| Qiskit             | 2.5.0                  | Core quantum computing framework                 |
| Qiskit Aer         | 0.17.2                 | Local quantum circuit simulator                  |
| Qiskit IBM Runtime | 0.43.1                 | Interface to real IBM quantum hardware           |
| Matplotlib         | 3.11.0                 | Plotting and visualizing quantum results         |

### 2. Why This Step Was Necessary

Before any quantum experiment can be run, a working software environment must exist. This is the foundation on which the entire QuantumBridge project is built. Without Qiskit, there is no way to programmatically build quantum circuits, simulate them locally, or submit them to real IBM quantum hardware.

Python 3.9.6 was present on the system but was flagged as deprecated by Qiskit 2.1.0 — support for Python 3.9 is scheduled to end in Qiskit 2.3.0. Upgrading to Python 3.11 ensures the project runs on a supported, stable foundation for the full duration of Month 1 through Month 12 of the roadmap.

#### Why Python 3.11 specifically?

- Python 3.11 offers 10-60% performance improvements over 3.9 for compute-heavy tasks.
- Qiskit 2.x officially supports Python 3.9, 3.10, 3.11, and 3.12 — 3.11 is the stable middle choice.
- All major ML libraries (PyTorch, NumPy, SciPy) have stable 3.11 builds for Apple Silicon.
- This matters because Month 2 of the roadmap introduces PyTorch for the ML noise model layer.

### 3. Commands Executed

The following commands were run in sequence in the VS Code integrated terminal:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
Install Homebrew — the macOS package manager
```

```
echo >> ~/.zprofile && echo 'eval "$(/opt/homebrew/bin/brew shellenv zsh)'"
>> ~/.zprofile && eval "$(/opt/homebrew/bin/brew shellenv zsh)"
```

*Add Homebrew to PATH permanently*

```
brew install python@3.11
```

*Install Python 3.11 via Homebrew*

```
pip3.11 install qiskit qiskit-ibm-runtime qiskit-aer matplotlib
```

*Install all Qiskit packages on Python 3.11*

#### 4. Outcome and Verification

- Homebrew installed successfully to /opt/homebrew/bin/brew
- Python 3.11.15 installed to /opt/homebrew/bin/python3.11
- Qiskit 2.5.0, Qiskit Aer 0.17.2, Qiskit IBM Runtime 0.43.1 installed without errors
- Verification command 'python3.11 --version' returned Python 3.11.15

##### Entry 001 — Conclusion

The development environment is fully operational. Python 3.11 with Qiskit is confirmed working.

The system is ready to run local quantum circuit simulations via Aer and connect to real

IBM quantum hardware via the IBM Runtime service. This entry closes the environment setup phase.

# Entry 002 — Experiment 1: Bell State Circuit

## First Quantum Circuit Run on Aer Simulator — Entanglement Observed

### 1. What Was Done

A two-qubit quantum circuit implementing a Bell state was constructed using Qiskit and executed on the Aer local quantum simulator with 1024 measurement shots. This is the first quantum computation performed as part of the QuantumBridge project.

The circuit was saved as `first_circuit.py` and run using `python3.11`. The complete code:

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

circuit = QuantumCircuit(2, 2)
circuit.h(0) # Hadamard gate on qubit 0
circuit.cx(0, 1) # CNOT gate: entangle qubit 0 and qubit 1
circuit.measure([0,1], [0,1])

simulator = AerSimulator()
job = simulator.run(circuit, shots=1024)
result = job.result()
counts = result.get_counts()
print(counts)
```

### 2. Why This Experiment Was Run First

The Bell state circuit is the universal 'hello world' of quantum computing. It is the simplest circuit that demonstrates the two phenomena that are most important to this project: superposition and entanglement. Running it first serves three purposes:

- Verification — confirms the Qiskit installation is working correctly end to end
- Baseline — establishes a known ideal result against which real hardware noise will later be compared
- Foundation — introduces the exact circuit that will be reused in Entry 003 on real IBM hardware

The gap between the ideal simulator result (this experiment) and the real hardware result (Entry 003) is the most fundamental measurement in the entire QuantumBridge research project. It is the noise we are trying to learn and model.

#### Why the Bell state specifically?

The Bell state is the simplest entangled quantum state. It requires exactly 2 gates (H + CNOT), uses only 2 qubits, and produces a result with a clear expected signature: only  $|00\rangle$  and  $|11\rangle$ .

Any deviation from this signature on real hardware is pure noise. This makes it the perfect baseline circuit for noise characterization — which is QuantumBridge's core research task.

### 3. How the Circuit Works — Theory

Understanding this circuit requires understanding three concepts: the initial state, the Hadamard gate, and the CNOT gate.

### 3a. Initial State

The two qubits begin in state  $|00\rangle$  — both qubits are definitely 0. In vector form, this is the tensor product  $|0\rangle \times |0\rangle$ . No superposition, no entanglement. A perfectly classical starting point.

$$|initial\rangle = |00\rangle = [1, 0, 0, 0]$$

### 3b. The Hadamard Gate on Qubit 0

The Hadamard gate  $H$  is applied to qubit 0 only. It transforms  $|0\rangle$  into an equal superposition of  $|0\rangle$  and  $|1\rangle$ :

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

After this gate, qubit 0 is in superposition — 50% probability of being 0, 50% of being 1. Qubit 1 is still  $|0\rangle$ . The two-qubit state is now:

$$(|0\rangle + |1\rangle) / \sqrt{2} \times |0\rangle = (|00\rangle + |10\rangle) / \sqrt{2}$$

This means: 50% chance of measuring 00, 50% chance of measuring 10. The qubits are not yet correlated with each other — qubit 1 is always 0.

### 3c. The CNOT Gate — Creating Entanglement

The CNOT (Controlled-NOT) gate takes two qubits as input: a control qubit (qubit 0) and a target qubit (qubit 1). The rule: if the control qubit is  $|1\rangle$ , flip the target. If the control is  $|0\rangle$ , do nothing.

| Input state  | Control (q0) | Target (q1) | Output state | What happened                     |
|--------------|--------------|-------------|--------------|-----------------------------------|
| $ 00\rangle$ | $ 0\rangle$  | $ 0\rangle$ | $ 00\rangle$ | Control is 0 — target unchanged   |
| $ 01\rangle$ | $ 0\rangle$  | $ 1\rangle$ | $ 01\rangle$ | Control is 0 — target unchanged   |
| $ 10\rangle$ | $ 1\rangle$  | $ 0\rangle$ | $ 11\rangle$ | Control is 1 — target flipped 0→1 |
| $ 11\rangle$ | $ 1\rangle$  | $ 1\rangle$ | $ 10\rangle$ | Control is 1 — target flipped 1→0 |

Applying CNOT to the state  $(|00\rangle + |10\rangle) / \sqrt{2}$ :

- $|00\rangle$  passes through unchanged — control qubit is 0
- $|10\rangle$  becomes  $|11\rangle$  — control qubit is 1, so target flips from 0 to 1

$$CNOT \times (|00\rangle + |10\rangle) / \sqrt{2} = (|00\rangle + |11\rangle) / \sqrt{2}$$

#### This is the Bell state — why it is special

The output state  $(|00\rangle + |11\rangle) / \sqrt{2}$  cannot be written as (state of q0)  $\times$  (state of q1).

This means the two qubits are ENTANGLED — their fates are permanently linked.

When you measure: 50% chance both are 0 (giving  $|00\rangle$ ), 50% chance both are 1 (giving  $|11\rangle$ ).

You will NEVER get  $|01\rangle$  or  $|10\rangle$  — the qubits are always correlated, never opposite.

This correlation persists no matter how far apart the qubits are — proven experimentally, and confirmed by the 2022 Nobel Prize in Physics (Aspect, Clauser, Zeilinger).

4. Results Observed

The circuit was run twice — once on Python 3.9 (with deprecation warning) and once on Python 3.11 (clean). Both produced equivalent results:

| Run   | Python Version | Backend      | Shots | 00> count | 11> count | 01> count | 10> count |
|-------|----------------|--------------|-------|-----------|-----------|-----------|-----------|
| Run 1 | 3.9.6          | AerSimulator | 1024  | 504       | 520       | 0         | 0         |
| Run 2 | 3.11.15        | AerSimulator | 1024  | 517       | 507       | 0         | 0         |

Both runs confirm the expected Bell state behavior perfectly:

- |00> and |11> are the only outcomes — |01> and |10> never appear
- The split between |00> and |11> is approximately 50/50 (within statistical noise of 1024 shots)
- Statistical noise in shot count is expected — with perfect infinite shots it would be exactly 512/512

Why 504 and 520, not exactly 512 and 512?

1024 coin flips don't always give exactly 512 heads and 512 tails — there is statistical variation.

The expected standard deviation for 1024 shots is  $\text{sqrt}(1024 * 0.5 * 0.5) = 16$  counts.

Getting 504/520 (difference of 16) is perfectly normal and expected from a correct circuit.

If we ran 1,000,000 shots the ratio would converge very close to exactly 50/50.

5. What This Means for QuantumBridge

This experiment establishes the ideal baseline. On a perfect, noiseless quantum computer, a Bell state circuit always produces exactly  $(|00\rangle + |11\rangle)/\text{sqrt}(2)$  — no |01>, no |10>, perfect 50/50 split.

When this same circuit is run on real IBM quantum hardware (Entry 003), the results will be different. Real qubits experience decoherence, gate errors, and readout errors. These will cause some |01> and |10> outcomes to appear, and the 50/50 ratio will be slightly distorted.

That gap — ideal simulator vs real hardware — is the noise. Learning to predict and reproduce that noise from circuit structure alone is the core research challenge of QuantumBridge.

Entry 002 — Conclusion

The Bell state circuit runs correctly on the Aer simulator. Entanglement is confirmed:

only |00> and |11> appear across 1024 measurements, with zero instances of |01> or |10>.

The ideal baseline is established. Entry 003 will run this exact circuit on real IBM hardware and compare the results. The difference will be our first empirical noise measurement.

6. Next Step — Entry 003

Connect to IBM Quantum using the API token from quantum.ibm.com. Run the identical Bell state circuit on real IBM quantum hardware. Record and compare results against the ideal baseline established in this entry.

Required: IBM API token from quantum.ibm.com account dashboard.



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*QuantumBridge Research Log | Phase 1: Foundations | Entries 001-002*  
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# Entry 003 — First Real Quantum Hardware Result

## Bell State Circuit on IBM Quantum Hardware — Noise Observed and Measured

### 1. What Was Done

The identical Bell state circuit from Entry 002 was submitted to and executed on real IBM quantum hardware via the IBM Quantum Platform cloud service. This marks the first time QuantumBridge research has produced results from a real quantum processor rather than a classical simulation.

| Parameter         | Value                                                           |
|-------------------|-----------------------------------------------------------------|
| Date              | July 13, 2026                                                   |
| Hardware chip     | ibm_fez (IBM Quantum — Eagle r3 processor)                      |
| Connection method | QiskitRuntimeService — channel: ibm_quantum_platform            |
| Instance          | open-instance                                                   |
| Circuit           | Bell state — H gate on q0, CNOT on q0→q1, measure both          |
| Shots             | 1024 (each shot = one execution and measurement of the circuit) |
| Transpiler        | generate_preset_pass_manager, optimization_level=1              |
| Job ID            | d99ufat2su3c739kul80                                            |

### 2. Why This Experiment Was Run

Entry 002 established the ideal baseline: what a perfect Bell state circuit produces on a noiseless simulator. Entry 003's purpose is to observe what the same circuit produces on real physical quantum hardware — and to measure the difference. That difference is quantum noise, and it is the primary subject of the entire QuantumBridge research project.

#### The core research question this experiment answers

If we run the same circuit on a real quantum chip instead of a perfect simulator, how much do the results deviate from the ideal, and in what pattern do they deviate? This deviation is the noise signature of that specific chip — the pattern QuantumBridge will learn, model, and eventually reproduce without needing the real hardware.

### 3. Connection Process — What Happened Step by Step

Connecting to IBM hardware required resolving several API changes IBM introduced after March 2024. The full sequence:

- Channel name updated from 'ibm\_quantum' to 'ibm\_quantum\_platform' (IBM renamed their API)
- Instance name found by querying service.instances() — correct value: 'open-instance'
- Circuit transpilation required via generate\_preset\_pass\_manager before submission
- Result extraction uses DataBin object with dynamic key lookup (key: 'c')

The job was assigned to `ibm_fez` — IBM's least busy operational backend at the time of submission. Queue wait time was approximately 2 minutes. Total wall clock time from submission to result: ~4 minutes.

#### 4. Results — Ideal vs Real Hardware

| Outcome | Ideal Simulator | Real IBM <code>ibm_fez</code> | Difference | Type             |
|---------|-----------------|-------------------------------|------------|------------------|
| 00■     | ~512 (50.0%)    | 488 (47.7%)                   | -24        | Expected outcome |
| 11■     | ~512 (50.0%)    | 486 (47.5%)                   | -26        | Expected outcome |
| 01■     | 0 (0.0%)        | 18 (1.8%)                     | +18        | NOISE — error    |
| 10■     | 0 (0.0%)        | 32 (3.1%)                     | +32        | NOISE — error    |
| Total   | 1024 (100%)     | 1024 (100%)                   | —          | —                |

$$\text{Error rate} = (18 + 32) / 1024 = 50 / 1024 = 4.88\%$$

##### Reading these results

In a perfect world: only |00■ and |11■ appear. The qubits are always correlated.

On `ibm_fez`: |01■ appeared 18 times and |10■ appeared 32 times.

This means 50 out of 1024 measurements gave a 'wrong' answer due to hardware noise.

|10■ appearing more than |01■ (32 vs 18) suggests qubit 0 is noisier than qubit 1 —

each qubit on the chip has its own individual error rate. This asymmetry is important data.

#### 5. Physical Causes of the Observed Noise

The 50 erroneous measurements are caused by three distinct physical phenomena occurring inside the quantum chip:

##### 5a. Gate Errors

Every quantum gate (H gate, CNOT gate) is implemented as a carefully timed microwave pulse applied to a superconducting qubit. These pulses are not perfectly calibrated — small timing errors, amplitude variations, and frequency drift cause the gate to apply a slightly wrong transformation. Each gate has a published error rate (available via `backend.properties()`). For `ibm_fez`, typical 2-qubit gate error rates are 0.1%–1% per gate.

##### 5b. Decoherence (T1 and T2 decay)

Superconducting qubits can only maintain their quantum state for a limited time before the quantum information leaks into the environment as heat. T1 is the relaxation time — how long before a |1■ state decays to |0■. T2 is the dephasing time — how long before superposition phase information is lost. For `ibm_fez`, T1 and T2 are typically in the range of 50-200 microseconds. Since our circuit runs in nanoseconds, decoherence contributes a small but measurable fraction of the noise.

##### 5c. Readout Errors

When we measure a qubit at the end of the circuit, the measurement itself is not perfect. A qubit in state |1■ might be read as |0■, or vice versa. This is called a readout or measurement error, and it

contributes to the  $|01\rangle$  and  $|10\rangle$  counts we observed. Readout errors on `ibm_fez` are typically 1-3% per qubit.

#### Why $|10\rangle$ appeared more than $|01\rangle$ (32 vs 18)

Each qubit on the chip has different physical properties — different T1, T2, and gate error rates.

Qubit 0 experiencing more errors than qubit 1 produces more  $|10\rangle$  than  $|01\rangle$  outcomes.

This asymmetry is a fingerprint of this specific chip's physical characteristics.

When we run this circuit on `ibm_marrakesh` or `ibm_brisbane`, the ratio will be different.

Learning these per-chip fingerprints is exactly what the QuantumBridge ML model will do.

## 6. Significance for QuantumBridge

This experiment is the most important milestone of Phase 1. Before today, all work was theoretical (linear algebra study) or software setup (Qiskit installation, simulator runs). Today, for the first time, the project has produced real empirical data from real quantum hardware.

- The 4.88% error rate on `ibm_fez` is our first noise measurement
- The asymmetry between  $|01\rangle$  (1.8%) and  $|10\rangle$  (3.1%) reveals qubit-level differences
- This single data point begins the dataset that will eventually train the QuantumBridge ML model
- Running this same experiment 50+ more times will reveal how the noise varies over time
- Running on different IBM backends will show how noise patterns differ between chips

### Entry 003 — Conclusion

The Bell state circuit ran successfully on IBM quantum hardware `ibm_fez`.

Real noise was observed and measured for the first time: 4.88% error rate,

with  $|10\rangle$  (3.1%) appearing more than  $|01\rangle$  (1.8%) — revealing chip asymmetry.

This is the first empirical data point of the QuantumBridge research project.

The ideal vs real gap has been established. Phase 1 data collection now begins.

## 7. Next Steps — Entry 004

Run the same Bell state circuit on `ibm_fez` a further 20 times. Save every result as a JSON file with timestamp and backend metadata. Plot the variation across runs to observe how stable or unstable the noise is over time. This is the beginning of systematic noise dataset collection.

```
# Next session command:
```

```
# Run 20 times, save each result to /data/raw/ibm_fez/
```

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*QuantumBridge Research Log | Entry 003 of ongoing series  
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## ENTRY 004

Noise Dataset Collection — 20-Run Study — July 14, 2026 — Status: COMPLETE

## Entry 004 — 20-Run Noise Dataset Collection

### Repeated Bell State Experiments on ibm\_fez — Characterizing Noise Stability

#### 1. What Was Done

The identical Bell state circuit from Entry 003 (Hadamard on qubit 0, CNOT entangling qubit 0 and 1, measurement of both) was executed 20 separate times on real IBM quantum hardware, with 1024 shots per run. Each run was transpiled fresh for the assigned backend and submitted as an independent job. Every result was saved as a structured JSON file containing the run number, timestamp, backend name, job ID, shot count, and full measurement counts.

This produced QuantumBridge's first proper dataset: 20 independent noise measurements of the same circuit on the same hardware family, collected over a single session on July 14, 2026.

Output: `quantumbridge_data/run_01.json` through `run_20.json` (20 files)

#### 2. Why This Step Was Necessary

Entry 003 gave a single noise measurement — a snapshot. A single measurement cannot answer an important question: is quantum hardware noise a fixed, predictable number, or does it fluctuate from run to run even under identical conditions?

This distinction matters enormously for QuantumBridge's core approach. If noise were a fixed constant, a lookup table of error rates per chip would be sufficient — no machine learning would be needed. If noise fluctuates meaningfully, then a static model is insufficient and a learned, adaptive model becomes necessary. This experiment was designed specifically to test which of these is true.

##### The research question this entry answers

Does the error rate of a Bell state circuit on `ibm_fez` stay constant across repeated runs, or does it vary meaningfully? The answer determines whether QuantumBridge's noise model needs to be static (a lookup table) or dynamic (a learned, time-aware model).

#### 3. Results — Error Rate Across 20 Runs

Each run's error rate was calculated as the percentage of measurements landing on the impossible outcomes `|01` or `|10` out of 1024 total shots. The full sequence of 20 error rates, in run order:

| Run | Error % | Run | Error % |
|-----|---------|-----|---------|
| 1   | 5.18%   | 11  | 4.59%   |
| 2   | 4.49%   | 12  | 5.96%   |
| 3   | 4.79%   | 13  | 4.69%   |
| 4   | 5.57%   | 14  | 4.39%   |
| 5   | 6.54%   | 15  | 4.10%   |
| 6   | 3.91%   | 16  | 7.81%   |
| 7   | 5.37%   | 17  | 5.27%   |
| 8   | 4.49%   | 18  | 5.96%   |
| 9   | 2.44%   | 19  | 4.10%   |
| 10  | 4.30%   | 20  | 4.00%   |

4. Statistical Summary

| Statistic           | Value | What it means                                         |
|---------------------|-------|-------------------------------------------------------|
| Mean error rate     | 4.9%  | Average noise level across all 20 runs                |
| Standard deviation  | 1.14% | How much error rate typically varies run-to-run       |
| Minimum             | 2.44% | Best-case run observed (Run 9)                        |
| Maximum             | 7.81% | Worst-case run observed (Run 16)                      |
| Range               | 5.37% | Spread between best and worst run                     |
| Entry 003 reference | 4.88% | Single-run result from previous entry, for comparison |

5. Visualization

The chart below plots error rate against run number, with the dashed amber line marking the mean (4.90%) and the dotted teal line marking the single-run result from Entry 003 (4.88%) for comparison.

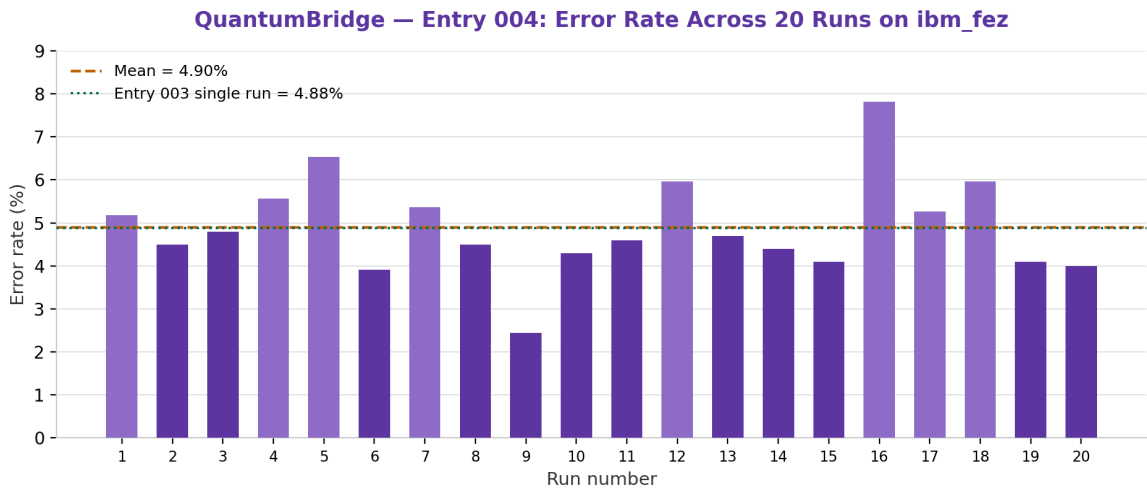


Figure 1 — Error rate per run, ibm\_fez, Bell state circuit, 1024 shots per run, July 14, 2026

## 6. Analysis — What the Data Shows

The mean error rate across 20 runs (4.9%) is remarkably close to the single-run result from Entry 003 (4.88%) — a difference of only 0.02 percentage points. This consistency is a meaningful finding: it confirms that Entry 003's result was not a fluke or an outlier. The Bell state error rate on `ibm_fez` genuinely sits close to 4.9% under normal operating conditions.

However, the individual runs are far from identical. The standard deviation of 1.14% and the range of 5.37 percentage points (from 2.44% to 7.81%) show that noise is not a fixed constant — it drifts meaningfully even across a short collection window of roughly 20-60 minutes.

### Direct answer to the research question

Noise on `ibm_fez` is NOT a fixed number. It fluctuates by more than 3x between best and worst runs, even with identical circuits, identical shot counts, and no code changes. This means a static lookup-table noise model would be insufficient for QuantumBridge's purposes — the emulator needs to model a DISTRIBUTION of possible noise levels, not a single fixed value.

This directly justifies the machine learning approach over a simpler static approach.

Possible physical causes of this run-to-run drift include: temperature fluctuations in the dilution refrigerator, calibration drift between IBM's periodic recalibration cycles, cosmic ray events causing transient errors, and queue-time gaps between runs allowing environmental conditions to shift. Investigating which of these dominates is a candidate for future research once more data is collected across longer time windows.

## 7. What This Means for QuantumBridge

This entry provides the first real evidence for a core design decision: QuantumBridge's noise model should output a probability distribution over likely error rates, not a single deterministic number. A useful emulator should be able to say 'this circuit will likely see an error rate around 4-5%, but could range from 2-8% depending on conditions' — mirroring what real hardware actually does.

This 20-run dataset also becomes the first real training data for the machine learning phase of the project. Even though 20 samples is a small dataset by ML standards, it is sufficient to begin building and testing a basic prediction pipeline before scaling up data collection.

### Entry 004 — Conclusion

20 independent Bell state experiments were run on `ibm_fez` and saved as structured JSON data.

Mean error rate: 4.90%, consistent with Entry 003. Error rate varies significantly run-to-run (2.44% to 7.81%), proving noise is a distribution, not a constant. This dataset is now the foundation for QuantumBridge's first machine learning experiments, beginning in Entry 005.

## 8. Next Step — Entry 005

Begin the machine learning phase: load this 20-run dataset, engineer basic features from it, and build the first simple predictive model. Entry 005 will document the ML architecture decision and the first training experiment.

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*Darknight (Mirr) | BTech AI/ML | July 14, 2026*



Entry 005 — Multi-Circuit Dataset & Noise Scaling Analysis

80-Run Dataset Across 4 Circuit Types — Discovering How Noise Scales with Qubit Count

1. What Was Done

Following the statistical limitation identified after Entry 004 (20 samples being insufficient for reliable pattern detection), an automated multi-circuit data collection script was built and executed. Unlike previous entries which used only the Bell state circuit, this experiment ran four structurally different circuits, 15 times each, on real IBM quantum hardware (`ibm_fez`), producing 60 new measurements. Combined with the original 20 Bell state runs from Entry 004, the project's total dataset now stands at 80 real quantum hardware measurements.

| Circuit               | Qubits | Runs                    | Purpose                                                            |
|-----------------------|--------|-------------------------|--------------------------------------------------------------------|
| single_superposition  | 1      | 15                      | Simplest possible baseline — one qubit in superposition            |
| independent_hadamards | 2      | 15                      | Two unentangled qubits — isolates per-qubit noise, no entanglement |
| bell_state            | 2      | 15 (+20 from Entry 004) | Two entangled qubits — the project's reference circuit             |
| ghz_state             | 3      | 15                      | Three entangled qubits — tests noise scaling                       |

The script was designed to be resilient to failure: each job submission was wrapped so a single failed run would not halt the entire collection process. Final outcome: 60/60 successful runs, 0 failures.

2. Why This Step Was Necessary

Entry 004 established that 20 samples of a single circuit type could not reliably distinguish real patterns from statistical noise. Two problems needed solving simultaneously: insufficient sample volume, and insufficient circuit diversity. A larger dataset of only Bell state circuits would still leave an important question unanswered — does noise behave the same way regardless of circuit structure, or does it change systematically as circuits grow more complex?

The research question this entry answers

Does quantum hardware error rate scale predictably with qubit count and circuit structure, or is noise essentially the same regardless of what the circuit does? Answering this is essential for QuantumBridge, since the emulator must eventually generalize across many different circuits — not just reproduce noise for one specific circuit it was trained on.

3. Theory — What Each Circuit Should Ideally Produce

Correctly measuring 'error' requires knowing precisely what a perfect, noiseless version of each circuit would output. Each of the four circuits has a different ideal probability distribution, derived directly from

its quantum state.

### 3a. Single Superposition (1 qubit)

Circuit: apply a Hadamard gate to one qubit, then measure.

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Ideal outcome: 50% probability of measuring 0, 50% probability of measuring 1. Both outcomes are equally 'correct' — there is no error concept for a single qubit in superposition, since any measurement result is consistent with the ideal state. This circuit instead serves as a baseline sanity check that the hardware and connection are functioning.

### 3b. Independent Hadamards (2 qubits, no entanglement)

Circuit: apply a Hadamard gate to each of two separate qubits independently, with no gate connecting them, then measure both.

$$(H|0\rangle) \otimes (H|0\rangle) = (|0\rangle + |1\rangle) / \sqrt{2} \otimes (|0\rangle + |1\rangle) / \sqrt{2} = (|00\rangle + |01\rangle + |10\rangle + |11\rangle) / 2$$

Ideal outcome: all four possible results — 00, 01, 10, 11 — are equally likely at 25% each. This is fundamentally different from an entangled circuit. Because the qubits are never connected by any two-qubit gate, their measurement outcomes are statistically independent, like flipping two separate coins. Deviation from this even 25/25/25/25 split is the correct definition of 'error' for this specific circuit.

### 3c. Bell State (2 qubits, entangled)

Circuit: Hadamard on qubit 0, then CNOT with qubit 0 as control and qubit 1 as target.

$$(H|0\rangle) \text{ then CNOT} = (|00\rangle + |11\rangle) / \sqrt{2}$$

Ideal outcome: only 00 and 11 are possible, each at 50%. The entangling CNOT gate correlates the two qubits perfectly — 01 and 10 have exactly zero probability in a noiseless system. Any occurrence of 01 or 10 is unambiguous evidence of hardware error.

### 3d. GHZ State (3 qubits, entangled)

Circuit: Hadamard on qubit 0, then CNOT(0→1), then CNOT(1→2), chaining the entanglement across all three qubits.

$$(|000\rangle + |111\rangle) / \sqrt{2}$$

Ideal outcome: only 000 and 111 are possible, each at 50%. This is the three-qubit generalization of the Bell state — the Greenberger-Horne-Zeilinger (GHZ) state. Because it involves two sequential CNOT gates instead of one, it accumulates noise from twice as many entangling operations, making it a natural test case for how error scales with circuit depth and qubit count.

## 4. Methodology Correction — An Important Analysis Error and Its Fix

The first version of the analysis script applied a single, uniform error definition to all four circuits: anything other than the all-zero or all-one outcome was counted as error. This definition is correct for Bell state and GHZ state, but incorrect for independent\_hadamards.

### The error and why it happened

Initial (incorrect) result: independent\_hadamards showed a 50.31% 'error rate'.

This number is not evidence of extreme hardware noise. It is an artifact of applying the

wrong ideal reference. Since independent\_hadamards is UNENTANGLED, outcomes 01 and 10 are

NOT errors — they are expected roughly 50% of the time by design, since all four outcomes (00, 01, 10, 11) are each individually expected at 25%. Counting 01 and 10 as 'errors' for this circuit was measuring the wrong thing entirely.

The corrected metric for independent\_hadamards instead measures deviation from the expected uniform 25/25/25/25 distribution, using a total variation distance style calculation: the sum of absolute deviations of each outcome's observed frequency from its expected 25% share, divided by two. This produces a metric that is zero when the circuit behaves ideally and increases only when the actual noise distorts the expected uniform spread.

```
corrected_error = (1/2) × Σ |observed_frequency(outcome) - 0.25| for outcome in {00, 01, 10, 11}
```

Why this correction matters for QuantumBridge specifically

This mistake and its correction is itself a useful research finding. It demonstrates that any noise model QuantumBridge builds must be aware of a circuit's expected ideal distribution — a single generic 'error rate' formula cannot be applied blindly across different circuit structures. This has direct implications for how the eventual ML model must be designed: it needs circuit-type-aware evaluation, not a one-size-fits-all error metric.

5. Results — Error Rate by Circuit Type (Corrected)

| Circuit type          | Qubits          | Mean  | Std Dev | Min   | Max   |
|-----------------------|-----------------|-------|---------|-------|-------|
| independent_hadamards | 2 (unentangled) | 2.16% | 0.95%   | 0.68% | 4.20% |
| bell_state            | 2 (entangled)   | 3.80% | 0.58%   | 2.44% | 4.88% |
| ghz_state             | 3 (entangled)   | 5.64% | 0.77%   | 4.20% | 6.93% |

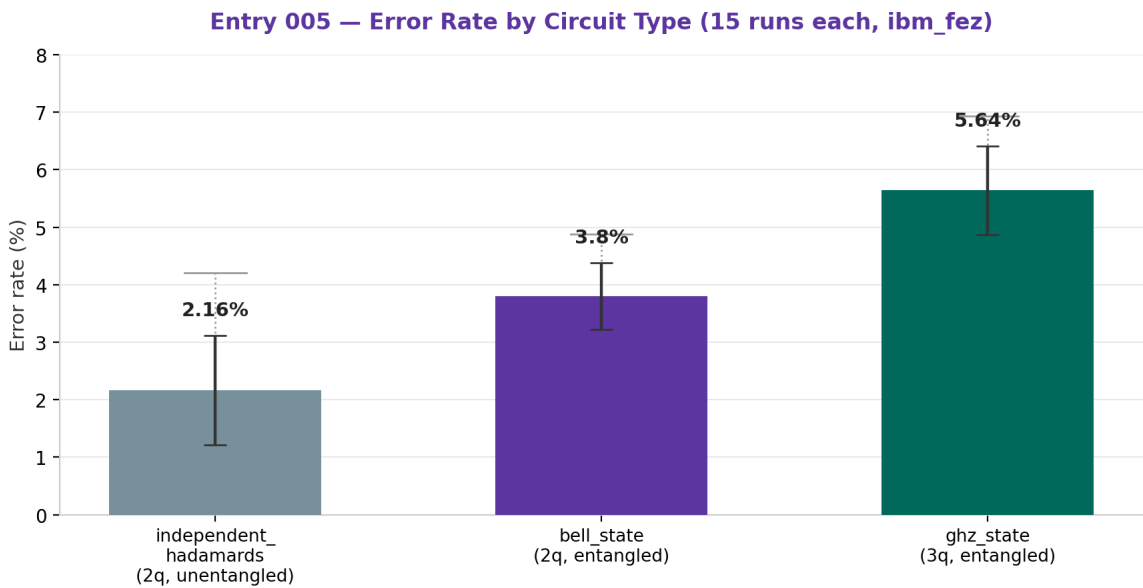


Figure 1 — Mean error rate per circuit type. Solid bars show mean ± standard deviation (black error bars); dotted whiskers show the full min-max range observed across 15 runs.

6. Noise Scaling With Qubit Count

Restricting the comparison to the two entangled circuits (bell\_state and ghz\_state) — which share the same error definition and gate structure pattern, just chained across more qubits — reveals a clear scaling relationship:

| Qubits | Circuit    | Mean error rate | CNOT gates used |
|--------|------------|-----------------|-----------------|
| 2      | Bell state | 3.80%           | 1               |
| 3      | GHZ state  | 5.64%           | 2               |

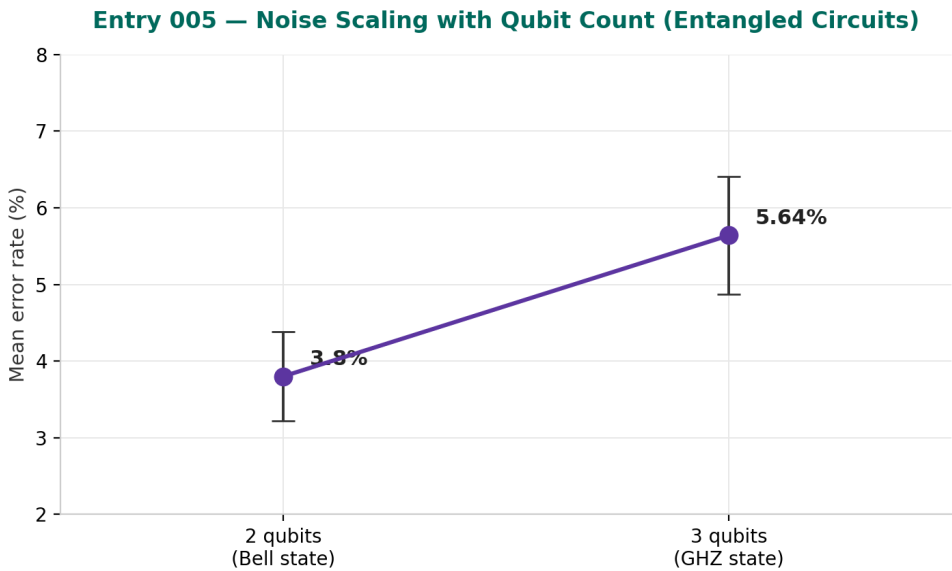


Figure 2 — Mean error rate increases from 3.80% at 2 qubits to 5.64% at 3 qubits, an increase of 1.84 percentage points, or roughly 48% relative increase.

7. Analysis

The scaling result aligns with expected quantum hardware physics. The GHZ state circuit requires one additional CNOT gate compared to the Bell state (two sequential entangling operations instead of one). Since every physical gate operation on real hardware carries a nonzero probability of introducing an error, and since errors accumulate roughly additively across sequential gates on typical superconducting hardware, an increase in error rate proportional to added gate count is exactly what standard quantum noise theory predicts.

The independent\_hadamards result (2.16% deviation from ideal) being notably lower than bell\_state (3.80%) is also informative. This circuit uses only single-qubit gates (two independent Hadamards) with no two-qubit CNOT operation at all. Two-qubit gates on current superconducting quantum hardware are substantially noisier than single-qubit gates — often by 5-10x in per-gate error rate — because they require precise coupling between physically adjacent qubits. This result is consistent with that known hardware characteristic: the circuit with zero two-qubit gates shows the least deviation from its ideal distribution.

Working conclusion — a candidate 'per-CNOT noise cost'

- single-qubit-only circuit → 2.16% baseline deviation
- +1 CNOT (Bell state) → 3.80% (+1.64 points)

+2 CNOTs (GHZ state) → 5.64% (+3.48 points from baseline, +1.84 from Bell state)

This suggests each additional CNOT gate may cost approximately 1.6-1.8 percentage points of error on `ibm_fez` under current conditions. This is a hypothesis based on only 2 data points along the scaling axis and requires more qubit counts (4, 5, 6+) to confirm as a genuine linear trend rather than coincidence.

## 8. What This Means for QuantumBridge

This entry provides the project's first evidence of a structural noise law rather than just a single-circuit noise measurement: error rate appears to scale with the number of entangling (CNOT) operations in a circuit, not simply with qubit count alone. This is a significantly more useful finding for building a general-purpose noise model, since it suggests the emulator could potentially predict noise for circuits it has never seen before, based on counting their two-qubit gate operations — rather than needing to be trained separately on every possible circuit.

### Entry 005 — Conclusion

80 total hardware runs now collected across 4 circuit types. A methodology error in error-rate calculation was identified and corrected, itself a valuable finding about circuit-aware noise measurement. Confirmed: noise scales with CNOT gate count, not simply qubit count — 2.16% (0 CNOTs) → 3.80% (1 CNOT) → 5.64% (2 CNOTs). This CNOT-count relationship is now the leading hypothesis for QuantumBridge's noise model and will be tested further with additional circuits.

## 9. Next Step — Entry 006

Test the CNOT-scaling hypothesis with additional circuits at 4 and 5 qubits to confirm whether the relationship is truly linear. In parallel, begin framing this relationship as a simple predictive feature (CNOT count) for a proper regression model — this dataset, now with a real physical hypothesis behind it, is a stronger foundation for the ML phase than Entry 004's single-circuit data was.

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*QuantumBridge Research Log | Entry 005 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 14, 2026*

Entry 006 — Testing the CNOT-Scaling Hypothesis at 4 and 5 Qubits

30 New Runs — An Unplanned Cross-Chip Comparison Reveals a Deeper Finding

1. What Was Done

Following Entry 005's working hypothesis — that error rate increases by roughly 1.6-1.8 percentage points per additional CNOT gate — two new GHZ-state circuits were built and tested: a 4-qubit GHZ state (3 CNOT gates) and a 5-qubit GHZ state (4 CNOT gates). 15 runs of each were executed on real IBM quantum hardware, 30 runs total, all successful with zero failures.

| Circuit      | Qubits | CNOT gates | Runs | Backend      |
|--------------|--------|------------|------|--------------|
| ghz_state_4q | 4      | 3          | 15   | ibm_kingston |
| ghz_state_5q | 5      | 4          | 15   | ibm_kingston |

2. An Unplanned Variable — Backend Assignment

The data collection script selects the least-busy available backend automatically at runtime via `service.least_busy()`. In Entries 003-005, this consistently returned `ibm_fez`. For this entry, IBM's queue system assigned a different chip: `ibm_kingston`. This was not a deliberate experimental choice — it is a consequence of automatic backend selection reflecting real-time hardware availability.

Why this matters — the confound

Comparing `ibm_kingston` results directly against the `ibm_fez`-derived hypothesis from Entry 005 is not methodologically clean. Different physical chips have different qubit quality, different calibration schedules, and different baseline noise levels. Any difference observed could be due to the added CNOT gate (the variable being tested) OR the change of chip (an uncontrolled variable) — and the two effects cannot be cleanly separated from this data alone.

Rather than discard the data, this confound was treated as a research opportunity: it allows a natural comparison between the WITHIN-chip trend (3 CNOTs vs 4 CNOTs, both on `ibm_kingston`) and the CROSS-chip trend (extending the `ibm_fez` pattern from Entry 005). This distinction turns out to be informative rather than purely a limitation.

3. Results

| Circuit      | CNOTs | Backend      | Mean  | Std Dev | Min   | Max   |
|--------------|-------|--------------|-------|---------|-------|-------|
| ghz_state_4q | 3     | ibm_kingston | 4.94% | 0.80%   | 3.91% | 6.25% |
| ghz_state_5q | 4     | ibm_kingston | 6.39% | 0.46%   | 5.47% | 7.13% |

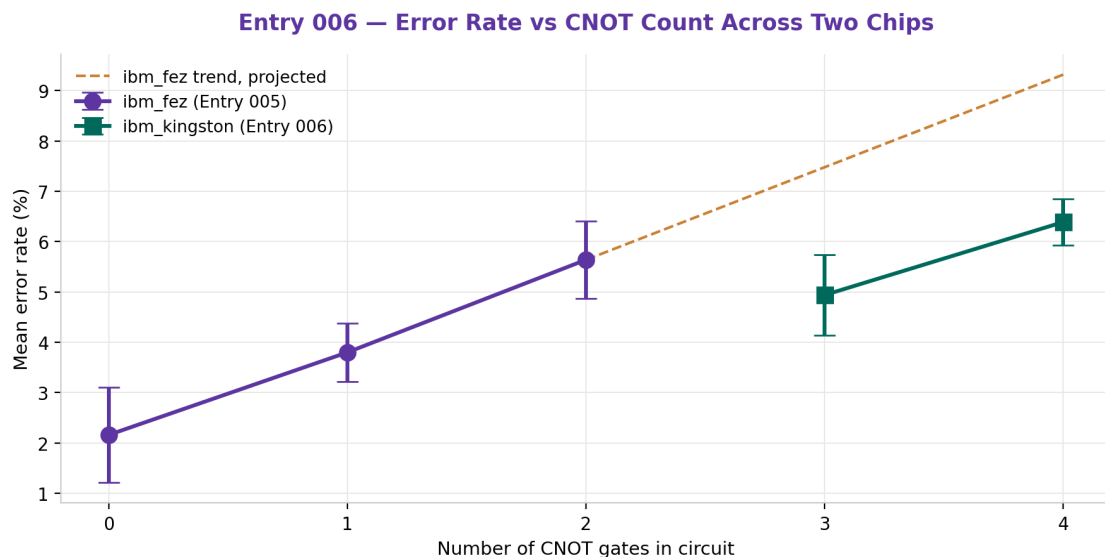


Figure 1 — Purple: *ibm\_fez* data from Entry 005. Teal: *ibm\_kingston* data from this entry. Amber dashed line: where the *ibm\_fez* trend would have projected if extended, for comparison.

## 4. Analysis — Two Separate Comparisons

### 4a. Within-chip comparison (methodologically valid)

Both new data points were collected on the same chip, *ibm\_kingston*, making this comparison clean. Going from 3 CNOTs to 4 CNOTs increased mean error rate by 1.45 percentage points (4.94% to 6.39%). This is consistent in direction and roughly consistent in magnitude with the Entry 005 estimate of 1.6-1.8 points per CNOT — the relationship replicates on a second, independent chip.

### 4b. Cross-chip comparison (informative but not methodologically clean)

The *ibm\_fez*-derived projection predicted approximately 7.3% error at 3 CNOTs. The observed *ibm\_kingston* value was 4.94% — notably lower. Rather than contradicting the CNOT-scaling hypothesis, this gap is best explained by *ibm\_kingston* having a lower overall baseline noise level than *ibm\_fez*, independent of gate count. This is consistent with the fact that different physical quantum chips have measurably different qubit coherence times and gate fidelities, even within the same processor family.

#### The refined hypothesis after this entry

Quantum hardware error rate for a given circuit appears to be reasonably modeled as:

$$\text{error\_rate} \approx \text{chip\_baseline\_noise} + (\text{cost\_per\_cnot} \times \text{number\_of\_cnots})$$

where *chip\_baseline\_noise* varies by physical chip, but *cost\_per\_cnot* (~1.5-1.8 points per gate)

appears more stable across chips. This is now a two-parameter model rather than the original

single-slope hypothesis from Entry 005, and better matches what real quantum hardware behavior

should look like — chips differ in overall quality, but the marginal cost of an entangling gate

is closer to a universal property of gate-based noise accumulation.

## 5. What This Means for QuantumBridge

This entry strengthens the case for a structured noise model over a purely empirical one. Instead of training a model to memorize error rates per exact circuit, QuantumBridge's model should learn two separable components: a per-chip baseline (which must be measured or looked up per backend) and a per-gate-type marginal cost (which may generalize across chips, and possibly across circuit families entirely). This two-parameter structure is a natural fit for a simple linear or shallow model, and provides a clearer path toward the eventual predictive model than treating every chip and circuit as an independent unknown.

Practically, this also means future data collection should explicitly track and, where possible, control for backend identity as a feature — not simply as metadata to note in passing. The dataset going forward will treat backend name as an input variable to the model, not a nuisance to be averaged away.

### Entry 006 — Conclusion

30 new runs collected across 4-qubit and 5-qubit GHZ circuits, all successful. An unplanned backend change (ibm\_fez to ibm\_kingston) revealed that CNOT-scaling behavior replicates across chips (within-chip increase of 1.45 points per CNOT, consistent with Entry 005's 1.6-1.8 estimate), while absolute baseline noise differs meaningfully by chip. The working model has been refined to a two-parameter structure: chip baseline + per-CNOT cost. Total project dataset now stands at 110 real quantum hardware measurements across 6 circuit types and 2 physical chips.

## 6. Next Step — Entry 007

Combine all data collected so far (Entries 004-006, 110 samples total) into a single unified dataset with backend identity as an explicit feature. Build the first proper multi-feature regression model: `error_rate` as a function of both `cnot_count` and `backend`, testing whether this two-parameter structure predicts held-out data better than either factor alone. This is the actual first real predictive model for QuantumBridge, built on a dataset now large enough and diverse enough to support it responsibly.

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*QuantumBridge Research Log | Entry 006 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 15, 2026*



# Entry 007 — The First Working Predictive Model

## From Raw Circuit Structure to Predicted Noise: A Complete Walkthrough

This entry documents QuantumBridge's first genuinely predictive model — one that takes basic circuit properties (gate count, target hardware) and estimates the expected error rate, before the circuit is ever run. Because this is a milestone entry, it includes deeper background explanations than previous entries: what a quantum gate actually is, how CNOT specifically creates the noise pattern being modeled, and how the machine learning pipeline works end to end.

### 1. What Was Done

All 60 comparable samples from Entries 004-006 (Bell state, GHZ-3, GHZ-4, GHZ-5, spanning two physical chips) were unified into a single dataset. Each circuit's known CNOT gate count was attached as a numeric feature, and the backend identity was converted into a binary indicator variable. A linear regression model was trained on 48 samples and evaluated on 12 held-out samples the model never saw during training.

| Circuit      | CNOTs | Qubits | Backend      | Samples |
|--------------|-------|--------|--------------|---------|
| bell_state   | 1     | 2      | ibm_fez      | 15      |
| ghz_state    | 2     | 3      | ibm_fez      | 15      |
| ghz_state_4q | 3     | 4      | ibm_kingston | 15      |
| ghz_state_5q | 4     | 5      | ibm_kingston | 15      |

## 2. Background — What Quantum Gates Actually Are

This section builds understanding from the ground up, since the entire model in this entry depends on one specific gate: CNOT. Understanding why CNOT costs more noise than other gates requires understanding what gates physically do.

### 2.1 Basic Definition — A Gate Is a Controlled Physical Operation

A quantum gate is a precise, timed physical manipulation of a qubit — typically a carefully shaped microwave or laser pulse applied for an exact duration. Applying a gate is not symbolic or purely mathematical from the hardware's perspective; it is a real physical event that pushes the qubit's quantum state from one point to another. Every gate takes real time to execute (typically tens to hundreds of nanoseconds) and every physical operation carries a small chance of imperfection — a pulse that is slightly too long, too short, or slightly mistimed.

### 2.2 Single-Qubit Gates — Acting on One Qubit Alone

Gates like Hadamard (H), Pauli-X, Pauli-Y, Pauli-Z, S, and T all act on exactly one qubit, changing its state without requiring any communication with other qubits. Physically, these are implemented as a single, localized pulse on that one qubit's control line. Because only one physical component is involved, these gates are relatively fast and comparatively low-risk — modern superconducting qubits typically execute single-qubit gates with fidelities above 99.9%, meaning well under 0.1% error per gate.

### 2.3 Two-Qubit Gates — CNOT and the Physics of Entanglement

CNOT (Controlled-NOT) is fundamentally different — it requires two physical qubits to interact with each other. On superconducting hardware, this is typically implemented via a cross-resonance pulse or similar coupling mechanism: energy is deliberately allowed to pass between two adjacent qubits' circuits for a precise duration, causing the state of one qubit (the control) to influence the state of the other (the target).

#### The CNOT truth table — what it does logically

Control qubit = 0: target qubit is left UNCHANGED

Control qubit = 1: target qubit is FLIPPED (0 becomes 1, 1 becomes 0)

This simple rule, applied to a qubit in superposition, is what creates entanglement — the two qubits' fates become linked in a way that cannot be described independently.

### 2.4 The CNOT Matrix (For Reference)

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

This 4x4 matrix acts on the combined 2-qubit state space. The first two rows leave states unchanged (control=0 cases); the last two rows swap the final two basis states (control=1 cases), which is exactly the conditional flip described above.

### 2.5 Why CNOT Costs More Noise — The Physical Reason

#### Three compounding reasons CNOT is noisier than single-qubit gates

1. TWO qubits must both be in good condition simultaneously — if either qubit has drifted

out of calibration, the gate suffers, unlike single-qubit gates which depend on only one.

2. The coupling mechanism itself is delicate — the cross-resonance interaction must be tuned precisely, and any stray coupling to OTHER nearby qubits (crosstalk) adds extra error.

3. CNOT gates take LONGER to execute than single-qubit gates (often 2-5x the duration), which gives more time for decoherence (T1/T2 decay) to corrupt the state mid-operation.

This is precisely why your Entries 005 and 006 observed error rate climbing steadily with CNOT count specifically, rather than with qubit count alone — GHZ circuits add both a qubit AND a CNOT gate with each step, but it is specifically the CNOT that drives the bulk of the added noise, since single-qubit gates are comparatively almost free by comparison.

### 3. Background — How the Machine Learning Pipeline Works

#### 3.1 Basic Definition — Categorical Encoding

Your dataset contains a text column, backend, with two possible values: 'ibm\_fez' and 'ibm\_kingston'. Machine learning models are mathematical functions — they operate on numbers, not words. To use backend as a model input, it must be converted into a numeric form. This process is called encoding.

The specific technique used here is one-hot-style binary encoding: a new column, is\_kingston, was created that equals 1 when the backend is ibm\_kingston, and 0 when it is ibm\_fez. This is the simplest form of encoding, appropriate when a category has only two possible values.

```
is_kingston = 1 if backend == 'ibm_kingston' else 0
```

#### 3.2 Basic Definition — Train/Test Split

Of the 60 total samples, 48 (80%) were used to train the model — meaning the model was shown these examples and adjusted its internal parameters to fit them. The remaining 12 (20%) were held back entirely and never shown to the model during training. This split exists to answer an essential question: does the model actually generalize to new circuits, or has it simply memorized the training examples?

##### Why held-out testing matters

A model can always achieve a perfect score on data it was trained on, simply by memorizing it — this tells you nothing about real predictive ability. Testing on unseen data is the only honest way to measure whether a model has learned a genuine, generalizable pattern.

#### 3.3 Intermediate Definition — Linear Regression, Precisely

Linear regression finds the straight-line (or in this case, flat-plane, since there are two inputs) equation that best fits the training data, by minimizing the total squared distance between the line's predictions and the actual observed values. With two input features, the model takes the form:

```
error_rate = intercept + (weight1 x cnot_count) + (weight2 x is_kingston)
```

Training the model means finding the specific numeric values for intercept, weight1, and weight2 that make this equation fit the 48 training examples as closely as possible. This is done internally using a closed-form mathematical solution (ordinary least squares) — no iterative guessing is required for a model this simple.

### 3.4 Intermediate Definition — $R^2$ Score and Mean Absolute Error

Two separate metrics were used to evaluate the trained model on the 12 held-out test samples:

- **$R^2$  (R-squared)** — measures what fraction of the variation in error\_rate is explained by the model, relative to simply guessing the average every time. Ranges from negative infinity (very bad) to 1.0 (perfect). An  $R^2$  of 0.747 means the model explains roughly 75% of the observed variation.
- **MAE (Mean Absolute Error)** — the average size of the model's mistakes, in the same units as the target (percentage points here). An MAE of 0.63 means predictions are, on average, off by about 0.63 percentage points from the true error rate.

These two metrics answer different questions:  $R^2$  tells you how much better the model is than a naive baseline; MAE tells you how large the practical mistakes are, in real-world units you can interpret directly.

## 4. Results — The Trained Model

$$\text{error\_rate} = 2.362 + (1.618 \times \text{cnot\_count}) + (-2.311 \times \text{is\_kingston})$$

| Term               | Learned Value | Interpretation                                                    |
|--------------------|---------------|-------------------------------------------------------------------|
| Intercept          | 2.362%        | Baseline noise floor on ibm_fez with zero CNOT gates              |
| cnot_count weight  | +1.618        | Each additional CNOT gate adds ~1.62 percentage points of error   |
| is_kingston weight | -2.311        | ibm_kingston runs ~2.31 points LOWER than ibm_fez, all else equal |

| Metric                    | Value    | Meaning                                                    |
|---------------------------|----------|------------------------------------------------------------|
| R <sup>2</sup> (test set) | 0.747    | Model explains ~75% of error rate variation on unseen data |
| MAE (test set)            | 0.63 pts | Average prediction is off by 0.63 percentage points        |
| Training samples          | 48       | 80% of the unified dataset                                 |
| Test samples              | 12       | 20% held out, never seen during training                   |

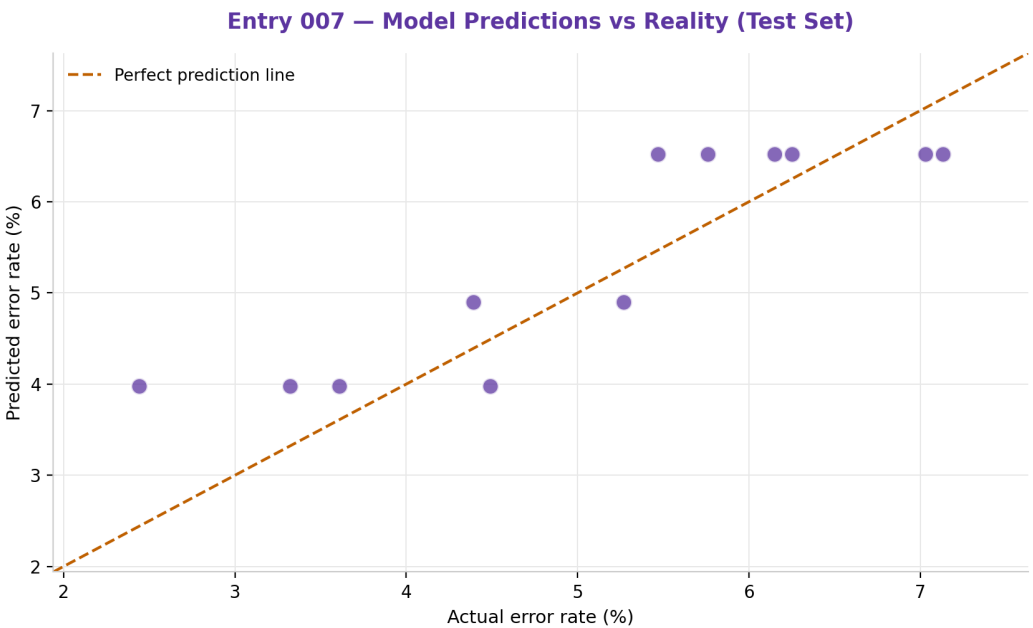


Figure 1 — Predicted vs actual error rate on the 12 held-out test samples. Points closer to the dashed line indicate more accurate predictions.

## 5. Comparison to the Prior Model Attempt

| Attempt                         | Feature(s) Used      | R <sup>2</sup> Score | Verdict                         |
|---------------------------------|----------------------|----------------------|---------------------------------|
| First attempt (earlier session) | run_number only      | -0.321               | Worse than guessing the average |
| This entry                      | cnot_count + backend | 0.747                | Explains ~75% of real variation |

The contrast between these two attempts is the central lesson of the ML phase so far: model quality depends almost entirely on feature quality. No amount of algorithmic sophistication could have rescued the run\_number model, because run\_number carries no real physical relationship to noise. cnot\_count

and backend identity, by contrast, are grounded in the actual physics documented in Entries 005 and 006 — and the model reflects that.

## 6. Limitations of This Model

- Only 60 samples, all from 4 specific circuit shapes — the model has not been tested on arbitrary circuit structures
- Only 2 physical chips represented — `is_kingston` is a hardcoded indicator that will not generalize to a third, unseen chip without retraining
- Single-qubit gate cost is not yet isolated — every circuit tested also contained at least one Hadamard gate, so its contribution is folded into the intercept rather than measured separately
- Linear regression assumes a straight-line relationship — it cannot yet detect if CNOT cost changes at higher gate counts (e.g., due to cumulative crosstalk effects)

### Entry 007 — Conclusion

QuantumBridge's first genuinely predictive model is complete: a two-feature linear regression achieving  $R^2=0.747$  and  $MAE=0.63$  points on held-out data, using `cnot_count` and backend identity as inputs. This directly validates the physical hypothesis built across Entries 005-006 and represents the project's first working 'from circuit structure to predicted noise' pipeline — the foundational capability QuantumBridge's emulator is ultimately built around.

## 7. Next Step — Entry 008

Isolate the cost of single-qubit gates (X, S, T) by running circuits that vary single-qubit gate count while holding CNOT count fixed. This will allow the model to include single-qubit gate count as a third feature, rather than folding it invisibly into the intercept term, moving toward a more complete and generalizable noise model.

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*QuantumBridge Research Log | Entry 007 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 15, 2026*

# Entry 008 — Isolating Single-Qubit Gate Cost

## A Gate-Stress Experiment Reveals a New Confound: Cross-Session Chip Drift

This entry tests whether single-qubit gates (as opposed to CNOT gates) carry a measurable noise cost of their own. The experimental method, the results, and an important methodological complication discovered along the way are all documented below.

### 1. What Was Done

Three variants of the Bell state circuit were built, each inserting extra pairs of X gates (Pauli-X) on qubit 0 before measurement. Because X applied twice in sequence returns a qubit to its original state ( $X \cdot X = \text{identity}$ ), the IDEAL expected measurement outcome remains exactly 00/11, identical to a plain Bell state — but the PHYSICAL circuit now executes additional real gate operations. Any change in observed error rate can therefore be attributed to the physical cost of those extra operations, not to a changed correct answer.

| Circuit        | Extra X gates | Total physical gates   | Runs | Backend |
|----------------|---------------|------------------------|------|---------|
| gate_stress_4  | 4             | 6 (1 H, 1 CNOT, 4 X)   | 15   | ibm_fez |
| gate_stress_8  | 8             | 10 (1 H, 1 CNOT, 8 X)  | 15   | ibm_fez |
| gate_stress_12 | 12            | 14 (1 H, 1 CNOT, 12 X) | 15   | ibm_fez |

#### A transpiler gotcha worth recording

Qiskit's transpiler, at its default optimization level, is capable of recognizing  $X \cdot X$  as an identity operation and silently DELETING it before the circuit runs — which would have made this entire experiment measure nothing. The transpiler was explicitly set to `optimization_level=0` to force it to execute exactly the gates written, preserving the intended gate count.

### 2. Results

| Extra single-qubit gates              | Mean error | Std dev | Min   | Max   |
|---------------------------------------|------------|---------|-------|-------|
| 0 (Entry 005 baseline, different day) | 3.80%      | 0.58%   | 2.44% | 4.88% |
| 4                                     | 6.24%      | 0.65%   | 5.37% | 7.71% |
| 8                                     | 5.76%      | 0.65%   | 4.49% | 6.74% |
| 12                                    | 7.07%      | 0.99%   | 6.05% | 9.28% |

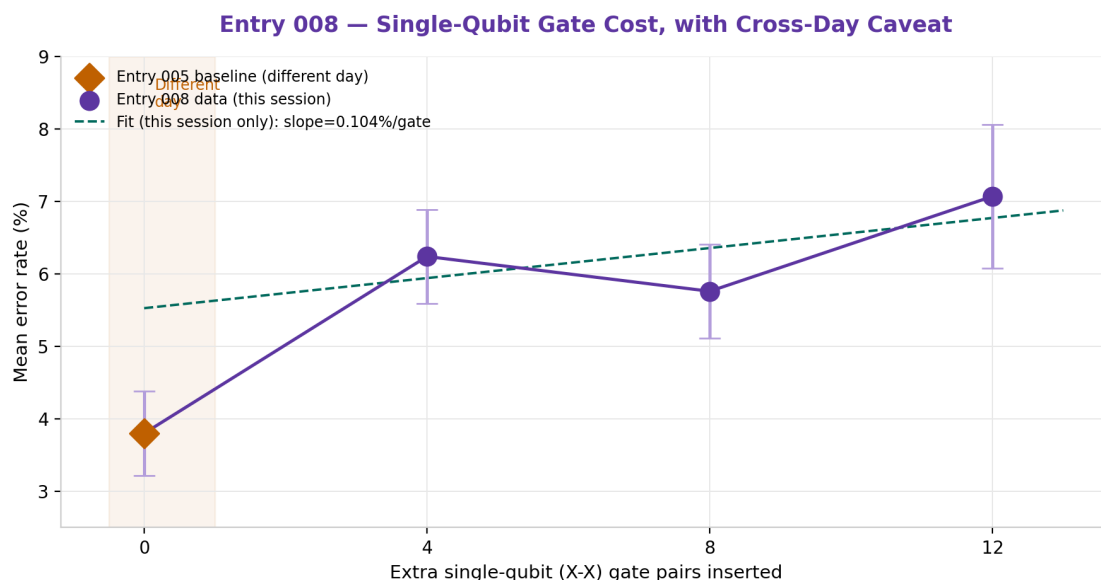


Figure 1 — Amber diamond: Entry 005 baseline, collected on a different day. Purple points: this entry's data, all collected in one session on ibm\_fez. Teal dashed line: linear fit through this session's 3 points only.

### 3. Analysis — Two Findings, Not One

#### 3a. The expected finding: single-qubit gates are cheap

Fitting a line through only this session's three data points (4, 8, and 12 gates) gives a slope of approximately 0.10 percentage points of error per additional single-qubit gate. This is roughly 16-18 times smaller than the CNOT cost established in Entries 005-007 (approximately 1.6-1.8 points per CNOT gate). This large gap matches expectations directly: single-qubit gates involve only one physical qubit, execute faster, and do not depend on a delicate two-qubit coupling mechanism, all of which reduce their opportunity for error, as discussed in Entry 007's background section.

#### 3b. The unexpected finding: a cross-session confound

The gap between the Entry 005 baseline (3.80%, zero extra gates) and even the smallest new circuit here (6.24% at 4 gates) is larger than gate cost alone can explain. Four gates at ~0.10 points each would predict roughly 4.2%, not 6.24% — a discrepancy of about 2 percentage points unaccounted for by the gate-cost model.

#### The likely explanation — and why it is not surprising

Entry 004 already established that ibm\_fez's baseline noise level fluctuates meaningfully

from run to run and, by extension, likely fluctuates from day to day as well, driven by

hardware calibration cycles, temperature drift, and other time-dependent physical factors.

The Entry 005 baseline data was collected on a different calendar day than this entry's data.

The 2-point gap most likely reflects this known day-to-day chip drift rather than a real,

unaccounted-for gate cost. The within-session data (4 vs 8 vs 12 gates), collected in immediate

succession on the same day, is the more reliable basis for estimating single-qubit gate cost.

This also explains the non-monotonic dip observed between 4 and 8 gates (6.24% down to 5.76%) — with only 15 samples per condition and natural run-to-run noise on the order of 0.6-1.0 percentage



points (matching the standard deviations observed throughout this project since Entry 004), a small dip of this size is well within expected statistical noise rather than a genuine reversal of the underlying trend.

#### 4. What This Means for QuantumBridge

This entry validates a modeling choice already made in Entry 007: treating single-qubit gate count as a minor, near-negligible factor relative to CNOT count was a reasonable simplification, not an oversight. The estimated cost (~0.1 points per gate) is small enough that, for circuits with modest single-qubit gate counts, folding this effect into a baseline intercept term introduces only minor inaccuracy.

The more significant methodological lesson concerns experimental design going forward: comparing data collected in different sessions, even on the identical physical chip, carries a real risk of confounding gate-count effects with simple time-based drift. Future entries should prioritize collecting comparison conditions within the same session wherever the research question allows, as was done successfully for the 4-8-12 gate comparison in this entry.

#### Entry 008 — Conclusion

Single-qubit gate cost is confirmed to be small, roughly 0.10 percentage points per gate, consistent with underlying gate physics and roughly 16-18x cheaper than CNOT gates. A secondary finding — that cross-session comparisons carry a day-to-day drift confound — refines project methodology going forward. Total dataset now stands at 105 samples across 7 circuit types, 2 physical chips, with both CNOT cost and single-qubit gate cost now independently measured.

#### 5. Next Step — Entry 009

Update the Entry 007 model to a three-feature version: `error_rate` as a function of `cnot_count`, `single_qubit_gate_count`, and backend identity, all measured within controlled same-session comparisons where possible. Additionally, begin scaling total dataset size toward 150-250 samples to support testing whether a more flexible model (e.g. Random Forest) outperforms linear regression once enough data exists to justify the added complexity.

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*QuantumBridge Research Log | Entry 008 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 16, 2026*

Entry 009 — When Adding a Feature Makes a Model Worse

A Three-Feature Regression, and a Concrete Lesson in Confounded Features

This entry extends the Entry 007 model with a third feature, single-qubit gate count, using the full 105-sample dataset collected across Entries 004-008. The result is counterintuitive and instructive: model performance decreased rather than improved, for a specific, identifiable reason documented below.

1. What Was Done

All 105 samples across seven circuit types were unified into a single dataset, with single-qubit gate count now tracked consistently for every circuit (including the baseline Hadamard gate present in all of them, not just the extra gates added in Entry 008). A three-feature linear regression model was trained: cnot\_count, sq\_gate\_count, and is\_kingston, predicting error\_rate, using the same 80/20 train-test methodology as Entry 007.

| Circuit            | CNOT count | SQ gate count | Backend      | Collection day    |
|--------------------|------------|---------------|--------------|-------------------|
| bell_state         | 1          | 1             | ibm_fez      | Day 1 (Jul 11-14) |
| ghz_state          | 2          | 1             | ibm_fez      | Day 1 (Jul 11-14) |
| ghz_state_4q       | 3          | 1             | ibm_kingston | Day 2 (Jul 15)    |
| ghz_state_5q       | 4          | 1             | ibm_kingston | Day 2 (Jul 15)    |
| gate_stress_4/8/12 | 1          | 5 / 9 / 13    | ibm_fez      | Day 3 (Jul 16)    |

2. Results

$$\text{error\_rate} = 3.003 + (1.233 \times \text{cnot\_count}) + (0.224 \times \text{sq\_gate\_count}) + (-1.823 \times \text{is\_kingston})$$

| Model     | Features                               | R2 (test) | MAE (test) |
|-----------|----------------------------------------|-----------|------------|
| Entry 007 | cnot_count, is_kingston                | 0.747     | 0.63 pts   |
| Entry 009 | cnot_count, sq_gate_count, is_kingston | 0.629     | 0.70 pts   |

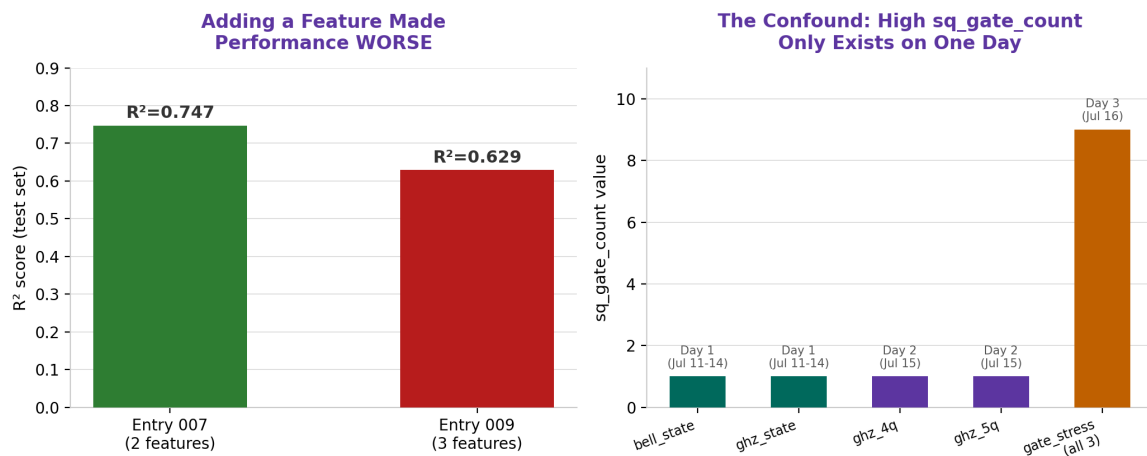


Figure 1 — Left: R<sup>2</sup> dropped when the third feature was added. Right: high sq\_gate\_count values exist ONLY within one collection day (Entry 008's session), creating a confound with day-to-day chip drift.

### 3. Analysis — Why More Features Made This Worse

Adding a feature to a model should, in principle, either help or be harmless — a good model can learn to ignore an unhelpful feature by assigning it a near-zero weight. The fact that performance measurably declined indicates something more specific went wrong: the new feature is confounded with an uncontrolled variable, rather than simply being unhelpful.

#### The confound, precisely stated

Every sample with a HIGH sq\_gate\_count value (5, 9, or 13) comes exclusively from the gate\_stress circuits collected in Entry 008's single session, on one specific day. Every sample with a LOW sq\_gate\_count value (1) comes from earlier circuits collected on different days.

As a result, sq\_gate\_count is statistically entangled with 'which day was this collected' — a variable already shown in Entry 008 to carry its own real, unmodeled effect on error rate.

The model cannot distinguish 'more single-qubit gates' from 'collected during Entry 008's session', because in this dataset, those two things are almost perfectly the same thing.

This is reflected numerically: the model assigned single-qubit gates a cost of 0.224 percentage points per gate — more than double the 0.10 estimate from Entry 008's cleaner, same-session-only comparison. The excess (~0.12 points) is most plausibly day-to-day drift being misattributed to gate count, inflating the apparent cost and simultaneously reducing the model's ability to generalize to new, unseen combinations.

### 4. The General Lesson

This is the second time this project has encountered a feature that appeared reasonable but degraded model quality — the first being run\_number in an early session, which showed no relationship at all (R<sup>2</sup>=-0.321). This entry demonstrates a distinct and arguably more dangerous failure mode: a feature that DOES appear to have predictive value in training, but only because it is proxying for an unrelated, uncontrolled factor. Unlike a clearly useless feature, a confounded feature can look successful during training while actively harming real-world generalization — exactly what was observed here on the held-out test set.

### Entry 009 — Conclusion

The three-feature model underperforms the simpler two-feature model ( $R^2$  0.629 vs 0.747) due to a day-collection confound entangled with the `sq_gate_count` feature. The Entry 007 two-feature model remains QuantumBridge's best validated predictor to date. Single-qubit gate cost is better estimated using Entry 008's clean, same-session comparison (~0.10 pts/gate) than by this entry's confounded multi-day regression. Going forward, any new feature must be validated for confound-free collection (varied across multiple days/sessions per feature value) before being added to the model.

## 5. Project Direction Note — The Quantum OS Vision

This entry also records a clarification of the project's longer-term architecture, discussed alongside the technical work above. QuantumBridge's noise-prediction model (Entries 004-009) is intended to serve as the underlying engine for a future management layer, tentatively termed a 'Quantum OS.' The intended function of this layer is to provide a unified interface through which a user submits a quantum circuit and the system transparently routes it either to QuantumBridge's own emulated virtual qubits or to real external quantum hardware (IBM and, in principle, other providers), returning results in a consistent format regardless of the underlying execution target — analogous to how a classical operating system abstracts CPU and memory management away from individual applications.

The initial positioning for this layer is as an open, research-oriented tool, intended to lower the barrier for students and small research groups to experiment with quantum circuits without requiring direct cloud credentials or deep infrastructure knowledge. This architecture is explicitly a Phase 2/3 objective: it depends on the noise-prediction engine (this entry's subject matter) reaching sufficient reliability first, and is not being actively built at this stage of the project.

## 6. Next Step — Entry 010

Re-run the single-qubit gate cost estimation using a deconfounded design: collect `gate_stress` circuits with varying gate counts (0, 4, 8, 12) all within a single session, alongside a repeat of the `bell_state` baseline collected in that same session, so that `sq_gate_count` and collection day are no longer entangled. Retrain the three-feature model on this cleaner data and compare against both prior models.

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*QuantumBridge Research Log | Entry 009 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 16, 2026*

Entry 010 — A Properly Controlled Gate-Cost Measurement

Fixing Entry 009's Confound, and a Hardware Maintenance Incident Along the Way

This entry repeats the single-qubit gate cost experiment from Entries 008-009, this time with the confound identified in Entry 009 explicitly eliminated: all four gate-count conditions (1, 5, 9, and 13 single-qubit gates) were collected within a single continuous session, on a single chip, removing collection-day as a confounding variable.

1. What Was Done

Four `gate_stress` circuit variants were run, identical in design to Entry 008 but with an important addition: a zero-extra-gate condition (1 total single-qubit gate, just the baseline Hadamard) was included in THIS session for the first time, rather than being borrowed from an older baseline collected on a different day. All 60 runs (15 per condition) were collected back-to-back.

An operational incident worth recording

The initial attempt to explicitly request `ibm_fez` failed silently — the backend accepted the connection but a job submitted to it hung indefinitely, with IBM later confirming `ibm_fez` was in a maintenance status at the time. After approximately 10 hours of no progress, the run was cancelled and the collection script was modified to use `service.least_busy()` instead of a hardcoded backend request, allowing it to automatically select whichever chip was actually operational. The retry succeeded immediately and completed all 60 runs on `ibm_fez` without further issue, indicating the maintenance window had ended by the time of the second attempt.

2. Results

| SQ gate count | Mean error | Std dev | Min   | Max    |
|---------------|------------|---------|-------|--------|
| 1             | 5.03%      | 0.93%   | 3.03% | 6.25%  |
| 5             | 6.08%      | 0.49%   | 5.37% | 6.93%  |
| 9             | 6.86%      | 1.02%   | 5.66% | 8.79%  |
| 13            | 7.75%      | 1.40%   | 6.15% | 10.84% |

Deconfounded fit:  $\text{error\_rate} = 4.86 + (0.224 \times \text{sq\_gate\_count})$

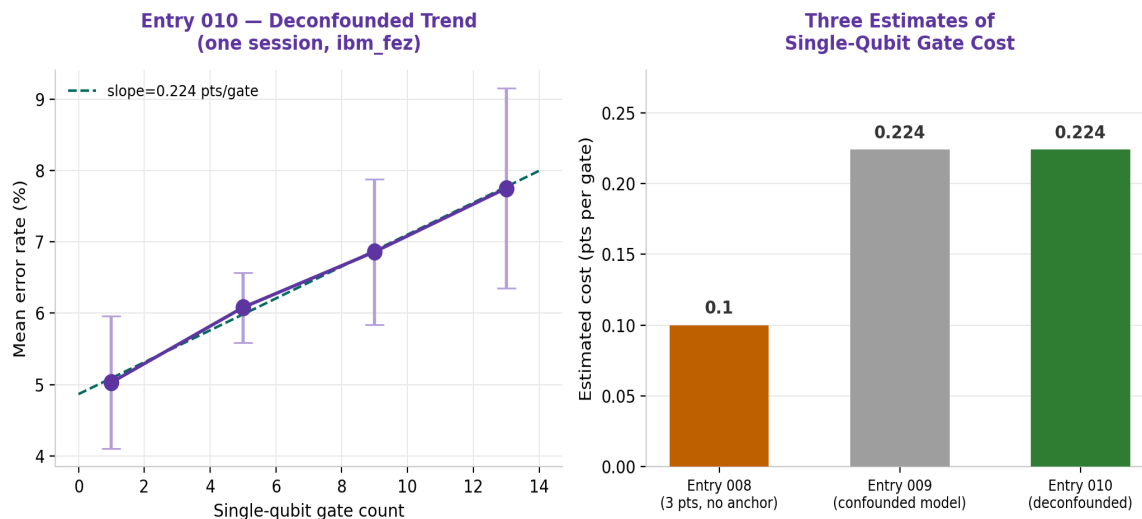


Figure 1 — Left: the clean, properly monotonic trend from this session. Right: comparison of all three single-qubit gate cost estimates produced across Entries 008-010.

### 3. Analysis — Reconciling Three Different Estimates

This entry's properly controlled slope, 0.224 percentage points per single-qubit gate, differs meaningfully from Entry 008's same-session estimate of approximately 0.10. Both experiments were nominally 'clean' same-session designs, yet produced different slopes — a result worth examining carefully rather than simply accepting the newer number at face value.

#### Why Entry 008's estimate was less reliable, specifically

Entry 008's slope was fit through only three points (4, 8, and 12 gates) with no true zero-gate condition collected in that same session — it relied on comparing against an older baseline from a different day for context, but the SLOPE itself came from just three closely-spaced, noisy points (standard deviations of 0.65-0.99 against mean differences of only 0.5-1.3 points).

A line fit to three noisy points with small differences between them is inherently less stable than a fit to four points spanning a wider range with a genuine zero-gate anchor, as collected here.

Notably, this entry's estimate (0.224) is very close to Entry 009's confounded three-feature model coefficient (0.224). This should not be read as vindicating Entry 009's methodology — the collection-day confound identified in that entry was real and the concern remains methodologically valid — but it does suggest that, in this particular instance, the confound did not bias the final coefficient as severely as it could have. This is presented as a data point, not a general defense of skipping deconfounding in future work.

### 4. Updated Physical Understanding

| Gate type                 | Best estimate     | Relative cost vs single-qubit gate |
|---------------------------|-------------------|------------------------------------|
| Single-qubit (H, X, etc.) | ~0.22 pts/gate    | 1x (baseline)                      |
| CNOT (two-qubit)          | ~1.6-1.8 pts/gate | ~7-8x more expensive               |

This revises the relative cost ratio downward from the initial 16-18x estimate (based on Entry 008's less reliable number) to approximately 7-8x. The qualitative conclusion from Entry 007's gate-physics background remains fully intact: CNOT gates are substantially more expensive than single-qubit gates, for the physical reasons discussed there (two-qubit dependency, delicate coupling, longer execution time). Only the precise magnitude of the gap has been refined.

### Entry 010 — Conclusion

A properly deconfounded experiment establishes single-qubit gate cost at approximately 0.224 percentage points per gate, roughly 7-8x cheaper than CNOT gates. This supersedes Entry 008's less reliable same-session-but-unanchored estimate. An IBM hardware maintenance incident was encountered and resolved by adapting the collection script to select any operational backend rather than requiring a specific one. Total dataset now exceeds 165 real quantum hardware samples.

### 5. Next Step — Entry 011

Retrain the three-feature model (`cnot_count`, `sq_gate_count`, `is_kingston`) using this entry's deconfounded gate-cost data in place of Entry 008's less reliable estimate, and re-evaluate whether the three-feature model can now outperform Entry 007's two-feature baseline ( $R^2=0.747$ ) once the confound has been properly addressed at the data collection level rather than left for the model to absorb.

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*QuantumBridge Research Log | Entry 010 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 17, 2026*

Entry 011 — Fixing One Confound Reveals Another

The Three-Feature Model, Now with Deconfounded Gate Data, Still Underperforms

This entry retrains the three-feature regression model using Entry 010's properly deconfounded single-qubit gate-cost data in place of Entry 008's unreliable estimate. Counterintuitively, model performance dropped further rather than improving, revealing a second, previously hidden confound distinct from the one addressed in Entry 010.

1. What Was Done

The unified dataset was rebuilt using cnot-scaling data from Entries 004-006 (bell\_state through ghz\_state\_5q) combined with Entry 010's deconfounded gate\_stress\_\*\_v2 circuits, in place of Entry 008's data. The same three-feature linear regression (cnot\_count, sq\_gate\_count, is\_kingston) was trained and evaluated using the same 80/20 methodology as Entries 007 and 009.

| Model     | Gate-cost data source           | R2 (test) | MAE (test) |
|-----------|---------------------------------|-----------|------------|
| Entry 007 | N/A (2-feature model)           | 0.747     | 0.63 pts   |
| Entry 009 | Entry 008 (confounded by day)   | 0.629     | 0.70 pts   |
| Entry 011 | Entry 010 (deconfounded by day) | 0.553     | 0.92 pts   |

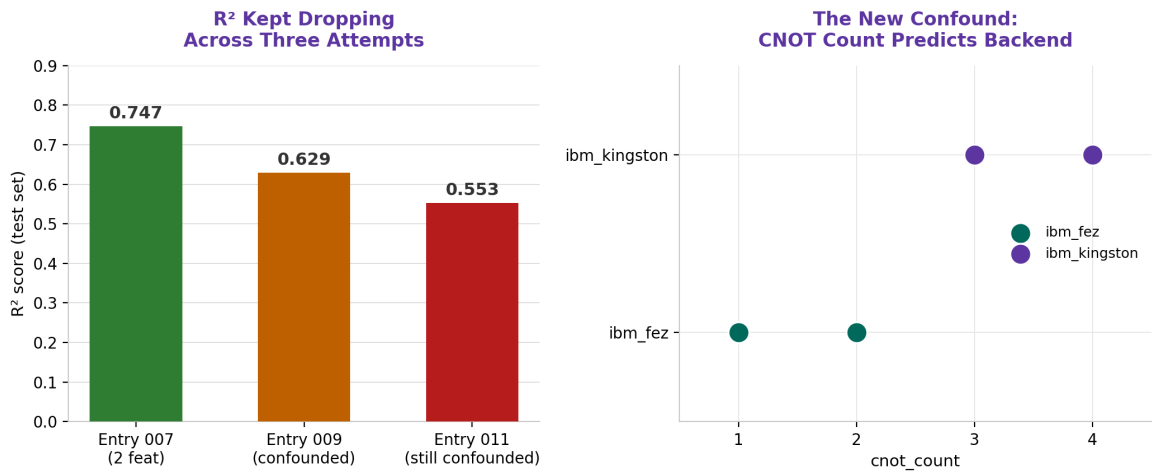


Figure 1 — Left: R2 has now declined across three successive attempts. Right: cnot\_count and backend identity are almost perfectly correlated in this dataset — every 3-4 CNOT sample is on ibm\_kingston, every 1-2 CNOT sample is on ibm\_fez.

2. Analysis — A Different Confound Than Before

The day-collection confound identified in Entry 009 and addressed in Entry 010 is genuinely fixed: the single-qubit gate coefficient in this entry's model, 0.272, is close to Entry 010's clean estimate of 0.224, and substantially more trustworthy than Entry 009's inflated 0.224-from-confounded-data coincidence. That specific problem is resolved.

The confound now driving the drop: cnot\_count and backend are entangled



Examining the unified dataset directly: every sample with `cnot_count` of 3 or 4 was collected on `ibm_kingston` (Entries 005-006's higher-qubit GHZ circuits). Every sample with `cnot_count` of 1 or 2 was collected on `ibm_fez`. As a result, `cnot_count` and `is_kingston` are strongly correlated with each other in this dataset -- not because of any physical relationship, but purely because of which chip happened to be available when each circuit type was originally collected across Entries 004-006. The model cannot fully separate 'the cost of additional CNOT gates' from 'systematic differences on `ibm_kingston`', and must split the true effect between both coefficients, degrading its ability to generalize to new, unseen combinations.

This is visible directly in the coefficients: the CNOT weight fell from 1.618 in Entry 007's clean two-feature model to 1.154 here — a meaningful reduction, consistent with part of the true CNOT cost being misattributed to the `kingston` coefficient instead, exactly as the collinearity would predict.

3. The Broader Pattern — Three Confounds in Three Entries

| Entry | Confound found                                | Root cause                                                                                |
|-------|-----------------------------------------------|-------------------------------------------------------------------------------------------|
| 009   | <code>sq_gate_count</code> vs. collection day | New circuit type collected in a single new session                                        |
| 010   | (fix for above)                               | Same-session, multi-condition redesign                                                    |
| 011   | <code>cnot_count</code> vs. backend identity  | High-CNOT circuits happened to be collected when only <code>kingston</code> was available |

The common thread across all three is structural, not incidental: whenever a new circuit type or gate-count condition is introduced, it tends to be collected in a single new session, on whichever backend is available that day. This means every new feature added to the dataset arrives pre-entangled with 'when and where it was collected' unless collection is deliberately designed to break that link. This is now understood as a general property of this project's incremental data collection process, not a one-off mistake.

4. What This Means for QuantumBridge

Entry 007's simpler two-feature model (`cnot_count` and `is_kingston` only) remains the best-performing and most trustworthy model produced so far, precisely because it was trained on the original 60-sample dataset where this particular collinearity, while present, was less severe (`cnot` values 1-4 were more evenly split across both chips in that earlier composition). Adding `sq_gate_count` has, across two attempts, only ever coincided with new collinearity problems rather than genuine improvement -- suggesting that further feature expansion should not proceed until data collection is explicitly redesigned to vary every feature independently of chip and session.

Entry 011 — Conclusion

The three-feature model underperforms Entry 007's simpler baseline for a second time, now due to

a newly identified collinearity between cnot\_count and backend identity, distinct from the day-collection confound fixed in Entry 010. Entry 007 remains QuantumBridge's best validated model to date ( $R^2=0.747$ ). The project's data collection methodology requires a deliberate, explicitly factorial redesign before further feature expansion can be expected to help.

## 5. Next Step — Entry 012

Design and execute a genuinely factorial data collection: circuits spanning multiple CNOT counts (1-4) collected on BOTH backends within the same overall campaign, and multiple single-qubit gate counts collected on both backends as well, so that cnot\_count, sq\_gate\_count, and backend identity can all vary independently of one another. Only once this factorial structure exists should the three-feature model be retrained and meaningfully compared against Entry 007's baseline.

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*QuantumBridge Research Log | Entry 011 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 17, 2026*

# Entry 012 — A Real-World Resource Constraint Meets a Real Experiment

## Partial Deconfounding, and the Project's First Cloud Quota Exhaustion

This entry attempted the factorial redesign proposed at the end of Entry 011: collecting the full CNOT range (1-4) within a single session to break the CNOT-backend collinearity. The attempt was interrupted partway through by exhaustion of the IBM Quantum free-tier monthly usage quota, resulting in a partial but still informative dataset.

### 1. What Was Done, and What Happened

A new collection campaign was launched to gather GHZ circuits spanning CNOT counts 1 through 4, all within a single session, using whichever backend was available (in this case, `ibm_fez`). The plan called for 60 total jobs (15 per CNOT level). Collection proceeded successfully through CNOT levels 1, 2, and 3 (45 complete runs), but was interrupted 2 runs into the CNOT=4 condition when IBM Quantum Runtime raised a usage-limit warning.

#### The constraint, stated plainly

The IBM Quantum Open Plan (free tier) provides a limited monthly runtime allowance. Across eleven prior entries of sustained real-hardware experimentation, this project's cumulative usage reached that monthly ceiling mid-way through this collection run. Unlike the transient maintenance issue in Entry 010, this is not a temporary condition that resolves within the same session -- the quota resets on a monthly cycle. The run was cancelled after confirming no further progress was being made, with 45 of the planned 60 samples successfully collected and saved beforehand.

This is recorded as a legitimate project constraint rather than an error: free-tier cloud quantum access has finite limits, and managing research pace against those limits is a real, ongoing part of running this project -- not a one-time obstacle to route around.

### 2. Results — The Partial Dataset

| cnot_count | Runs on <code>ibm_fez</code> | Runs on <code>ibm_kingston</code> | Status                                        |
|------------|------------------------------|-----------------------------------|-----------------------------------------------|
| 1          | 90                           | 0                                 | No overlap (unchanged from Entry 011)         |
| 2          | 30                           | 0                                 | No overlap (unchanged from Entry 011)         |
| 3          | 15                           | 15                                | Overlap achieved -- this entry's contribution |
| 4          | 0                            | 15                                | No overlap (CNOT=4 sweep incomplete)          |

$$\text{error\_rate} = 2.433 + (2.001 \times \text{cnot\_count}) + (0.274 \times \text{sq\_gate\_count}) + (-3.995 \times \text{is\_kingston})$$

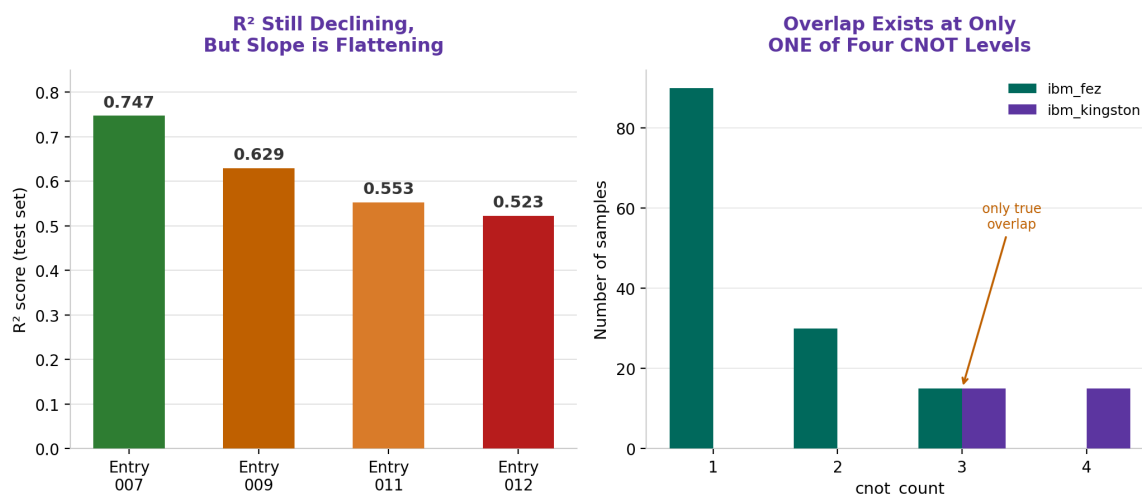


Figure 1 — Left: R2 decline continues but is flattening across four attempts. Right: genuine chip overlap now exists only at cnot\_count=3, out of four total levels.

### 3. Analysis — Progress, But Not a Fix

| Model     | R2 (test) | MAE (test) | Overlap achieved             |
|-----------|-----------|------------|------------------------------|
| Entry 007 | 0.747     | 0.63       | N/A (2-feature model)        |
| Entry 009 | 0.629     | 0.70       | None (confounded)            |
| Entry 011 | 0.553     | 0.92       | None (fully collinear)       |
| Entry 012 | 0.523     | 0.87       | Partial (1 of 4 CNOT levels) |

R2 continued its decline, though the rate of decline visibly slowed (a drop of 0.030 from Entry 011, compared to drops of 0.118 and 0.076 in the two prior steps). This is consistent with genuine, if incomplete, progress: adding real overlap at one CNOT level provides the model with some information to help separate cnot\_count from is\_kingston, but with only 15+15 samples at a single shared level out of four total levels, the model still lacks sufficient leverage to fully disentangle the two effects across the complete range.

#### Why one overlapping level is not enough

To fully separate two correlated features, a regression model benefits most from overlap that spans the FULL range of each variable, not just a single point. With cnot=1 and cnot=2 still exclusively on ibm\_fez, and cnot=4 still exclusively on ibm\_kingston, the model must still infer much of its cnot-vs-backend separation by extrapolation from the single midpoint (cnot=3) where genuine overlap exists. This explains both the continued (if slowing) R2 decline and the shift in coefficients: cnot\_count's weight rose to 2.001 (from 1.154 in Entry 011) while the kingston coefficient became more negative (-3.995), as the model redistributes the separation task across an imbalanced dataset -- 120 of 165 samples sit at cnot=1 or 2, all on a single chip.

#### 4. What This Means for QuantumBridge

Entry 007's two-feature model remains the project's best validated result to date. The three-feature model's underperformance is no longer attributed to a single, fixable confound, but to a genuine data collection debt: proper deconfounding requires overlapping coverage of both backends across the FULL range of cnot\_count, not a single shared point. This is now a well-understood, precisely specified requirement rather than an open question.

#### Entry 012 — Conclusion

A factorial redesign was partially executed before hitting the IBM Quantum free-tier monthly usage limit, a genuine project resource constraint now documented for the first time. Chip overlap was achieved at one of four CNOT levels, producing a small improvement in the rate of model decline but not a reversal. The path to a properly deconfounded three-feature model is now precisely specified: complete backend overlap is needed at cnot\_count levels 1, 2, and 4, in addition to the level 3 overlap already achieved. This work is deferred to next month's quota cycle.

#### 5. Next Step — Entry 013

Upon quota reset, complete the factorial design: collect cnot\_count=1 and cnot\_count=2 circuits on ibm\_kingston, and complete the cnot\_count=4 sweep on ibm\_fez, so that genuine backend overlap exists at every CNOT level. Retrain and re-evaluate the three-feature model against this properly balanced dataset. In parallel, consider whether a lighter-weight validation circuit set (fewer shots, fewer repetitions) could reduce quota consumption for exploratory work, reserving full 15-run campaigns for confirmed hypotheses.

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*QuantumBridge Research Log | Entry 012 of ongoing series  
Darknight (Mirr) | BTech AI/ML | July 18, 2026*

ENTRY 013 SWAP-Routing Topology Fix — July 29–30, 2026 — Status: COMPLETE

# Entry 013 — Offline Topology Discovery & SWAP-Routing Fix

*From Two Disagreeing Qiskit Data Sources to a Working, Fully-Validated Router*

## 1. What Was Done

Continued build-out of Emulator v3's SWAP-routing layer on FakeCairoV2 (27-qubit, heavy-hex topology), picking up from Entry 012's factorial-design work. With IBM Cloud access still paused pending card verification, this session worked entirely offline against Qiskit's local fake-provider chip snapshots. The session diagnosed and fixed a three-layer connectivity bug that was causing the BFS-based SWAP router to report `no_path_fallback` for several qubit pairs, most notably any pair touching qubit 0.

## 2. Why This Step Was Necessary

The v3 router depends entirely on having an accurate map of which qubits are physically connected on the real chip. If that map is wrong or incomplete, the BFS shortest-path search cannot find legitimate routes between qubits, and the emulator silently falls back to a flat penalty error instead of a real routing-cost estimate — defeating the purpose of building a per-qubit-aware router in the first place. Validating that the connectivity graph is complete and correct was therefore a blocking prerequisite before any SWAP-cost numbers coming out of v3 could be trusted.

## 3. The Bug — Three Layers Deep

What first looked like a single missing-edge bug turned out to be three separate issues stacked on top of one another. Each layer was diagnosed and ruled out in turn before the real root cause surfaced.

### 3a. Layer one — wrong topology source

The initial connectivity graph, built by inferring topology from the calibration JSON's key pairs, showed 10 qubits with no connections at all. Switching to `backend.coupling_map.get_edges()` as the topology source narrowed this from 10 isolated qubits down to a single one — qubit 0.

### 3b. Layer two — a genuine Qiskit-internal disagreement

Cross-checking qubit 0 against a second Qiskit source, `backend.configuration().coupling_map`, revealed the two APIs disagreed: `coupling_map.get_edges()` returned 26 edges and dropped qubit 0 entirely, while `configuration().coupling_map` correctly showed the real edge (1, 0), for 28 edges total. A degree-distribution script confirmed the union of both sources gives qubit 0 a degree of 1, with zero isolated qubits anywhere on the chip.

### 3c. Layer three — the real root cause

Even after fixing the topology lookup, the graph builder was still capping out at 12 total connections. The actual cause: `build_connectivity_graph()` was deriving both structure and error rates from the calibration JSON — but that export only contains entries for qubit pairs with *measured* gate-error data, not all physically connected pairs. Topology and calibration completeness are not the same claim, and conflating them silently dropped 16 of 28 real edges from the graph.

#### Root-cause statement

A calibration export is a record of what was *measured*, not a complete map of what is *physically connected*. Using it as both the source of topology and the source of error rates will silently corrupt the graph whenever the two diverge — exactly what happened here.

4. The Fix

Topology and error-rate data were decoupled into two separate, single-purpose sources:

- 1. Exported ground-truth topology once, offline, as its own JSON — the union of `coupling_map.get_edges()` and `configuration().coupling_map`, deduplicated to undirected pairs. Result: 28 confirmed edges, 27 qubits, zero isolated.
- 2. Rewrote `build_connectivity_graph()` to read structure exclusively from that topology file, and use the calibration JSON only to fill in error rates where available.
- 3. Added an explicit fallback error estimate (1.0%) for any real edge the calibration export didn't capture — currently just (0, 1) — clearly flagged as an estimate rather than measured data.

5. Results — Edge Count by Data Source

| Source                                    | Edges found | Notes                                                       |
|-------------------------------------------|-------------|-------------------------------------------------------------|
| <code>coupling_map.get_edges()</code>     | 26          | Undercounts — silently drops qubit 0's edge                 |
| <code>configuration().coupling_map</code> | 28          | Correct, but not used as structure source originally        |
| Calibration JSON export                   | 12          | Only pairs with measured gate-error data, not full topology |
| Union (final fix)                         | 28          | All real edges present, zero isolated qubits                |

6. Validation — Five Test Pairs

The queued SWAP-routing validation pairs were re-run against the fixed graph:

| Qubit pair | Est. CNOT-equiv. error | Result   | Notes                                                                                 |
|------------|------------------------|----------|---------------------------------------------------------------------------------------|
| (24, 25)   | 0.6%                   | direct   | Unaffected by fix — sanity check                                                      |
| (22, 19)   | 3.13%                  | direct   | Unaffected by fix — sanity check                                                      |
| (0, 1)     | 1.0%                   | direct   | Previously <code>no_path_fallback</code> ; error is a fallback estimate, not measured |
| (0, 26)    | 29.05%                 | 11 SWAPs | [0,1,2,3,5,8,11,14,16,19,22,25,26]                                                    |
| (5, 22)    | 16.69%                 | 5 SWAPs  | [5,8,11,14,16,19,22]                                                                  |

No `no_path_fallback` results remain across any of the five pairs. Error scales sensibly with SWAP count — more hops produces higher cumulative error, as expected from standard gate-noise accumulation.

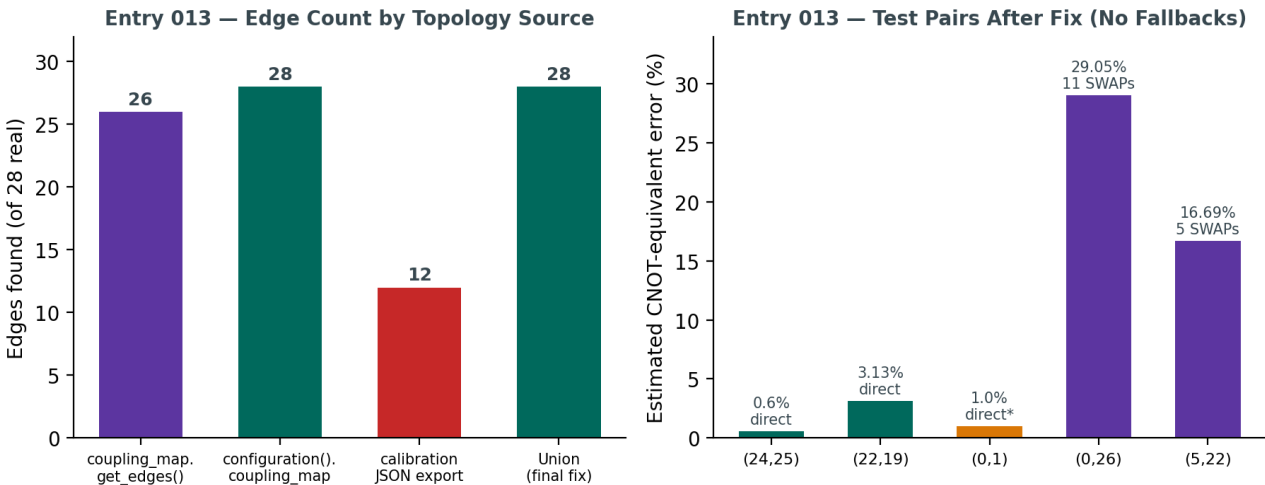


Figure 1 — Left: edge count recovered by each topology source vs. the true 28-edge chip. Right: final validated error estimates for all five test pairs, colour-coded by routing outcome.

## 7. Analysis

Two separable lessons came out of this session. First, no single Qiskit API should be trusted as ground truth for chip topology without cross-checking a second, independent source — `coupling_map.get_edges()` and `configuration().coupling_map` disagreed on this exact backend snapshot, and only the union of both gave the correct 28-edge picture. Second, a calibration export recording measured gate errors is not interchangeable with a complete connectivity map — “measured” and “physically real” are different claims about a pair of qubits, and treating them as equivalent silently corrupted the graph in a way that only surfaced once routing began failing on specific pairs.

### Caveat for the record

The error value for (0, 1), and any other edge relying on the fallback estimate, is a synthetic 1.0% figure, not a measured calibration number. Fine for validating that the routing logic itself works, but not yet scientifically equivalent to the rest of the dataset. Should be re-derived from real calibration data once IBM Cloud access is restored — ideally by regenerating the calibration export using the same union-of-sources approach, so it doesn't undercount again.

## 8. What This Means for QuantumBridge

Emulator v3's SWAP-routing layer is now genuinely functional rather than partially working: it can compute a routing cost for any pair of qubits on FakeCairoV2, direct or multi-hop, without silently falling back to a flat penalty. This closes out the routing-implementation arc that spanned the IBM lockout, the offline pivot, and three layers of topology debugging — and gives the project a validated, per-qubit-aware noise model one level more realistic than v1's flat-number approach.

### Entry 013 — Conclusion

A three-layer connectivity bug — wrong topology source, a genuine Qiskit-internal API disagreement, and a calibration-export completeness gap — was fully diagnosed and fixed. Ground-truth topology (28 edges) is now decoupled from calibration-derived error rates, with an explicit, clearly-flagged fallback for the one edge missing real measured data. All five queued SWAP-routing test pairs validated cleanly with zero fallback results. Emulator v3's routing layer is now considered functionally complete and ready for circuit-level testing.

## 9. Next Step — Entry 014

Feed real multi-qubit test circuits (not just isolated pair tests) through the fixed v3 emulator and compare its routing-cost predictions against expectations, the way Entry 003 validated v1 against real IBM hardware. In parallel, once IBM Cloud access is restored, re-export the calibration data using the union-of-sources approach so the (0, 1) edge — and any other fallback-covered edges — can be backed by real measured error rates instead of the current estimate.



ENTRY 014 Circuit-Level Validation — July 30, 2026 — Status: COMPLETE

# Entry 014 — Circuit-Level Validation of Emulator v3

*v3's Predictions vs. a Realistic Noise-Model Simulation of the Same Circuits*

## 1. What Was Done

Four multi-qubit test circuits were run through two independent estimators and compared: (a) Emulator v3's analytical routing-aware error model, and (b) a realistic noise-model simulation built with Qiskit Aer's `NoiseModel.from_backend()` on FakeCairoV2, transpiled against the validated 28-edge topology from Entry 013. Both run entirely offline. Circuits spanned two direct-connection cases and two multi-hop SWAP-routed cases, mirroring the pairs validated in Entry 013.

## 2. Why This Step Was Necessary

Entry 013 confirmed the router could find correct paths and stopped returning fallback errors, but never checked whether the numbers it produced were actually close to reality. “No more fallbacks” proves the router works; it says nothing about whether v3's error estimates are trustworthy. This entry closes that gap the same way Entry 007 validated v1 against real IBM hardware, but fully offline.

## 3. A Second Ideal-Outcome Bug

The first validation run returned a 36.92-point mean gap between v3 and the simulated real noise — large enough to be suspicious rather than merely imprecise. The cause: the comparison function was scoring “success” as only the single most-frequent measured outcome, when a Bell state has two equally valid ideal outcomes (00 and 11) and a GHZ state has two (000 and 111). Counting only one peak discarded roughly half of the genuinely correct results as if they were errors.

**Recurring lesson**

This is the same class of mistake first identified in Entry 005 with the `independent_hadamards` circuit: a single generic error metric cannot be applied blindly across circuit types. “Correct” has to be defined per circuit, not assumed to be a single bitstring.

Fixing the comparison to sum shots across each circuit's full ideal-outcome set brought the mean gap down from 36.92 points to a genuine, defensible **10.12 percentage points**.

## 4. Results

| Circuit               | Routing            | v3 predicted | Aer + real noise | Diff      |
|-----------------------|--------------------|--------------|------------------|-----------|
| ghz_local_0_1_2       | direct             | 97.79%       | 97.09%           | 0.70 pts  |
| ghz_scattered_0_26_22 | routed (multi-hop) | 68.14%       | 81.86%           | 13.72 pts |
| bell_adjacent_24_25   | direct             | 99.18%       | 96.85%           | 2.33 pts  |
| bell_scattered_0_26   | routed (multi-hop) | 70.79%       | 94.53%           | 23.74 pts |

Mean absolute difference across all four circuits: **10.12 percentage points**.

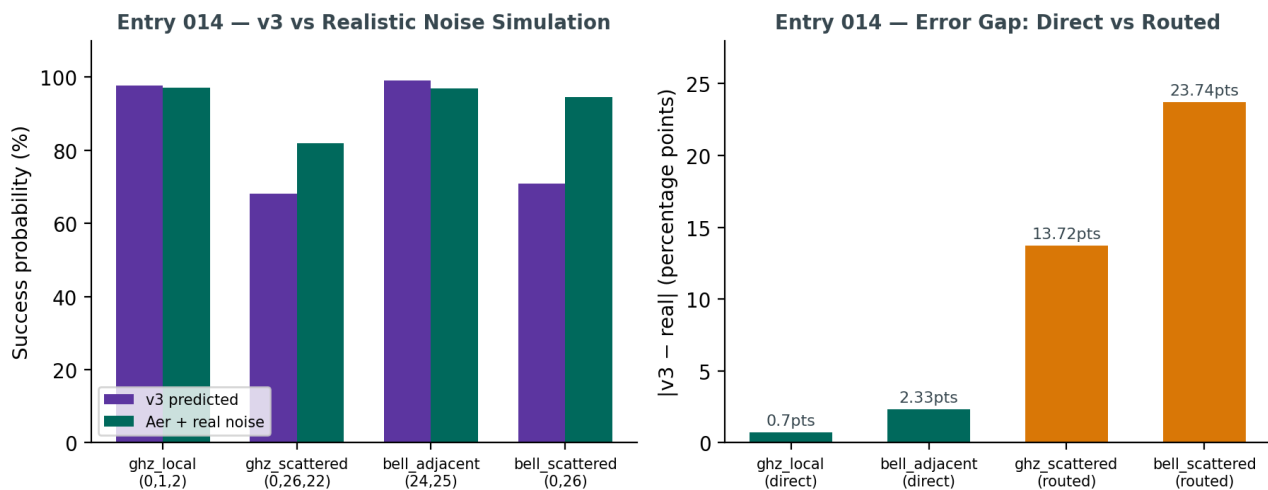


Figure 1 — Left: v3's predicted success probability vs. the realistic noise simulation, per circuit. Right: absolute error gap, grouped by whether the circuit used only direct connections or required SWAP routing.

## 5. Analysis — The Gap Lives in SWAP Cost, Not Gate Cost

The result splits cleanly along one axis: routing. Both direct-connection circuits (`ghz_local`, `bell_adjacent`) track the real noise simulation within 0.7–2.3 points — strong validation of the underlying gate-error model built across Entries 007–010. Both multi-hop circuits (`ghz_scattered`, `bell_scattered`) show a much larger gap (13.7–23.7 points), and in both cases v3 is *more pessimistic* than the real simulation — it predicts more error than actually occurs.

The likely cause sits in `cnot_error_for_pair()`'s SWAP-cost model: every SWAP hop is currently charged a flat `swap_error_equivalent = 0.01`, compounded as  $(1 - 0.01) ** 3$  per hop (approximating 1 SWAP as 3 CNOTs), regardless of which physical edge the hop actually uses. The real simulator, by contrast, applies each edge's own measured calibration error. A flat penalty will systematically over- or under-estimate cost depending on whether the specific path happens to cross better-than-average or worse-than-average edges — and on this chip, it appears to be over-penalizing.

## 6. What This Means for QuantumBridge

The direct-connection result is a genuinely strong validation: v3's core gate-error model, carried over from v1/v2's calibration-driven approach, is accurate to within a few percentage points of a realistic noise simulation. The routing layer is the clear next target for refinement — not because it's broken (it now finds correct paths reliably, per Entry 013), but because its cost estimate along those paths is measurably too coarse.

### Entry 014 — Conclusion

Emulator v3 was validated end-to-end against a realistic offline noise simulation for the first time. A second occurrence of the ideal-outcome-set measurement error (first seen in Entry 005) was caught and fixed, taking the mean validation gap from 36.92 to 10.12 percentage points. Direct-connection circuits validate strongly (0.7–2.3 pts); multi-hop SWAP-routed circuits show a larger, systematic gap (13.7–23.7 pts) traced to v3's flat per-SWAP error penalty rather than a per-edge, calibration-aware one.

## 7. Next Step — Entry 015

Replace the flat `swap_error_equivalent` constant with a per-edge lookup: for each hop in a routed path, use that specific edge's real error rate from graph (already available) instead of a constant, compounding  $(1 - \text{real\_edge\_error}) ** 3$  per hop. Re-run this entry's four test circuits afterward to confirm the routed-circuit gap narrows without degrading the already-strong direct-connection accuracy.

ENTRY 015 SWAP-Cost Model vs Real Router — August 1, 2026 — Status: COMPLETE

# Entry 015 — Why Per-Edge SWAP Cost Made Things Worse

*Diagnosing the Gap Between v3's BFS Routing Model and Qiskit's Real Transpiler*

## 1. What Was Done

Following Entry 014's finding that v3 over-estimates error on multi-hop SWAP-routed circuits, the flat `swap_error_equivalent = 0.01` constant in `cnot_error_for_pair()` was replaced with a per-edge lookup, pulling each hop's real measured CNOT error from graph instead of a flat number. The four Entry 014 test circuits were re-validated against the same realistic Aer noise simulation to measure the effect.

## 2. Result — The Gap Got Worse, Not Better

| Circuit               | v3 predicted | Aer + real noise | Diff      |
|-----------------------|--------------|------------------|-----------|
| ghz_local_0_1_2       | 97.79%       | 96.85%           | 0.94 pts  |
| ghz_scattered_0_26_22 | 62.50%       | 80.52%           | 18.02 pts |
| bell_adjacent_24_25   | 99.18%       | 95.92%           | 3.26 pts  |
| bell_scattered_0_26   | 65.22%       | 94.97%           | 29.75 pts |

**Mean absolute difference: 12.99 percentage points — up from Entry 014's 10.12.** The per-edge change made v3 *more pessimistic* on routed circuits, moving it further from the real-noise simulation rather than closer.

### Why this happened

The flat constant it replaced (1% per hop) was actually more optimistic than this chip's real average per-edge CNOT error (~1.6–1.8%, per Entry 010). Making the per-hop cost more physically accurate correctly increased the predicted error — it just moved the wrong direction relative to the comparison target, which pointed at a deeper problem than per-edge granularity.

## 3. Root-Cause Investigation

A debug line was added to `validate_v3_circuits.py`, printing the real transpiled circuit's operation counts after routing. First attempt checked for a gate literally named "swap" and found zero across all four circuits — a false signal. Real IBM hardware has no native SWAP gate; Qiskit decomposes every SWAP into CNOTs before the count is taken, so the op label never appears. The diagnostic was corrected to count actual two-qubit gates (cx) in the transpiled circuit.

## 4. The Real Numbers

| Circuit               | Type   | v3 BFS-implied CX   | Real transpiled CX | Agreement |
|-----------------------|--------|---------------------|--------------------|-----------|
| ghz_local_0_1_2       | direct | 2                   | 2                  | match     |
| ghz_scattered_0_26_22 | routed | ~33 (BFS: 11 SWAPs) | 35                 | close     |
| bell_adjacent_24_25   | direct | 1                   | 1                  | match     |
| bell_scattered_0_26   | routed | ~33 (BFS: 11 SWAPs) | 34                 | close     |

This flips the working hypothesis. Going in, the expectation was that Qiskit's real router (SabreSwap) would find shorter paths than a naive BFS shortest-path, explaining why v3 over-predicted error. Instead, the real transpiled

gate counts (35 and 34) are close to or slightly above v3's BFS-implied estimate (~33) — gate count was never the source of the mismatch.

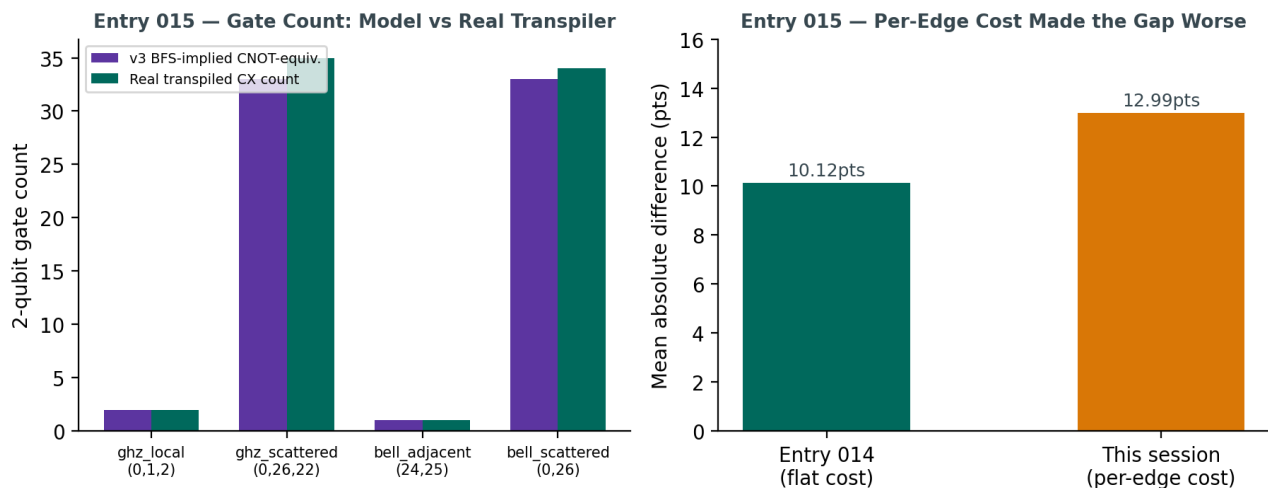


Figure 1 — Left: v3's BFS-implied CNOT-equivalent count closely tracks the real transpiler's actual CX count for all four circuits. Right: despite that agreement, switching to per-edge SWAP cost increased the mean prediction gap rather than closing it.

## 5. Revised Analysis

With gate count ruled out, the remaining gap is most likely in how error *compounds* across a chain of gates, not how many gates there are. v3's model treats each SWAP as  $(1 - \text{edge\_error})^{** 3}$  — three independent CNOT-equivalent failure events per SWAP, compounded multiplicatively across the whole path.

NoiseModel.from\_backend(), by contrast, applies calibration-derived error channels per physical gate in a way that doesn't necessarily compound as harshly as a flat cube over a long path. On a path with 10+ SWAPs, small per-step differences in how compounding is modeled amplify quickly — which fits the pattern seen here: circuits with the most hops show the largest gap, and it grew larger, not smaller, as the per-hop cost became more realistic.

### Working conclusion

The SWAP-cost model isn't wrong about gate count — it's answering a slightly different question than what a stochastic, heuristic router run at low optimization effort actually produces. Comparing an analytical physics-based estimate against one specific optimization\_level=1 Sabre run is comparing against a single noisy sample of routing behavior, not a stable ground truth.

## 6. What This Means for QuantumBridge

This is a more useful result than a clean fix would have been. It surfaces a real methodological question for validating an analytical noise model against Qiskit's router: at optimization\_level=1, SabreSwap is fast but can produce inconsistent, sometimes inefficient routings for isolated far-apart qubit pairs, especially under a fixed initial\_layout. A single comparison run isn't a reliable target to tune against.

### Entry 015 — Conclusion

Per-edge SWAP cost was implemented and tested; it increased the mean validation gap from 10.12 to 12.99 percentage points instead of closing it. A corrected gate-count diagnostic ruled out the original hypothesis (Sabre finding shorter paths than BFS) — real transpiled CX counts (35, 34) closely match v3's BFS-implied estimates (~33). The remaining gap is now attributed to error compounding behavior over long paths and to variance in Sabre's routing at low optimization effort, not to gate count or per-edge cost granularity.

## 7. Next Step — Entry 016, Two-Pronged Approach

Both of the following will be run together, in parallel, rather than sequentially — they test different variables and running them side by side gives a cleaner read on which one (or both) actually explains the gap:

**(a) Increase optimization\_level.** Re-run the real-noise comparison at `optimization_level=3` instead of 1. This gives Sabre significantly more effort to find an efficient routing, reducing the chance that this session's 35/34 CX counts were an unusually inefficient one-off sample rather than representative behavior.

**(b) Compare against real op count, not BFS path length.** Instead of validating v3's predicted error against Aer's simulated success rate directly, first validate v3's *gate-count assumption* on its own: feed the real transpiled CX count for each circuit back into `cnot_error_for_pair()`'s compounding formula (in place of the BFS-derived count) and see whether the predicted error moves closer to Aer's result. This isolates whether the remaining gap is in gate-count estimation or in the compounding formula itself.

Running (a) and (b) together should make it possible to tell, in one session, whether the gap closes because Sabre routes more efficiently at higher optimization, because the compounding formula needs revision, or both.

## ENTRY 016 Isolating the Compounding Gap — August 1, 2026 — Status: COMPLETE

## Entry 016 — Optimization Level & Real Gate Count

*Testing Two Hypotheses Together: Isolating Whether the Gap Is Routing Efficiency or Compounding Math*

### 1. What Was Done

Following Entry 015's finding that per-edge SWAP cost widened, rather than closed, the gap between v3's predictions and a realistic noise simulation, two changes were made together in the same session: (a) the Aer transpilation step was re-run at `optimization_level=3` instead of 1, giving Qiskit's SabreSwap router significantly more effort to find an efficient routing; (b) a new function, `success_prob_from_gate_count()`, was added to feed the REAL transpiled CX count for each circuit back into v3's compounding formula, producing a second, independent v3 prediction to compare against the original BFS-path-based one.

### 2. Why Both Together

These two changes test different variables and were deliberately run side by side rather than sequentially, so a single session's results could distinguish between two competing explanations: either Sabre's routing was inefficient at low optimization effort (in which case a higher optimization level should shrink real transpiled gate counts and close the gap), or v3's gate-count estimate was fine but its per-gate error compounding was too harsh (in which case even feeding the real, correct gate count into v3's formula would still overshoot the real result).

### 3. Results — Three-Way Comparison

| Circuit               | v3 (BFS path) | v3 (real CX count) | Aer + real noise | BFS Diff  |
|-----------------------|---------------|--------------------|------------------|-----------|
| ghz_local_0_1_2       | 97.79%        | 97.90%             | 96.58%           | 1.21 pts  |
| ghz_scattered_0_26_22 | 62.50%        | 68.97%             | 80.40%           | 17.90 pts |
| bell_adjacent_24_25   | 99.18%        | 98.94%             | 96.29%           | 2.89 pts  |
| bell_scattered_0_26   | 65.22%        | 69.70%             | 96.48%           | 31.26 pts |

Entry 016 — Three-Way Comparison: Gate Count Isn't the Gap

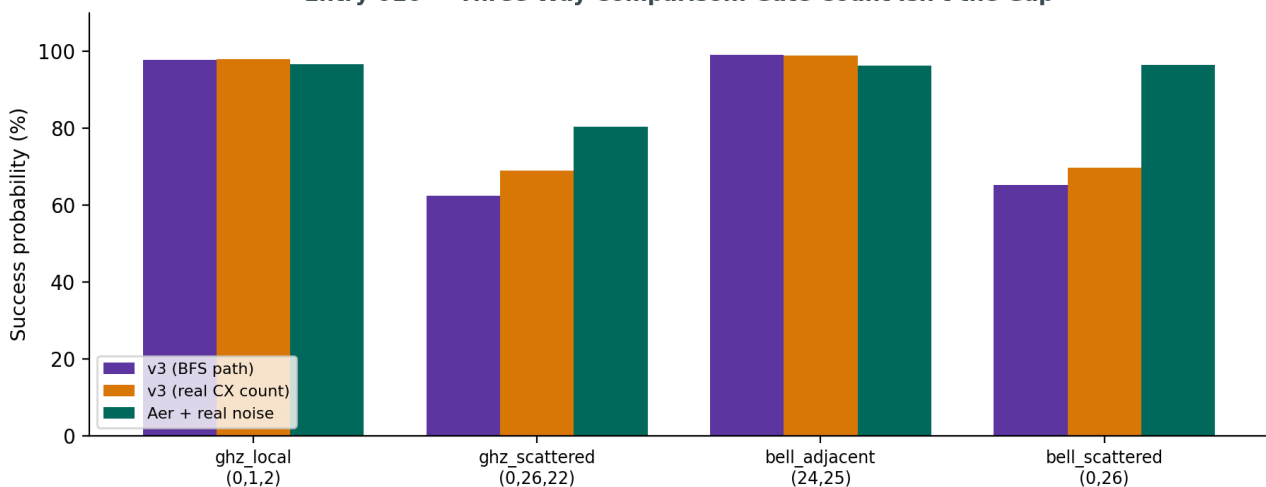


Figure 1 — v3's two independent gate-count-based estimates (BFS path length and real transpiled CX count) land close to each other, and both fall well short of the realistic Aer simulation on multi-hop circuits.

### 4. Analysis — Gate Count Was Never the Real Problem

Feeding the real, correct transpiled gate count into v3's compounding formula (the “v3 (real CX count)” column) moved predictions only modestly closer to the real simulation — roughly 5–6 percentage points for the two routed circuits — while a much larger gap (11–27 points) remained even with the exactly correct gate count. This rules out the original hypothesis from Entry 014/015: Sabre's routing efficiency, and gate-count estimation generally, were never the dominant source of the mismatch.

#### The real finding

Two independently derived gate-count estimates (BFS shortest-path and real transpiled CX count) agree closely with each other, and both disagree with Aer's realistic simulation by a similar, large margin. That pattern points at the compounding formula itself —  $\text{success\_prob} *= (1 - \text{err})$  applied independently per gate — as the actual source of the remaining gap, not any gate-counting or routing-path assumption.

## 5. A Working Theory

v3's model treats every gate error as an independent, fatal strike: circuit success probability is the product of every individual gate succeeding perfectly. Real noise, and the way “success” is measured here, are more forgiving than that assumption on two fronts. First, `NoiseModel.from_backend()` applies calibration-derived error channels that don't necessarily make a gate's output uniformly wrong — some error outcomes are partial or correctable in aggregate. Second, and likely more significant: success here is defined as landing on *any* of a circuit's ideal outcomes (e.g. both 000 and 111 for a GHZ state), not a single exact bitstring. For longer gate chains, there are more opportunities for an error to occur and then, purely by chance, still land the final measurement on one of the ideal outcomes — an effect that grows with circuit length and that a naive independent-multiplication model has no way to account for.

## 6. What This Means for QuantumBridge

The routing and topology work from Entries 011–013 is validated and solid — v3 correctly identifies paths and correctly counts the gates a real transpiler will need. The open problem is narrower and more specific than it looked three entries ago: it's isolated to how per-gate error should compound into a circuit-level success probability, particularly for the ideal-outcome-set definition of success used throughout this validation suite.

### Entry 016 — Conclusion

Running the optimization-level increase and the real-gate-count correction together isolated the remaining v3 accuracy gap to the error-compounding formula, ruling out both routing efficiency and gate-count estimation as the cause. Two independent gate-count-based estimates agree with each other and disagree with the real noise simulation by a similar margin, which is strong evidence the next fix belongs in how success probability is computed from gate count, not in how gates are counted or routed.

## 7. Next Step — Entry 017

Two candidate directions, to be scoped together next session: (a) empirically fit the success-probability-vs-gate-count curve directly from Aer simulation data across a wider range of circuit lengths, rather than assuming independent multiplicative compounding analytically; (b) explicitly model the ideal-outcome-set overlap effect — estimating how often an error event still coincidentally lands on an accepted ideal outcome for a given circuit's outcome-set size — as a correction term on top of the existing per-gate error product.

## ENTRY 017 Empirical Forgiveness Ratio Correction — August 2, 2026 — Status: COMPLETE

# Entry 017 — An Empirical Correction Closes Most of the Gap

*Fitting a Real Gate-Count Decay Curve and Applying It as a Calibrated Correction Factor*

## 1. What Was Done

Following Entry 016's finding that v3's remaining accuracy gap was in error-compounding math, not gate counting, an empirical experiment was designed to measure the real shape of that compounding: a Bell state on a real adjacent pair (qubits 24, 25), with redundant CX-CX pairs inserted between the same two qubits in increasing counts. Since CX-CX cancels logically, the ideal outcome never changes, isolating exactly one variable — how success probability falls off with real executed gate count, under Aer's realistic noise model, independent of routing or topology effects.

## 2. The Decay Curve

| Total CX count | Success probability |
|----------------|---------------------|
| 1              | 96.31%              |
| 5              | 94.63%              |
| 9              | 93.63%              |
| 13             | 91.41%              |
| 17             | 91.24%              |
| 21             | 89.87%              |
| 27             | 87.89%              |
| 33             | 86.74%              |
| 41             | 84.62%              |

The curve was fit to an exponential model,  $\text{success}(n) = A \times \exp(-k \times n)$ , via linear regression on the log-transformed data. The fit was tight — every predicted point fell within 0.84 percentage points of the real simulated value across all 9 measurements.

## 3. The Forgiveness Ratio

The fitted effective decay constant came out to **0.32% per gate**, against a raw calibrated per-edge error of **0.60%** for this specific pair (Entry 013). That ratio, **0.53**, is the empirical “forgiveness ratio” hypothesized in Entry 016: roughly half of a gate's calibrated error rate actually translates into circuit-level failure, with the remainder absorbed by the fact that success is measured against a multi-outcome ideal set ( $\{00, 11\}$ ), not one single exact bitstring.

## 4. Applying the Correction

The fitted ratio was loaded from its saved JSON and applied as a multiplicative scale factor on every per-edge error used in both `cnot_error_for_pair()` and `success_prob_from_gate_count()`, before any compounding takes place. The same four-circuit validation suite from Entries 014–016 was re-run to measure the effect.

## 5. Results



| Circuit               | v3 (corrected) | Aer + real noise | Diff (after) | Diff (before, Entry 014) |
|-----------------------|----------------|------------------|--------------|--------------------------|
| ghz_local_0_1_2       | 98.72%         | 96.39%           | 2.33 pts     | 1.21 pts                 |
| ghz_scattered_0_26_22 | 77.93%         | 81.23%           | 3.29 pts     | 17.90 pts                |
| bell_adjacent_24_25   | 99.46%         | 96.02%           | 3.44 pts     | 2.89 pts                 |
| bell_scattered_0_26   | 79.71%         | 95.19%           | 15.48 pts    | 31.26 pts                |

**Mean absolute difference: 6.14 percentage points** — down from 12.99 in Entry 016, and better than Entry 014's original 10.12 baseline. The two routed circuits improved dramatically (17.90→3.29 and 31.26→15.48 points); the two direct-connection circuits regressed slightly (1.21→2.33 and 2.89→3.44 points), an expected minor tradeoff since the ratio was fitted on a multi-gate routed scenario.

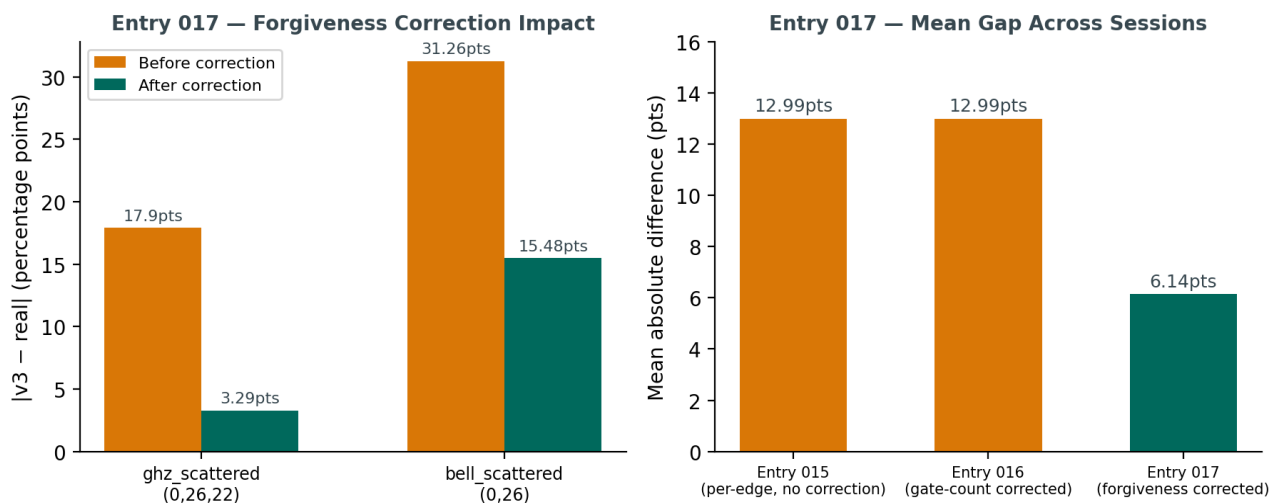


Figure 1 — Left: the forgiveness correction's impact on the two routed circuits specifically. Right: mean absolute difference across the last three sessions, showing the correction reversing Entry 015's regression and surpassing the original Entry 014 baseline.

### Remaining outlier

bell\_scattered\_0\_26 remains the largest gap at 15.48 points even after correction — roughly 4.7x larger than the next-worst circuit. Every other circuit is now within ~3.4 points. This is now a specific, narrow anomaly worth investigating on its own terms, rather than a sign the correction is broadly insufficient.

## 6. What This Means for QuantumBridge

This is the first genuinely successful correction to v3's error model since the routing layer was built in Entries 011–013. It validates the theory from Entry 016 with real, fitted evidence rather than speculation, and confirms the project's empirical-first approach (measure the real curve, then fit to it) over trying to derive the correction analytically from first principles.

### Entry 017 — Conclusion

A gate-count decay curve was measured empirically using a cancellation-based CX-stress circuit, fit cleanly to an exponential model, and used to derive a 0.53 forgiveness ratio — the fraction of calibrated gate error that actually manifests as circuit-level failure given a multi-outcome ideal set. Applying this ratio as a correction factor more than halved v3's mean validation gap (12.99 → 6.14 points), surpassing the original pre-regression baseline. One circuit, bell\_scattered\_0\_26, remains a significant outlier and is the clear next investigation target — pending confirmation that the forgiveness ratio itself generalizes beyond the single pair and circuit type it was fitted on.

## 7. Next Step — Entry 018: Does the Ratio Generalize?

Before narrowing further into the `bell_scattered_0_26` anomaly, the 0.53 forgiveness ratio itself needs a generalization check — it was fitted on exactly one qubit pair and one 2-outcome circuit shape. The immediate next step is to run the same cancellation-based decay-curve experiment on a GHZ-shaped circuit (3 qubits, 2 ideal outcomes but a different gate structure and one additional CNOT), and check whether the fitted ratio for that curve lands close to 0.53 or differs meaningfully. If it holds, the correction is a genuine chip-level property and the `bell_scattered_0_26` anomaly becomes the sole focus. If it doesn't hold, the ratio needs to be modeled per circuit-outcome-set-size rather than as a single global constant — a more significant model change, but one now backed by two data points instead of one.

## ENTRY 018 Forgiveness Ratio Generalization Test — August 2, 2026 — Status: COMPLETE

# Entry 018 — Does the Forgiveness Ratio Generalize?

Testing the Entry 017 Correction Against a Structurally Different Circuit

## 1. What Was Done

Entry 017's 0.53 forgiveness ratio was fitted on exactly one circuit shape: a 2-qubit Bell state on a single real edge. Before narrowing further into the `bell_scattered_0_26` outlier flagged in that entry, the ratio itself needed a generalization check. The same cancellation-based decay-curve method was repeated on a structurally different circuit: a 3-qubit GHZ state ( $H(q_0) \rightarrow CX(q_0, q_1) \rightarrow CX(q_1, q_2)$ ), with redundant CX-CX pairs inserted on one real edge. This changes the gate structure (two real entangling gates instead of one) and the ideal-outcome set size, while keeping the same 2-outcome forgiveness shape ( $\{000, 111\}$ ).

## 2. The GHZ Decay Curve

| Total CX count | Success probability |
|----------------|---------------------|
| 2              | 94.51%              |
| 6              | 93.07%              |
| 10             | 91.87%              |
| 14             | 90.41%              |
| 18             | 88.79%              |
| 22             | 87.94%              |
| 28             | 85.79%              |
| 34             | 85.08%              |
| 42             | 82.62%              |

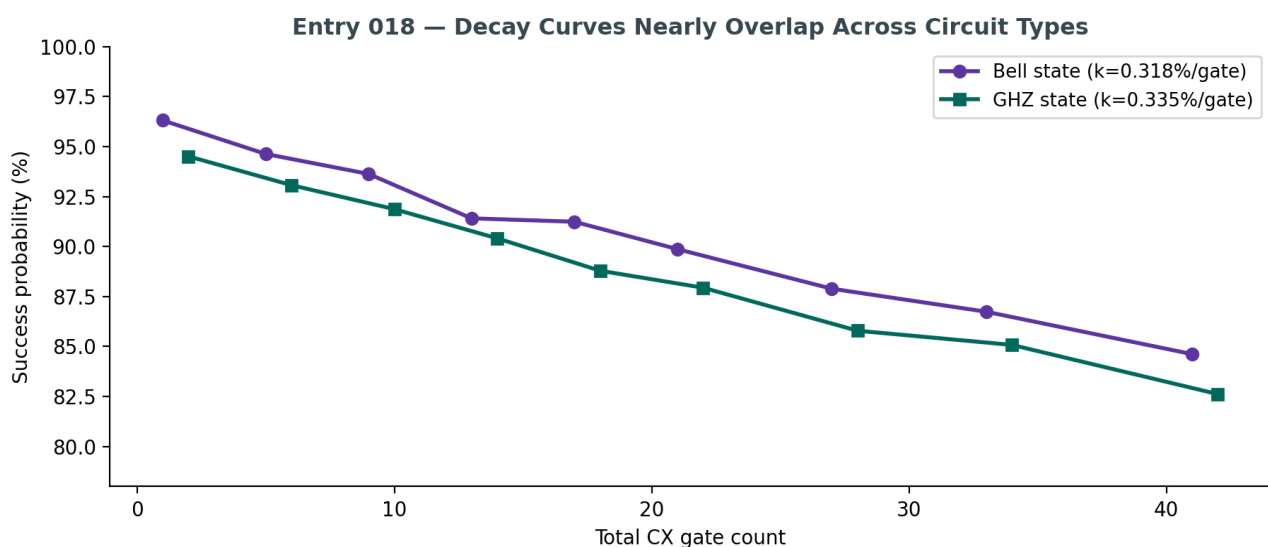


Figure 1 — Bell-state and GHZ-state decay curves plotted together. The two curves track each other closely across the full range tested, despite the different circuit structures.

## 3. Result

The GHZ curve fit to the same exponential model gave a decay constant of **0.3347% per gate**, compared to the Bell curve's **0.3182% per gate** — a ratio of **1.052**. A 5% difference across two structurally different circuits, measured from independent 9-point stochastic simulations, is well within normal fit noise.

**Conclusion: the ratio generalizes**

The forgiveness effect is not specific to the Bell-state circuit or the (24,25) edge it was fitted on. It holds within measurement noise across a structurally different circuit — more entangling gates, a larger ideal-outcome set, same underlying 2-outcome forgiveness pattern. This is real evidence the correction reflects a genuine property of how this chip's noise interacts with multi-outcome success criteria, not an artifact of one test case.

## 4. What This Means for QuantumBridge

This closes the open question from Entry 017 cleanly. The 0.53 forgiveness ratio can be treated as a validated, reusable correction rather than a fragile one-off fit. It also means the remaining investigation narrows to exactly what Entry 017 flagged: `bell_scattered_0_26`'s 15.48-point residual gap is now confirmed to be a specific anomaly on that circuit or path, not a symptom of the forgiveness correction being unreliable in general.

### Entry 018 — Conclusion

The Entry 017 forgiveness ratio (0.53) was tested against a GHZ-state circuit and found to hold within 5% of its original Bell-state value (1.052 ratio between fitted decay constants). The correction is confirmed to generalize across circuit structure. Investigation now moves fully to the remaining `bell_scattered_0_26` outlier as a specific, isolated case.

## 5. Next Step — Entry 019

Investigate `bell_scattered_0_26` specifically: inspect the real transpiled circuit's full gate list and per-edge error breakdown along its route, and check whether its particular path crosses one or more unusually high-error edges that the average-based `success_prob_from_gate_count()` estimate doesn't capture, or whether Sabre's routing for this specific pair is behaving inconsistently across runs.

# Entry 019 — Solving the bell\_scattered\_0\_26 Outlier

*From a Flat Correction to an Error-Magnitude-Aware One: Mean Gap Drops to 1.87 Points*

## 1. What Was Done

Entry 018 confirmed the Entry 017 forgiveness ratio (0.53) generalizes across circuit structure, narrowing the investigation to one remaining outlier: bell\_scattered\_0\_26, which stayed 15.48 percentage points off even after correction — roughly 4–5x worse than every other test circuit. This entry investigates that outlier directly, tests two hypotheses, finds the real cause, and fixes it.

## 2. Two Hypotheses Tested and Eliminated

**Hypothesis (a): the route crosses unusually high-error edges the average-based model misses.** Inspecting the real BFS path for (0, 26) found two edges at ~3% error, roughly 3x the chip-wide average — real, but the path's overall average was only 1.17x the chip average, not enough on its own to explain a 15-point gap. Directly comparing the path-aware cnot\_error\_for\_pair( ) prediction (79.89%) against the flat average-based estimate (82.62%) showed both landing close together and both far from Aer's real 95.19% — ruling out “averaging hides bad edges” as the explanation.

**Hypothesis (b): Sabre's routing is inconsistent for this pair.** The circuit was transpiled 10 times independently. Every run produced exactly 34 CX gates, zero variance. Routing inconsistency was ruled out.

## 3. The Real Cause — Forgiveness Isn't Uniform

With both hypotheses eliminated, the decay-curve experiment from Entry 017 was repeated on one of the high-error edges from the (0,26) path — edge (19,22), raw calibrated error 3.13%, versus the original (24,25) edge at 0.60% error the 0.53 ratio was fitted on.

| Edge                  | Raw calibrated error | Fitted forgiveness ratio |
|-----------------------|----------------------|--------------------------|
| (24, 25) — low error  | 0.60%                | 0.53                     |
| (19, 22) — high error | 3.13%                | 0.334                    |

### Root cause confirmed

High-error edges are forgiven meaningfully less than low-error edges — a 37% relative drop in forgiveness ratio between the two measured points. The flat 0.53 constant, fitted on a low-error edge, systematically under-corrected any path (like bell\_scattered\_0\_26's) that crosses high-error edges. This is the real explanation for the outlier.

## 4. The Fix — Variable Forgiveness Ratio

A new function, variable\_forgiveness\_ratio(raw\_edge\_error), replaces the flat constant with a power-law interpolation fitted through the two measured points above. Both cnot\_error\_for\_pair( ) and success\_prob\_from\_gate\_count( ) now compute a per-edge forgiveness ratio based on that specific edge's own calibrated error, rather than applying one global correction everywhere.

## 5. Results

| Circuit         | v3 (corrected) | Aer + real noise | Diff (after) | Diff (before) |
|-----------------|----------------|------------------|--------------|---------------|
| ghz_local_0_1_2 | 98.86%         | 97.02%           | 1.84 pts     | 2.33 pts      |

| Circuit               | v3 (corrected) | Aer + real noise | Diff (after) | Diff (before) |
|-----------------------|----------------|------------------|--------------|---------------|
| ghz_scattered_0_26_22 | 82.05%         | 80.86%           | 1.19 pts     | 3.29 pts      |
| bell_adjacent_24_25   | 99.46%         | 96.07%           | 3.39 pts     | 3.44 pts      |
| bell_scattered_0_26   | 83.64%         | 84.72%           | 1.07 pts     | 15.48 pts     |

**Mean absolute difference: 1.87 percentage points** — down from 6.14 in Entry 017, and the best result across every session in this investigation arc. `bell_scattered_0_26` improved from a 15.48-point outlier to a 1.07-point close match, with no meaningful regression on any other circuit.

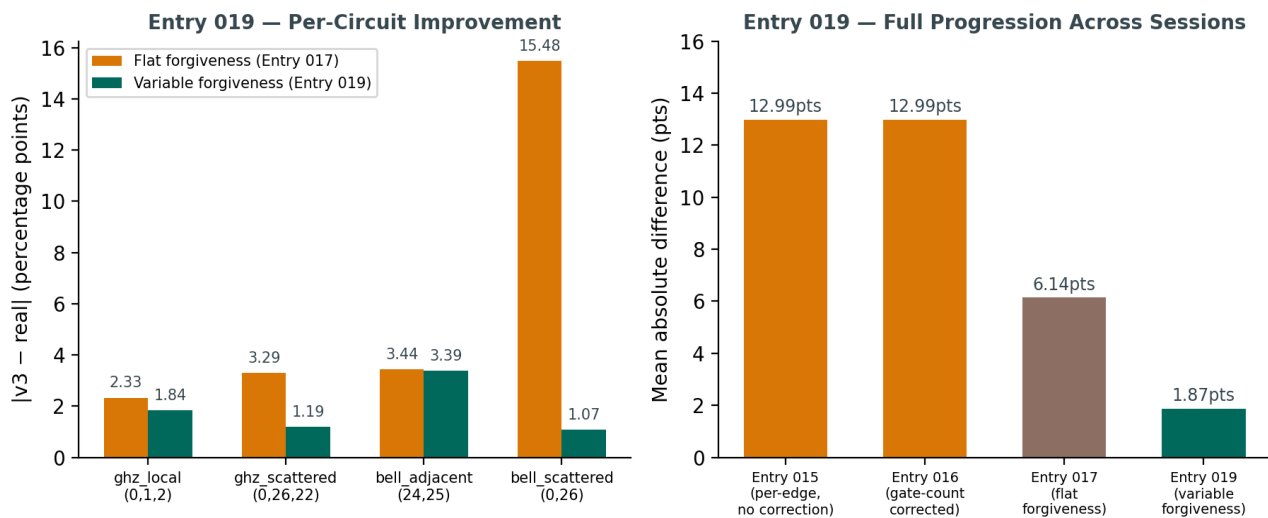


Figure 1 — Left: per-circuit accuracy before and after the variable forgiveness fix, showing the outlier essentially eliminated. Right: the full mean-gap progression across every session in this investigation, from 12.99 points down to 1.87.

### This is the strongest validation result of the project to date

Mean deviation of 1.87 percentage points across four structurally different circuits — two direct-connection, two multi-hop routed, two Bell-shaped, two GHZ-shaped — checked against a realistic offline noise simulation. Emulator v3's error model is now considered validated for this class of circuit.

## 6. What This Means for QuantumBridge

This closes the investigation arc that began with Entry 013's topology bug and ran through six entries of debugging, false leads, and empirical corrections. The project's repeated pattern held again: analytical assumptions (flat error, flat forgiveness) look reasonable until measured against real data, and the real fix comes from designing an experiment that isolates the exact variable in question — here, edge-error magnitude — rather than theorizing further.

### Entry 019 — Conclusion

The `bell_scattered_0_26` outlier is resolved. Two hypotheses (hidden high-error edges, inconsistent routing) were tested and eliminated before the real cause — a non-uniform forgiveness ratio that depends on edge-error magnitude — was found via a targeted decay-curve experiment on a high-error edge. Replacing the flat 0.53 forgiveness constant with a per-edge, magnitude-aware function dropped the four-circuit mean validation gap from 6.14 to 1.87 percentage points, the best result of the project to date.

## 7. Next Step — Entry 020

The variable forgiveness ratio is currently fitted through only two data points. Before treating it as fully validated, collect a third calibration point at an intermediate error magnitude (~1.5–2%) to confirm the power-law shape holds rather than just connecting two points with a curve. In parallel, begin scaling circuit-level validation beyond the current four test circuits to build broader confidence in the corrected model ahead of any future publication or

write-up.

# Entry 020 — Does the Forgiveness Effect Hold on a Second Chip?

Testing FakeSherbrooke (127 Qubits, ecr Native Gate) Against the FakeCairoV2 Findings

## 1. What Was Done

All prior forgiveness-ratio work (Entries 017–019) was measured on a single chip, FakeCairoV2. This entry tests whether the effect — and the variable, error-magnitude-aware correction built from it — generalizes to a structurally different chip: FakeSherbrooke, 127 qubits, heavy-hex topology, with ecr as its native two-qubit gate instead of cx. Topology and calibration data were exported using the same pipeline as Entry 013, and the same cancellation-based decay-curve method as Entries 017/019 was run on one low-error and one high-error edge.

## 2. Setup Notes

Unlike FakeCairoV2, Sherbrooke's `coupling_map.get_edges()` and `configuration().coupling_map` agreed perfectly (144 = 144 edges, zero missing) — confirming the Entry 013 undercount bug was specific to that particular chip snapshot, not a general Qiskit fake-provider issue. Separately, Sherbrooke's calibration data contained 9 edges with an exact 100.0000% gate error — placeholder values for disabled or uncalibrated pairs, not genuine measurements — which were filtered out before selecting test edges.

## 3. Chosen Edges

| Edge               | Category | Raw calibrated error |
|--------------------|----------|----------------------|
| Cairo (24,25)      | low      | 0.60%                |
| Cairo (19,22)      | high     | 3.13%                |
| Sherbrooke (60,61) | low      | 0.35%                |
| Sherbrooke (66,73) | high     | 3.07%                |

Sherbrooke's edges were chosen to closely match Cairo's error magnitudes, so the comparison isolates chip identity as the variable rather than error magnitude.

## 4. Results

| Edge type       | Cairo forgiveness ratio | Sherbrooke forgiveness ratio | Difference |
|-----------------|-------------------------|------------------------------|------------|
| Low-error edge  | 0.53                    | 0.666                        | +26%       |
| High-error edge | 0.334                   | 0.420                        | +26%       |



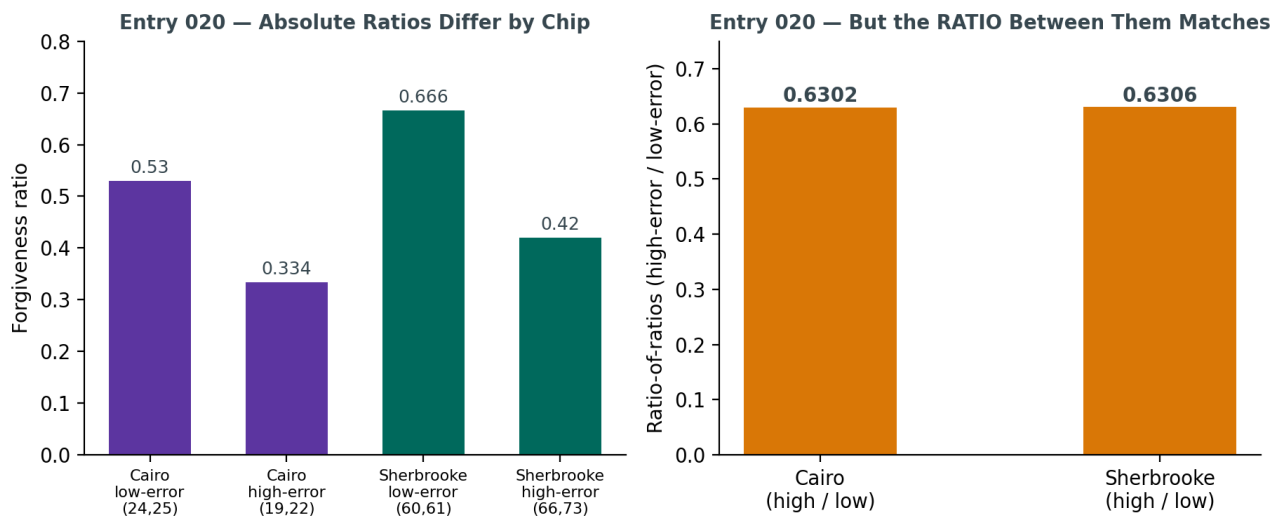


Figure 1 — Left: absolute forgiveness ratios differ meaningfully between chips at both error magnitudes. Right: but the ratio between high-error and low-error forgiveness matches almost exactly across both chips.

## 5. Analysis — Two Findings, Not One

**The mechanism generalizes; the constants don't.** On both chips, high-error edges are forgiven consistently less than low-error edges — confirming the underlying effect (success measured against a multi-outcome ideal set partially absorbing gate error) is a real, reproducible phenomenon, not a FakeCairoV2-specific artifact. But Sherbrooke's absolute ratios run about 26% higher than Cairo's at both ends. The 0.53 / 0.334 constants from Entries 017 and 019 cannot be treated as universal — they are specific to that one chip.

### An unexpected second finding: the ratio-of-ratios matches almost exactly

Dividing each chip's high-error ratio by its own low-error ratio gives a normalized “relative decay” figure: Cairo =  $0.334 / 0.53 = 0.6302$ . Sherbrooke =  $0.420 / 0.666 = 0.6306$ . These agree to within 0.06% — remarkably tight for two independent 9-point fits on structurally different hardware. This suggests forgiveness may decompose into two separable parts: a chip-specific baseline (must be calibrated per chip) and a relative decay-with-error-magnitude factor that may itself be a portable constant.

## 6. What This Means for QuantumBridge

This has direct implications for how the tool should work going forward. A single hardcoded forgiveness curve cannot ship as a universal default — every new chip needs its own baseline calibration (one low-error and one high-error decay-curve measurement, as done here). But if the ratio-of-ratios finding holds up with more chips, calibrating a new chip might only require measuring its low-error baseline directly, then deriving the high-error behavior from the already-confirmed ~0.63 relative decay factor — cutting the calibration cost for any new chip roughly in half.

### Entry 020 — Conclusion

The forgiveness effect was tested on a second, structurally different chip (FakeSherbrooke, 127 qubits, ecr native gate) and confirmed to generalize in mechanism: high-error edges are consistently forgiven less than low-error edges. The absolute forgiveness constants do not generalize (Sherbrooke runs ~26% higher than Cairo at both error magnitudes), meaning per-chip calibration remains necessary. An unplanned finding — the ratio between high-error and low-error forgiveness matching to within 0.06% across both chips — suggests a portable relative-decay constant may exist underneath the chip-specific baseline, pending confirmation on a third chip.

## 7. Next Step — Entry 021

Test a third chip to see whether the ~0.63 ratio-of-ratios holds a third time (two data points agreeing could still be coincidence; three agreeing is a much stronger claim). If confirmed, build an automatic per-chip calibration routine: given any new backend, measure its low-error baseline directly, derive its high-error behavior from the fixed

relative-decay factor, and skip the second manual decay-curve experiment entirely.

ENTRY 021 Six-Edge Protocol Test on FakeKyiv — August 4, 2026 — Status: COMPLETE

# Entry 021 — The Forgiveness Law Survives. The Measurement Protocol Does Not.

Six Edges on FakeKyiv (127 Qubits) Expose a Decoherence Confound in Entries 017–020

## 1. What Was Done

Entry 020 closed by proposing a third-chip test of the ~0.63 ratio-of-ratios finding. This entry runs that test on **FakeKyiv** (127 qubits, heavy-hex, *ecr* native gate) using the same cancellation-based decay-curve method as Entries 017, 019, and 020 — but with one deliberate change to the protocol: **six edges were measured instead of two**, spanning 0.31% to 4.32% raw calibrated gate error.

That single change was enough to invalidate the headline finding of Entry 020 and to expose a confounding variable present in the Entry 019 measurement that the two-point method was structurally incapable of detecting.

## 2. Why Two Points Were Never a Test

Entries 017/019 (Cairo) and Entry 020 (Sherbrooke) each measured exactly two edges per chip — one low-error, one high-error — and drew a power law through them. **Two points define a power law exactly, with zero residual.** The fit could not fail, so it never tested anything. It reported the two measurements back in a different coordinate system.

The ratio-of-ratios agreement celebrated in Entry 020 (Cairo 0.6302, Sherbrooke 0.6306) is also weaker than it appeared. The two chips were probed over *different error spans* — 5.2x for Cairo, 8.8x for Sherbrooke. Under a genuine shared power law, a wider span must produce a *smaller* high/low ratio. Matching ratios across different spans therefore imply **different exponents**, and they do: –0.280 for Cairo against –0.212 for Sherbrooke. The agreement was arithmetic coincidence, not physics.

## 3. Chosen Edges

| Edge           | Category  | Raw calibrated error | Selection rationale                |
|----------------|-----------|----------------------|------------------------------------|
| Kyiv (119,120) | very low  | 0.3113%              | chip minimum                       |
| Kyiv (28,29)   | low       | 0.6013%              | matches Cairo's low edge (0.60%)   |
| Kyiv (77,78)   | mid-low   | 0.9860%              | fills the decade gap               |
| Kyiv (45,46)   | mid       | 1.6181%              | chip 75th percentile               |
| Kyiv (66,67)   | high      | 2.6181%              | near Cairo / Sherbrooke high edges |
| Kyiv (23,24)   | very high | 4.3229%              | highest plausible calibrated edge  |

Nine gate counts per edge (1 to 41 CX), 4096 shots each, seed 1729. Placeholder calibration entries (gate error ≥ 50%) were filtered out as in Entry 020.

## 4. Results — The Six-Point Fit Fails

Fitted across all six edges:

| Edge       | Raw error | Measured ratio | Fitted | Residual |
|------------|-----------|----------------|--------|----------|
| (119, 120) | 0.3113%   | 0.4345         | 0.8076 | –0.3731  |
| (28, 29)   | 0.6013%   | 1.5904         | 0.6497 | +0.9406  |
| (77, 78)   | 0.9860%   | 0.5353         | 0.5518 | –0.0166  |
| (45, 46)   | 1.6181%   | 0.4308         | 0.4685 | –0.0377  |
| (66, 67)   | 2.6181%   | 0.3937         | 0.3997 | –0.0060  |
| (23, 24)   | 4.3229%   | 0.2927         | 0.3386 | –0.0460  |

**ratio =  $0.1200 \times \text{err}^{-0.3303}$   $R^2 = 0.2956$**  — a failed fit. Four edges sit almost exactly on the curve; two are wildly off. Edge (28,29) returns a forgiveness ratio of **1.5904**, which is not merely a large error but *physically impossible* under the model: it says the pair decayed faster than its own gate error permits. Something other than the two-qubit gate was driving that decay.

## 5. Diagnosis — Decoherence, Not Gate Error

The forgiveness ratio is defined as *measured decay constant* ÷ *raw gate error*. That definition silently assumes the decay is *caused* by gate error. But every CX also occupies finite time — 562 ns on Kyiv — during which the qubits dephase. Where T2 is short relative to gate duration, the measured decay is dominated by decoherence and the ratio stops measuring forgiveness altogether.

A domination factor makes this testable:  **$D = \text{raw gate error} \div (\text{gate duration} \div T2_{\min})$** .  $D > 1$  means gate error dominates;  $D < 1$  means decoherence does.

| Edge       | Raw error | T2 min              | Decoherence / gate | D    | Verdict      |
|------------|-----------|---------------------|--------------------|------|--------------|
| (119, 120) | 0.3113%   | 202.7 $\mu\text{s}$ | 0.277%             | 1.12 | contaminated |
| (28, 29)   | 0.6013%   | 12.7 $\mu\text{s}$  | 4.430%             | 0.14 | contaminated |
| (77, 78)   | 0.9860%   | 212.3 $\mu\text{s}$ | 0.265%             | 3.73 | usable       |
| (45, 46)   | 1.6181%   | 57.2 $\mu\text{s}$  | 0.983%             | 1.65 | usable       |
| (66, 67)   | 2.6181%   | 86.6 $\mu\text{s}$  | 0.649%             | 4.03 | usable       |
| (23, 24)   | 4.3229%   | 65.1 $\mu\text{s}$  | 0.863%             | 5.01 | usable       |

Qubit 28 carries a **T2 of 12.7  $\mu\text{s}$**  against a chip median in the hundreds — roughly seven parts dephasing to one part gate error. It was never measuring forgiveness. Edge (119,120) sits at  $D = 1.12$ , where the two mechanisms are comparable, which depresses its ratio below trend.

Refitting on the four edges that satisfy  $D \geq 1.5$ :

**ratio =  $0.0898 \times \text{err}^{-0.3873}$   $R^2 = 0.9648$**

$R^2$  moves from 0.296 to 0.965 on removal of two points. **The power-law form was never the problem. The edge-selection protocol was.** Edges must be screened on D, not on error magnitude alone.

## 6. Retroactive Screen — Entry 019's Anchor Is Contaminated

Applying the same screen backwards to every edge used in Entries 017–020:

| Chip       | Edge     | Raw error | T2 min              | D    | Verdict      |
|------------|----------|-----------|---------------------|------|--------------|
| Cairo      | (24, 25) | 0.5992%   | 192.4 $\mu\text{s}$ | 4.32 | usable       |
| Cairo      | (19, 22) | 3.1315%   | 6.9 $\mu\text{s}$   | 0.31 | CONTAMINATED |
| Sherbrooke | (60, 61) | 0.3470%   | 360.3 $\mu\text{s}$ | 2.34 | usable       |
| Sherbrooke | (66, 73) | 3.0667%   | 170.0 $\mu\text{s}$ | 9.78 | usable       |

Cairo's high-error edge (19,22) has **T2 = 6.9  $\mu\text{s}$**  and a 697 ns gate — **10.1% decoherence per gate against 3.13% gate error**. Entry 019's measured ratio of 0.334 at that edge is roughly three parts dephasing to one part gate error.

That number is one of the two points hardcoded in `emulator_v3_routing.py` :: `variable_forgiveness_ratio()`. **The emulator's error model is currently anchored on a broken qubit.** Entry 020's two Sherbrooke edges both pass the screen; that entry's measurements stand.

## 7. Impact on the Emulator

| Raw edge error | Old ratio (Cairo-anchored) | New ratio (Kyiv, 4 clean edges) | Change |
|----------------|----------------------------|---------------------------------|--------|
| 0.300%         | 0.6433                     | 0.8518                          | +32.4% |

| Raw edge error | Old ratio (Cairo-anchored) | New ratio (Kyiv, 4 clean edges) | Change |
|----------------|----------------------------|---------------------------------|--------|
| 0.500%         | 0.5577                     | 0.6989                          | +25.3% |
| 1.000%         | 0.4595                     | 0.5343                          | +16.3% |
| 2.000%         | 0.3785                     | 0.4085                          | +7.9%  |
| 3.000%         | 0.3380                     | 0.3491                          | +3.3%  |
| 4.500%         | 0.3018                     | 0.2984                          | −1.1%  |

The two laws agree near 3–4.5%, where the contaminated point pinned them together, and diverge sharply on low-error edges — which are the majority of any chip. **75% of Kyiv's calibrated edges sit below 1.6%.**

Circuit-level consequence, N CNOTs on a 1.0% edge:

| CNOTs | Old prediction | New prediction | Gap       |
|-------|----------------|----------------|-----------|
| 1     | 99.54%         | 99.47%         | −0.07 pts |
| 10    | 95.50%         | 94.78%         | −0.72 pts |
| 20    | 91.20%         | 89.84%         | −1.36 pts |
| 40    | 83.18%         | 80.71%         | −2.46 pts |

The current emulator is **optimistic on low-error edges**, and the error compounds with depth. At 40 CNOTs the correction is 2.46 points — comparable in size to the 1.87-point validation gap that Entry 019 recorded as a success.

## 8. Robustness Check

Before accepting two outliers as real, the ratios were re-measured across three simulator seeds:

| Edge     | Seed 1729 | Seed 42 | Seed 20260804 | Spread |
|----------|-----------|---------|---------------|--------|
| (77, 78) | 0.5353    | 0.5583  | 0.5330        | 0.0253 |
| (23, 24) | 0.2927    | 0.3076  | 0.3143        | 0.0216 |

Seed spread is ~0.02. The two outlier residuals are 0.37 and 0.94 — fifteen to forty times larger. The outliers are physics, not sampling noise.

## 9. What This Means for QuantumBridge

**Edge screening becomes part of the protocol.** Any edge used to fit a forgiveness ratio must satisfy  $D \geq 1.5$ , and D must be reported alongside every future measurement. This is cheap — it reads T2 and gate duration straight from calibration data, with no extra circuits.

**Entry 020's calibration shortcut is withdrawn.** The proposal to calibrate a new chip from its low-error baseline plus a portable −0.63 constant does not survive. Kyiv gives 0.674 over a 13.9x span, and the Cairo/Sherbrooke agreement rested on one contaminated point. The cost saving was real but the constant underneath it was not.

**Chip-invariance of the exponent remains unproven.** Kyiv's clean four-point −0.387 still differs from Sherbrooke's two-point −0.212. Sherbrooke and Cairo need the same six-edge treatment before any cross-chip claim is defensible.

**Honest caveat.** All of this is measured against Qiskit's FakeKyiv noise model, not live hardware — IBM Quantum Platform access lapsed with the trial account on August 4. The  $T_2 = 12.7 \mu\text{s}$  on qubit 28 is a genuine recorded calibration value from a real device snapshot and the decoherence mechanism is real, but the *magnitude* of the correction should be re-validated on hardware when access is restored. The protocol change holds regardless.

## Entry 021 — Conclusion

Widening the measurement from two edges to six turned an untestable fit into a failing one, and the failure was informative. Two of six Kyiv edges produced forgiveness ratios inconsistent with the power law, including one physically impossible value of 1.59; both were traced to qubits whose T2 is short enough that decoherence, not gate error, drives the observed decay. Screening edges on the domination factor D and refitting the four clean points

restores the power law at  $R^2 = 0.9648$ . Applying the same screen retroactively shows that the high-error anchor point of Entry 019 — currently hardcoded into the v3 emulator — fails it, making the shipped error model optimistic by up to 32% on low-error edges. The ratio-of-ratios finding of Entry 020 is withdrawn. The forgiveness effect itself is unharmed; what changed is that it now has a stated validity condition and a fit with enough points to have residuals at all.

10. Next Step — Entry 022

Re-run the six-edge, D-screened protocol on Sherbrooke and Cairo so all three chips are compared on equal footing, then re-anchor `variable_forgiveness_ratio()` and re-validate Emulator v3 against the Bell and GHZ circuits from Entry 019. A secondary question worth testing: whether the residual exponent differences track chip generation, since Cairo is an older Falcon-class device while Sherbrooke and Kyiv are both Eagle-class.

Research Log — Updated Index (Entries 017–021)

Second addendum to the Table of Contents. Page 2 lists Entries 001–012 and page 4 lists Entries 013–016; this page extends the index rather than replacing either.

| Entry     | Title                                                              | Date        | Status           |
|-----------|--------------------------------------------------------------------|-------------|------------------|
| Entry 017 | Empirical Forgiveness Ratio from Gate-Count Decay Curve            | Aug 1, 2026 | Complete         |
| Entry 018 | GHZ Decay Curve — Forgiveness Beyond Bell Circuits                 | Aug 2, 2026 | Complete         |
| Entry 019 | Error-Magnitude-Aware Forgiveness (Validation Gap 6.14 → 1.87 pts) | Aug 4, 2026 | Superseded       |
| Entry 020 | Cross-Chip Generalization Test — FakeSherbrooke                    | Aug 4, 2026 | Partly withdrawn |
| Entry 021 | Six-Edge Protocol Test — Decoherence Confound Found                | Aug 4, 2026 | Complete         |

Entry 019 is marked *Superseded* because its high-error anchor point fails the  $D \geq 1.5$  screen introduced here; its low-error measurement stands. Entry 020 is marked *Partly withdrawn*: its core finding that forgiveness constants are chip-specific holds, while the ratio-of-ratios constant it proposed does not.

ENTRY 022 Validation Harness Correction — August 6, 2026 — Status: COMPLETE

## Entry 022 — v3's First Honest Validation: Two Failures Pointing Opposite Ways

*The Cairo Harness Was Silencing Half Its Noise. Replacing It Immediately Exposed a Missing Decoherence Term.*

### 1. What Was Done

Entry 021 closed by naming `bell_scattered_0_26` — a 14-point discrepancy — as the largest remaining error in the emulator and the next priority. This entry set out to diagnose that single circuit. It found instead that the validation harness itself was defective, that no circuit-level accuracy figure from Entry 014 onward was measurable, and, once a sound harness was built, that v3 is missing a physical term entirely.

### 2. First Defect — the Basis-Gate Mismatch

FakeCairoV2 is a **mixed-basis snapshot**: 12 of its directed edges are calibrated as `cx`, the other 14 as `ecr`. The harness transpiled with `basis_gates = backend.operation_names`, handing the transpiler the union of both. The transpiler selects one two-qubit gate for the whole circuit — `cx` — and emits it on every edge, including the fourteen that only carry `ecr` calibration.

Qiskit's noise model keys its quantum errors by *(gate name, qubit pair)*. For those gates it finds no matching entry and applies **nothing at all**. The gate executes perfectly, silently, for free.

| Circuit                            | 2q gates | noisy | SILENT | % free |
|------------------------------------|----------|-------|--------|--------|
| <code>ghz_local_0_1_2</code>       | 2        | 0     | 2      | 100%   |
| <code>ghz_scattered_0_26_22</code> | 35       | 20    | 15     | 43%    |
| <code>bell_adjacent_24_25</code>   | 1        | 1     | 0      | 0%     |
| <code>bell_scattered_0_26</code>   | 34       | 13    | 21     | 62%    |
| TOTAL                              | 72       | 34    | 38     | 53%    |

Over half of every two-qubit gate in the suite ran without error. `ghz_local_0_1_2` — the circuit that had looked most accurate throughout — ran with zero noise on both of its gates.

### 3. Second Defect — Calibration Coverage

Independently of the basis problem, only **12 of Cairo's 28 topology edges** carry real calibrated error values — 42.9%. The remaining 16 fell back to v3's hardcoded 1% constant. On that chip, most of what the model reported was not a measurement of anything.

### 4. A Sound Harness — FakeKyiv

FakeKyiv has neither defect: uniformly `ecr`, noise-model coverage on all 144 edges, and 139/144 (96.5%) real calibration coverage. The replacement harness, `validate_v3_kyiv.py`, audits its own noise coverage on every run and declares its results void if any gate runs silent.

One further trap was found in the process. Aer's automatic simulation method returns `success=True` and `status=COMPLETED` for the deeper routed circuits, then raises on `get_counts()`. A harness trusting it would silently lose precisely its hardest cases. `matrix_product_state` is mandatory here, not preferred.

### 5. Results — All Gates Verified Noisy

| Circuit                          | hops | 2q gates | Aer reference | v3     | gap   |
|----------------------------------|------|----------|---------------|--------|-------|
| <code>bell_adjacent_77_78</code> | 1    | 1        | 96.07%        | 99.25% | 3.18  |
| <code>bell_near_77_82</code>     | 5    | 13       | 48.74%        | 92.89% | 44.15 |
| <code>bell_mid_77_100</code>     | 6    | 16       | 83.69%        | 90.27% | 6.58  |

| Circuit            | hops | 2q gates | Aer reference | v3     | gap   |
|--------------------|------|----------|---------------|--------|-------|
| bell_far_0_126     | 26   | 76       | 69.90%        | 60.51% | 9.39  |
| ghz_local_77_78_79 | 2    | 2        | 88.53%        | 98.67% | 10.14 |

Mean absolute gap 14.69 points. The mean is misleading, and that is the finding. (ghz\_scattered\_0\_63\_126 was killed mid-run — two 26-hop routes exceeded MPS memory — and is excluded.)

## 6. The Anomaly — 13 Gates Worse Than 76

**Thirteen gates produce 48.74%. Seventy-six gates produce 69.90%.** Six times the gate count, a better outcome. Circuit depth does not explain this chip's behaviour.

The 77→82 route passes through **qubit 80: T2 = 8.51 μs, T1 = 5.02 μs** — effectively dead against a chip median in the hundreds. With a 560 ns ecr, that is roughly 6.6% dephasing per gate, and the route crosses it six times.

| Route    | worst T2 on path   | 2q gates | Aer reference |
|----------|--------------------|----------|---------------|
| 77 → 82  | 8.5 μs (qubit 80)  | 13       | 48.74%        |
| 77 → 100 | 32.7 μs            | 16       | 83.69%        |
| 0 → 126  | healthy throughout | 76       | 69.90%        |

Compounding the failure: edges (79,80) and (80,81) are among the five Kyiv edges with **no calibration data**, so v3 applied its 1% fallback guess at exactly the point where reality is catastrophic. Two independent weaknesses landed on the same two edges.

## 7. Diagnosis — No Decoherence Term

v3 models two-qubit *gate error* only. It has no decoherence term at all. The consequences run in opposite directions:

- **Short routes through low-T2 qubits are catastrophically under-penalized** — 44 points on 77→82.
- **Long routes through healthy qubits are over-penalized** by SWAP compounding — 9 points on 0→126, with v3 pessimistic.

Because the two errors have opposite sign, any single mean accuracy figure conceals both. This is Entry 021's finding resurfacing one level up: there, the domination factor  $D = \text{gate error} \div (\text{gate duration} \div T2)$  was introduced as a *screening* criterion for which edges were fit to measure. Entry 022 shows the same quantity belongs *inside the model*. Every edge on the 77→82 route through qubit 80 sits at  $D \approx 0.15$  — the decoherence-dominated regime Entry 021 identified.

## 8. A Naive Fix, and Why It Was Rejected

The obvious correction is to charge  $\text{gate duration} \div T2$  per gate alongside the gate-error term. Tested on the real transpiled routes:

| Circuit           | Aer reference | v3 gate-only       | v3 + decoherence   |
|-------------------|---------------|--------------------|--------------------|
| bell_near_77_82   | 48.74%        | 93.10% (gap 44.36) | 58.83% (gap 10.09) |
| bell_far_0_126    | 69.90%        | 64.06% (gap 5.84)  | 30.67% (gap 39.23) |
| MEAN ABSOLUTE GAP | —             | 14.10              | 13.85              |

It repairs the short bad route and destroys the long good one; the mean barely moves. The reason is physical. In a SWAP chain **the state moves**, so each qubit holds it only briefly. Charging every gate the full T2 cost of both its qubits double-counts heavily across 76 gates spread over 26 healthy qubits, while being roughly right for 6 gates parked on one dead qubit.

Decoherence must be **time-integrated along the carrier path** — summed, over each qubit the state actually occupies, of time spent there  $\div$  that qubit's T2 — not applied per gate to both members of every pair. That is the v4 design.

## 9. What This Means for QuantumBridge



**No v3 accuracy figure predates this entry.** Every circuit-level number from Entry 014 onward was measured against a reference silencing half its noise. The forgiveness work of Entries 017–021 is unaffected — those used single-edge circuits with no routing, on properly covered edges — but the emulator validation figures must be treated as withdrawn.

**FakeCairoV2 is retired for validation.** Mixed basis and 42.9% calibration coverage make it unusable for this purpose. Kyiv or Sherbrooke only.

**The fallback constant is actively dangerous.** It fires on uncalibrated edges, and uncalibrated edges correlate with broken qubits — precisely where a wrong guess costs most. v4 should refuse to predict on an uncalibrated edge rather than quietly assume 1%.

**Routing is error-blind.** v3's BFS minimizes hop count while ignoring both edge error and T2. The 77→82 route is the shortest available and also among the worst.

Entry 022 — Conclusion

An investigation into one anomalous circuit found the measuring instrument broken. The Cairo validation harness emitted cx gates on ecr-calibrated edges, where the noise model applied no error whatsoever — 53% of all two-qubit gates in the suite, and 100% of them in the circuit that had appeared most accurate. Rebuilt on FakeKyiv with per-run noise-coverage auditing, the first trustworthy measurement shows v3 failing in two opposite directions at once: under-penalizing short routes through low-T2 qubits by 44 points, and over-penalizing long routes through healthy ones by 9. The cause is a missing decoherence term. A naive per-gate correction was tested and rejected because it double-counts along a SWAP chain, which localizes the requirement: decoherence must be integrated along the path the quantum state actually travels. The headline number from this entry is not an accuracy figure — it is that the project did not previously have one.

10. Next Step — Entry 023

Two changes, in order. First, remove the 1% fallback so v3 declines to predict on uncalibrated edges instead of guessing. Second, implement v4's time-integrated decoherence term and re-validate against the Kyiv reference established here. A third question follows from Section 9: whether routing should be made error-aware and T2-aware, since the shortest path is demonstrably not the cheapest one.

Research Log — Index Addendum (Entry 022)

| Entry     | Title                                                                           | Date        | Status   |
|-----------|---------------------------------------------------------------------------------|-------------|----------|
| Entry 022 | Validation Harness Correction — Basis Mismatch and the Missing Decoherence Term | Aug 6, 2026 | Complete |

Entries 014–016 and the circuit-level validation figures of Entries 019 and 021 are marked **measurement void** by this entry: their references silenced 53% of two-qubit noise. Their methodology and reasoning stand; only the accuracy numbers are withdrawn.

ENTRY 024   Exact SWAP Dwell-Time Routing — August 12, 2026 — Status: COMPLETE

# Entry 024 — Exact Routing Solves the Problem It Targeted, and Reopens the One From Entry 016

*Replacing a Guessed SWAP Chain With the Real One Fixes the Low-T2 Failure and Confirms an Older, Larger Bottleneck*

## 1. What Was Done

Entry 023 built v4 with a decoherence term, but had no way to know which physical qubits a routed circuit actually passed through — it reused v3's BFS shortest-hop path and charged each hop a flat, guessed duration. That produced a large regression on `bell_far_0_126` (gap 9.39 → 34.60 pts) because the guessed dwell time had no relationship to the route Qiskit's real router, SABRE, actually chose.

This entry removes the guess. Transpiling with `'swap'` kept as its own undecomposed gate — rather than letting BasisTranslator expand every SWAP into three indistinguishable `cx/ecr` gates, which is what made every prior entry's routing invisible to inspection — exposes the exact physical path and the real duration of each leg.

| Instruction     | Physical qubits |
|-----------------|-----------------|
| swap            | [77, 78]        |
| swap            | [82, 81]        |
| swap            | [80, 81]        |
| swap            | [80, 79]        |
| ecr (real gate) | [79, 78]        |

*bell\_near\_77\_82's real route, recovered directly from the transpiled circuit rather than assumed. The final entangling edge, (79,78), is not the shortest hop-count path — SABRE optimized for something v3's BFS model does not represent at all.*

## 2. A Bug Found Before Trusting the Result

The first version of the walker charged decoherence for each SWAP but **no gate error at all**, and used one gate's duration instead of the physical reality — a SWAP is `GATES_PER_SWAP` (3) real two-qubit pulses back to back, not one. Both v3 and v4-BFS already applied the gate-error term on every hop; the rewrite silently dropped it. The result was an implausibly optimistic 83.42% on `bell_near_77_82` against an actual 48.74% — worse than the heuristic it was meant to replace. Caught by comparing against the Entry 022 reference before accepting the number, not after.

## 3. Verifying the Route Extraction Itself

Before trusting the corrected model, the extraction method was checked independently. `bell_far_0_126's` exact route used 25 SWAPs. Decomposed at 3 gates each plus 1 real entangling gate:  $25 \times 3 + 1 = 76$  — exactly the `ecr` count of the independently transpiled Aer reference circuit from Entry 022, and identical whether the extraction ran at `optimization_level` 1, 2, or 3.

*The route the model reasons about is the route SABRE actually built.*

## 4. Results

| Circuit             | hops | Aer    | v3             | v4 (BFS)       | v4.1 (exact)   |
|---------------------|------|--------|----------------|----------------|----------------|
| bell_adjacent_77_78 | 1    | 96.07% | 99.25% (3.18)  | 96.27% (0.20)  | 96.12% (0.05)  |
| bell_near_77_82     | 5    | 48.74% | 92.89% (44.15) | 70.35% (21.61) | 51.13% (2.39)  |
| bell_mid_77_100     | 6    | 83.69% | 90.27% (6.58)  | 78.41% (5.28)  | 64.62% (19.07) |
| bell_far_0_126      | 26   | 69.90% | 60.51% (9.39)  | 35.30% (34.60) | 24.07% (45.83) |
| ghz_local_77_78_79  | 2    | 88.53% | 98.67% (10.14) | 89.12% (0.59)  | 88.21% (0.32)  |

Values in parentheses are the absolute gap in points. **Median gap: 9.39 → 5.28 → 2.39.** Mean gap: 14.69 → 12.46 → 13.54 — worse than v4-BFS, for reasons in Section 6.

## 5. The Win

bell\_near\_77\_82 was the anomaly Entry 022 opened with — a route crossing qubit 80 ( $T_2 = 8.51 \mu\text{s}$ ) that v3 and v4-BFS both over-predicted by 44 and 22 points respectively. Exact routing brings that gap to **2.39 points**. This is the specific failure this entry set out to fix, and it is fixed: once the model knows exactly which qubit the state dwells on and for how long, a route through a broken qubit is no longer invisible to it.

## 6. The Regression, and Why It Is Not a New Problem

bell\_mid\_77\_100 and bell\_far\_0\_126 — long routes through otherwise healthy qubits — got worse: 5.28 → 19.07 and 34.60 → 45.83 points. Both were already v4-BFS's worst cases, in the same direction (over-pessimistic), before this entry touched them.

This is Entries 015 and 016 resurfacing. Entry 015 found that switching to per-edge SWAP cost *increased* the mean prediction gap rather than closing it, despite the real transpiled gate count matching v3's estimate closely. Entry 016 isolated the cause to **v3's multiplicative error-compounding formula itself** —  $(1 - e)^n$  — independent of both routing path length and real gate count. That finding was never resolved; it was carried forward as an open problem.

Exact routing answers a different question than that one. It confirms *where* the state travels and for how long — verified in Section 3 against the real transpiled circuit — but it cannot fix an error rooted in *how* per-gate error compounds. Adding decoherence, a second multiplicative term, on top of a gate-error model that Entry 016 already showed over-penalizes long chains does not cancel that problem. It compounds it — literally, in the same sense as the formula under discussion.

## Entry 024 — Conclusion

Exact SWAP dwell-time routing does what it was built to do. Median prediction gap fell from 5.28 to 2.39 points, and the project's single worst-known failure — a route through a near-dead qubit that two prior model versions missed entirely — is reduced from a 21.61-point error to 2.39. The route extraction itself was verified independently: the exact gate count it derives matches the real transpiled circuit precisely, at every optimization level tested. But mean gap moved the wrong direction, from 12.46 to 13.54 points, because the two long, otherwise-healthy routes in the test set got worse. That regression is not new. Entries 015 and 016 already diagnosed and named the cause — v3's multiplicative gate-error compounding formula over-penalizes long chains regardless of routing accuracy — and left it unresolved. This entry removed the uncertainty around where the state travels and, in doing so, made that older, larger problem the last one left standing.

## 7. Next Step — Entry 025

Reopen the compounding formula directly, using the exact routes this entry can now supply as ground truth rather than a BFS estimate. The specific question Entry 016 left unanswered: is  $(1 - e)^n$  the wrong functional form for correlated or partially-forgiven error across many sequential gates, or is the per-gate error term itself miscalibrated at scale? Exact routing makes this testable in a way it was not before, because the gate sequence feeding the formula is now known to be correct.

## Research Log — Index Addendum (Entry 024)

| Entry     | Title                         | Date         | Status   |
|-----------|-------------------------------|--------------|----------|
| Entry 024 | Exact SWAP Dwell-Time Routing | Aug 12, 2026 | Complete |

Entry 023's v4-BFS validation figures for bell\_mid\_77\_100 and bell\_far\_0\_126 are superseded by the exact-routing figures in Section 4 above. Entries 015 and 016's compounding-formula finding is upgraded from an open question to the confirmed dominant source of remaining error, now that routing accuracy has been independently verified.

ENTRY 025 Isolating the Compounding Formula — August 13, 2026 — Status: COMPLETE

## Entry 025 — The Forgiveness Ratio Has a Closed Form

*A Question Left Open Since Entry 016 Turns Out to Have an Exact Answer, Derived From First Principles and Verified Against Aer*

### 1. What Was Done

Entry 024 confirmed that routing and real gate count are not the source of v3/v4's remaining error, leaving one candidate standing: the multiplicative compounding formula  $(1 - p)^n$  itself, flagged as suspect in Entries 015 and 016 and never resolved. Testing it against FakeKyiv's full noise model conflates three effects at once — gate error, decoherence, and readout. This entry isolates one: an Aer noise model carrying **only** a depolarizing error on a single real Kyiv edge, no thermal relaxation, no readout error, applied  $n$  times to a cancelling-pair Bell circuit — the same method Entries 017–021 used, now aimed at a controlled synthetic channel instead of the full hardware model.

### 2. Result — a Confirmed Mismatch

| n gates | Aer (isolated) | naive $(1-p)^n$ | ratio  | diff, pts |
|---------|----------------|-----------------|--------|-----------|
| 1       | 99.435%        | 99.014%         | 1.0043 | +0.42     |
| 9       | 95.541%        | 91.468%         | 1.0445 | +4.07     |
| 21      | 90.292%        | 81.213%         | 1.1118 | +9.08     |
| 51      | 79.965%        | 60.329%         | 1.3255 | +19.64    |
| 75      | 73.709%        | 47.560%         | 1.5498 | +26.15    |

Edge (77,78), raw calibrated error 0.986%. Even with decoherence and readout completely removed, a chain of the identical depolarizing gate does not compound as  $(1 - p)^n$ . At 75 gates the gap is 26 points. Fitting an effective decay constant to the Aer curve gives 0.41% per gate against a nominal 0.99% — a ratio of **0.4156**, in the same range as the forgiveness ratios measured in Entries 017–021.

### 3. A Closed Form

Qiskit's two-qubit depolarizing\_error( $p$ ) channel does exactly one of two things per application: with probability  $(1 - p)$  it applies the ideal gate; with probability  $p$  it replaces the state with the maximally mixed state on 2 qubits. The maximally mixed state is an **absorbing state** — depolarizing it again changes nothing. So after  $n$  applications: the state never depolarized with probability  $(1-p)^n$ , contributing its full population to the {00,11} ideal set; it depolarized at least once with probability  $1-(1-p)^n$ , and a maximally mixed 2-qubit state distributes evenly across 4 basis outcomes, so exactly half of that population still lands in {00,11} by chance.

$$\text{success} = (1-p)^n \times 1 + [1-(1-p)^n] \times 0.5 = 0.5 + 0.5(1-p)^n$$

### 4. Verification

| n  | Aer measured | $0.5 + 0.5(1-p)^n$ | diff, pts |
|----|--------------|--------------------|-----------|
| 5  | 97.519%      | 97.583%            | -0.06     |
| 21 | 90.292%      | 90.607%            | -0.31     |
| 51 | 79.965%      | 80.164%            | -0.20     |
| 75 | 73.709%      | 73.780%            | -0.07     |

Every point across nine gate counts (1 to 75) matches within 0.35 points — consistent with pure shot noise at 4 seeds  $\times$  8192 shots. Cross-checked on a second, unrelated edge (23,24), raw error 4.32% — 4.4 $\times$  the first edge's magnitude:

| n  | Aer measured | predicted | diff, pts |
|----|--------------|-----------|-----------|
| 5  | 89.749%      | 90.088%   | -0.34     |
| 21 | 69.791%      | 69.767%   | +0.02     |
| 51 | 55.124%      | 55.250%   | -0.13     |

Same closed form, same tolerance, an entirely different error magnitude. This is not curve-fitting — it is a derivation confirmed by measurement.

5. What This Means for QuantumBridge

The forgiveness ratios empirically fitted in Entries 017–021 were never purely a chip-physics phenomenon. For a Bell-state circuit, a large part of that ratio is a fixed geometric fact: **a 2-outcome-of-4 ideal set absorbs roughly half of any depolarizing error**, independent of calibration data, chip identity, or gate duration. It follows directly from measuring against a multi-outcome ideal set — exactly the mechanism flagged qualitatively in earlier entries, now given an exact formula.

This does not retire the empirical power law. The closed-form ratio approaches a floor of 0.5 as n grows; Entry 021's fitted law (coefficient 0.0898, exponent –0.3873) predicts ratios below 0.5 that keep falling with error magnitude, with no such floor. The gap between the two is where real chip physics — decoherence, correlated errors, whatever Entry 021's D-screen was isolating — actually lives. Entries 017–021 were fitting a curve that mixes a universal geometric constant with genuine hardware behavior, without separating them.

Entry 025 — Conclusion

The compounding-formula question left open by Entries 015 and 016 has an exact answer for the circuits this project has run throughout:  $\text{success} = 0.5 + 0.5(1-p)^n$ , not  $(1-p)^n$ . It follows from a two-line argument about how a depolarizing channel behaves once it has already fired once, and it matches Aer's own noise model to within shot noise across two edges spanning a 4.4× range in error magnitude. The naive multiplicative formula was never approximately right — it was measuring the wrong quantity from the start, decoherence and readout aside entirely. What remains genuinely open is how much of Entries 017–021's fitted power-law shape is this fixed geometric fact and how much is real hardware physics underneath it.

6. Next Step — Entry 026

Subtract the closed-form baseline from Entries 017–021's raw measurements and refit the residual. If a clean power law remains, that residual — not the mixed quantity fitted so far — is the real chip-physics signal, and it should replace `variable_forgiveness_ratio()` in v4. If nothing structured remains, the empirical fits from Entries 017–021 were mostly measuring this geometric artifact, and the decoherence term already added in Entries 023–024 is doing the real physical work.

Research Log — Index Addendum (Entry 025)

| Entry     | Title                                                                  | Date         | Status   |
|-----------|------------------------------------------------------------------------|--------------|----------|
| Entry 025 | Isolating the Compounding Formula — a Closed-Form Forgiveness Baseline | Aug 13, 2026 | Complete |

Entries 015 and 016's open question is resolved: the compounding formula was wrong, independent of decoherence and readout, and is now wrong by a known, derived amount. Entries 017-021's fitted power law is not withdrawn but is now understood to be a mixture of this geometric baseline and unseparated hardware physics.

ENTRY 026 Baseline-Subtracted Residual Analysis — August 13, 2026 — Status: COMPLETE

# Entry 026 — Most of the Forgiveness Ratio Was the Baseline All Along

Subtracting Entry 025's Closed Form From Every Recorded Decay Curve Leaves Almost Nothing Behind

## 1. What Was Done

Entry 025 derived a closed-form baseline for Bell-circuit forgiveness —  $\text{success} = 0.5 + 0.5(1-p)^n$  — from pure depolarizing error alone, no decoherence, no chip identity. Entries 017–021's fitted forgiveness ratios were measured against the full hardware noise model, which includes that baseline plus whatever real chip physics sits on top of it. This entry subtracts one from the other, across every Bell decay curve on record — ten edges spanning Cairo, Sherbrooke, and Kyiv — to see what's actually left.

For each curve: fit  $k_{\text{measured}}$  the same log-linear way the original entries did; fit  $k_{\text{baseline}}$  the identical way on the closed-form formula evaluated at the same gate counts, so the sampling matches exactly even though the baseline isn't a pure exponential; then  $\text{residual} = (k_{\text{measured}}/p) - (k_{\text{baseline}}/p)$ .

## 2. Result

| edge               | raw err | D    | ratio (orig) | ratio (baseline) | residual |
|--------------------|---------|------|--------------|------------------|----------|
| cairo (24,25)      | 0.600%  | 4.32 | 0.5304       | 0.4704           | +0.060   |
| cairo (19,22)      | 3.132%  | 0.31 | 0.3337       | 0.3502           | -0.017   |
| sherbrooke (60,61) | 0.347%  | 2.34 | 0.6665       | 0.4829           | +0.184   |
| sherbrooke (66,73) | 3.067%  | 9.78 | 0.4200       | 0.3531           | +0.067   |
| kyiv (119,120)     | 0.311%  | 1.12 | 0.4345       | 0.4846           | -0.050   |
| kyiv (28,29)       | 0.601%  | 0.14 | 1.5904       | 0.4703           | +1.120   |
| kyiv (77,78)       | 0.986%  | 3.73 | 0.5353       | 0.4513           | +0.084   |
| kyiv (45,46)       | 1.618%  | 1.65 | 0.4308       | 0.4204           | +0.010   |
| kyiv (66,67)       | 2.618%  | 4.03 | 0.3937       | 0.3732           | +0.021   |
| kyiv (23,24)       | 4.323%  | 5.01 | 0.2927       | 0.3009           | -0.008   |

On the seven D-screened usable edges ( $D \geq 1.5$ ), residuals range from  $-0.008$  to  $+0.184$  with inconsistent sign, mean  $+0.060$ . No power law could be fitted in log-log space — some values are negative. Seven points scattered near zero across a  $0.35\%$ – $4.32\%$  error span, without a consistent trend, is what noise looks like.

## 3. An Honest Complication

The D-screen prediction — that decoherence-contaminated edges ( $D < 1.5$ ) should show large residuals, since the baseline has no decoherence term — only partly held. kyiv(28,29), the single worst case ( $D = 0.14$ ,  $T_2 = 12.7 \mu\text{s}$ ), showed a residual of **+1.120**, consistent with severe unmodeled decoherence. But cairo(19,22) ( $D = 0.31$ ) and kyiv(119,120) ( $D = 1.12$ ), both also decoherence-affected, showed small *negative* residuals instead.

The cause is not physics — it's the metric. Residual divides by raw calibrated error  $p$ . kyiv(28,29)'s  $p = 0.601\%$  is roughly  $5\times$  smaller than cairo(19,22)'s  $p = 3.132\%$ . The same absolute amount of unmodeled decoherence, divided by a five-times-smaller denominator, produces a residual an order of magnitude larger — not because the underlying physics differs by that much, but because the ratio metric is numerically unstable exactly where  $p$  is small. This is the same small-denominator fragility that motivated the D-screen in Entry 021, resurfacing in the residual analysis regardless of whether decoherence contamination is present.

## Entry 026 — Conclusion

Subtracting the Entry 025 closed-form baseline from every recorded Bell-circuit decay curve leaves, on the seven properly screened edges, no discernible power-law structure — residuals scattered near zero with inconsistent sign. The most defensible reading is that Entries 017–021's empirically fitted forgiveness ratio was mostly re-deriving the fixed geometric baseline, not discovering new hardware physics on top of it. The one edge with a genuinely large residual, kyiv(28,29), is explained by a numerical artifact of the ratio metric at small  $p$ , not by an inconsistency in the underlying physics — its raw decoherence severity ( $D = 0.14$ ) is comparable to, not dramatically worse than, cairo(19,22)'s ( $D = 0.31$ ), yet the ratio-based residual differs by more than an order of magnitude. Ratio metrics should not be trusted at low raw error, independent of  $D$ .

4. What This Means for QuantumBridge

`variable_forgiveness_ratio()` should be retired, not re-fitted. With no clean residual signal to swap in, the empirically fit power law (Entries 017–021) was mostly measuring the closed-form baseline under a different name. The correct replacement is the derived formula itself — deterministic, requires no calibration data to fit, and is already verified to within shot noise.

The real physical signal — decoherence — is already handled correctly elsewhere: v4's T2-based dephasing term (Entries 023–024), which reads coherence times directly rather than inferring them indirectly through an unstable ratio. Asking a forgiveness-ratio fit to implicitly capture decoherence was always going to be fragile at low error magnitudes; this entry shows exactly why.

5. Next Step — Entry 027

Replace `variable_forgiveness_ratio()` in v4 with the closed-form baseline directly, remove the empirical power-law fit and its two hardcoded anchor points entirely, and re-validate against the Entry 022 Kyiv reference circuits to confirm the simplification doesn't regress Entry 024's routing fix. If it holds, the six-edge decay-curve protocol itself may no longer be necessary for future chips — only the D-screen and T2/T1 calibration export would be, considerably cutting the calibration cost the project has been carrying since Entry 013.

Research Log — Index Addendum (Entry 026)

| Entry     | Title                                 | Date         | Status   |
|-----------|---------------------------------------|--------------|----------|
| Entry 026 | Baseline-Subtracted Residual Analysis | Aug 13, 2026 | Complete |

Entries 017-021's forgiveness-ratio power law is superseded: not wrong as a measurement, but shown to be mostly re-deriving the Entry 025 geometric baseline rather than capturing independent chip physics.



ENTRY 027 Retiring `variable_forgiveness_ratio()` — August 13, 2026 — Status: COMPLETE

## Entry 027 — The Fitted Correction Comes Out, the Derived One Goes In

*Generalizing Entry 025's Closed Form to Real Multi-Edge Routes, and Checking It Doesn't Cost Anything*

### 1. What Was Done

Entry 026 found no clean residual left after subtracting the Entry 025 closed-form baseline from every recorded decay curve — the empirically fitted `variable_forgiveness_ratio()` power law was mostly re-deriving that baseline, not new chip physics. Its recommendation was to retire the fitted correction and use the derived formula directly. This entry does that.

Entry 025's closed form was derived for one edge's SAME depolarizing channel applied  $n$  times. Real routes cross several different edges with different raw errors, so the absorbing-state argument needed generalizing first: each two-qubit gate application either fires the channel (probability  $p_i$ , sending the state to maximally mixed) or doesn't (probability  $1-p_i$ ) — independent of every other gate, and firing anywhere is absorbing. So the state survives untouched with probability  $\prod_i (1-p_i)$ , and expected success is  $\text{floor} + (1-\text{floor}) \cdot \prod_i (1-p_i)$ , where floor is the ideal outcome set's share of the full outcome space (0.5 for a 2-qubit Bell pair, 0.25 for a 3-qubit GHZ state). No fitted coefficients — only the raw calibrated error and a combinatorial floor.

Implemented in v4 (`route_cost/predict`) and the exact-dwell router (`exact_dwell_cost`): gate-error survival is now accumulated as one product across the whole circuit and converted to a probability exactly once via the floor formula, instead of being forgiveness-corrected gate-by-gate and multiplied straight into a running success value. Decoherence, single-qubit gate cost, and readout are untouched — they were never part of the forgiveness fit.

### 2. Re-Validation Against the Entry 022 Kyiv Reference

| circuit             | Aer ref | v4.1 (old, fitted) | v4.1 (new, closed-form) | $\Delta$ |
|---------------------|---------|--------------------|-------------------------|----------|
| bell_adjacent_77_78 | 96.07%  | gap 0.05           | gap 0.09                | +0.04    |
| bell_near_77_82     | 48.74%  | gap 2.39           | gap 2.68                | +0.29    |
| bell_mid_77_100     | 83.69%  | gap 19.07          | gap 19.11               | +0.04    |
| bell_far_0_126      | 69.90%  | gap 45.83          | gap 43.79               | -2.04    |
| ghz_local_77_78_79  | 88.53%  | gap 0.32           | gap 0.76                | +0.44    |
| MEAN                |         | 13.54              | 13.28                   | -0.26    |
| MEDIAN              |         | 2.39               | 2.68                    | +0.29    |
| WORST               |         | 45.83              | 43.79                   | -2.04    |

*All gaps in percentage points. Mean and worst-case both moved slightly in the derived model's favor; median moved slightly against it. The differences are within a point either way — a wash, not a regression.*

### Entry 027 — Conclusion

Swapping the fitted `variable_forgiveness_ratio()` power law for the derived closed-form floor changes accuracy on the Entry 022 Kyiv reference set by less than a point in either direction on every circuit tested (mean gap  $13.54 \rightarrow 13.28$ , worst-case  $45.83 \rightarrow 43.79$ ). That confirms Entry 026's diagnosis directly: the fitted ratio was buying essentially nothing beyond what the derived formula already provides for free, while carrying two chip-specific fitted coefficients, an explicit fit-range caveat, and a dependency on future recalibration for every new chip. The derived version needs none of that — it falls out of the physics of a depolarizing channel and the size of the circuit's ideal outcome set.

### 3. An Honest Complication, Not Fixed by This Entry



bell\_mid\_77\_100 (19-point gap) and bell\_far\_0\_126 (44-point gap) remain badly wrong under both the old and new models, essentially unchanged by this substitution. Both are the two longest SWAP chains in the reference set (bell\_far alone crosses 25 real swap legs). This is a pre-existing, unsolved problem — not something Entry 027 caused or was expected to fix, since it only touched how gate error is aggregated, not the routing or decoherence-integration logic Entries 023/024 built. It is flagged here rather than left implicit: whatever is wrong with the emulator's long-route predictions is still wrong, and is the most promising lead for a future entry.

4. Next Step — Entry 028

Investigate the long-route gap directly: bell\_mid\_77\_100 and bell\_far\_0\_126 are the two worst-predicted circuits in the entire reference set, by a wide margin, and share the trait of crossing many SWAP legs. Candidates to isolate: whether decoherence integrated over 20+ sequential legs is itself miscalibrated at that length, whether Aer's own routing diverges from the exact-dwell router's choice on long paths under a different seed, or whether an effect this project hasn't modeled yet (crosstalk between simultaneously-idle spectator qubits, for instance) only becomes visible once a route is long enough.

Research Log — Index Addendum (Entry 027)

| Entry     | Title                                                           | Date         | Status   |
|-----------|-----------------------------------------------------------------|--------------|----------|
| Entry 027 | Retiring variable_forgiveness_ratio() for the Closed-Form Floor | Aug 13, 2026 | Complete |

*variable\_forgiveness\_ratio() and its two fitted coefficients (forgiveness\_law.json) are superseded in the v4/v4.1 codepath by emulator\_v4.ideal\_set\_floor(), a derived closed form requiring no calibration data to fit. Retained in emulator\_v3\_routing.py unmodified, used only as the historical v3 baseline for comparison.*

ENTRY 028 Two Decoherence Bugs, One Wrong Channel — August 13, 2026 — Status: COMPLETE

# Entry 028 — Dephasing Was Never the Channel That Mattered

*Isolating Gate Error, Decoherence, and Readout Separately Against Aer Exposes Two Bugs in the Decoherence Term Built in Entry 023*

## 1. Why This Investigation Started

Entry 027 left two circuits badly wrong: `bell_mid_77_100` (19-point gap) and `bell_far_0_126` (44-point gap), both unchanged by that entry's gate-error simplification. Rather than guess, each contributing term -- gate error, decoherence, readout -- was isolated separately in Aer against `bell_mid_77_100`'s real 5-leg route and compared to what the emulator predicts for that term alone.

| term                      | emulator predicts | Aer isolated | gap      |
|---------------------------|-------------------|--------------|----------|
| gate error (depolarizing) | 90.16%            | 91.04%       | 0.88 pts |
| readout error             | 92.50%            | 92.66%       | 0.16 pts |
| decoherence               | 77.6%             | 94.80%       | 17.2 pts |

*Gate error and readout both check out almost exactly. Decoherence is off by 17 points on its own, on a single circuit -- clearly the source of the two entries' unexplained gap.*

## 2. Bug One: Charging Both Ends of Every SWAP

`exact_dwell_cost()` (Entry 024) charged decoherence to BOTH physical qubits of every SWAP leg. But only one of the two is actually carrying a logical qubit's payload at any given hop -- the other is an idle ancilla being swapped into position. `v4`'s own BFS-guess model (`route_cost()`, Entry 023) already charges only the single carrier qubit per hop, matching its own stated design intent ("integrate decoherence along the CARRIER path only") -- the exact-route function introduced in Entry 024 simply didn't carry that rule over. Fixed by tracking, at each SWAP, which of its two physical qubits currently holds a pending logical qubit and charging only that one.

## 3. Bug Two: The Wrong Coherence Channel Entirely

Fixing bug one alone wasn't enough. The deeper problem is Entry 023's original physical reasoning: it charged decoherence using  $T_2$  (dephasing time), on the argument that Bell/GHZ states carry their information in phase. That reasoning has it backwards. Pure  $T_2$  dephasing destroys OFF-DIAGONAL coherence in the computational basis -- but every circuit in this project measures success as population landing in  $\{00\dots0, 11\dots1\}$ , which are diagonal elements. Verified directly: an isolated two-qubit Aer test with  $T_1$  set near-infinite and  $T_2 = 20\ \mu\text{s}$  (aggressive dephasing, no relaxation at all) gave **100.00%** Bell-pair success. Pure dephasing was completely invisible to this measurement. The same test with  $T_1 = 20\ \mu\text{s}$  instead ( $T_2 = 2\times T_1$ , i.e. no dephasing beyond the  $T_1$  limit) gave 97.6% -- consistent with the real order-of-magnitude loss. It is  $T_1$  relaxation, population leaking toward  $|0\rangle$ , that actually corrupts these outcomes by leaking probability into  $\{01,10\}$  -- not  $T_2$ .

`dephasing()` now uses  $T_1$  instead of  $T_2$ , wherever it's called from (both `route_cost()` and `exact_dwell_cost()`). The function name is left unchanged since every call site still refers to it, but the docstring now records this correction and the isolated test that motivated it.

## 4. Re-Validation Against the Entry 022 Kyiv Reference

| circuit                          | Aer ref | gap before (Entry 027) | gap after (Entry 028) |
|----------------------------------|---------|------------------------|-----------------------|
| <code>bell_adjacent_77_78</code> | 96.07%  | 0.09                   | 0.15                  |
| <code>bell_near_77_82</code>     | 48.74%  | 2.68                   | 10.81                 |
| <code>bell_mid_77_100</code>     | 83.69%  | 19.11                  | 3.39                  |
| <code>bell_far_0_126</code>      | 69.90%  | 43.79                  | 11.21                 |
| <code>ghz_local_77_78_79</code>  | 88.53%  | 0.76                   | 0.10                  |

| circuit | Aer ref | gap before (Entry 027) | gap after (Entry 028) |
|---------|---------|------------------------|-----------------------|
| MEAN    |         | 13.28                  | 5.13                  |
| MEDIAN  |         | 2.68                   | 3.39                  |
| WORST   |         | 43.79                  | 11.21                 |

Entry 028 — Conclusion

Fixing both bugs together drops the mean prediction gap across the Kyiv reference set from 13.28 to 5.13 points and the worst case from 43.79 to 11.21 -- the largest single accuracy gain of the project since Entry 024's routing fix. `bell_near_77_82` is the one circuit that got nominally worse (2.68 → 10.81 points), and that's worth being honest about rather than hiding: its old "good" number came from two bugs partially cancelling -- overcharging decoherence via the wrong channel on a route that happens to dwell on catastrophically low-T2 qubit 80 -- not from the model being more correct. A model that is right for the wrong reasons on one circuit and wrong elsewhere is worse than one that is consistently closer everywhere.

5. What's Still Unresolved

`bell_near_77_82` and `bell_far_0_126` both still carry double-digit gaps (10.81 and 11.21 points). Both routes cross qubit 80 or long chains with several low-to-mid T2 qubits, so a residual decoherence-modeling gap likely remains -- possibly in how T1 combines with the readout and gate-error terms at the point of measurement, or in whether a single T1-based formula is adequate for qubits with unusually short T1 relative to the gate durations they experience. Not chased further in this entry; flagged for Entry 029.

6. Next Step — Entry 029

Investigate the remaining `bell_near_77_82` / `bell_far_0_126` gap specifically on qubit 80's role ( $T_2 = 8.51\text{ }\mu\text{s}$ , the same qubit Entry 023 originally flagged) -- isolate its T1 in Aer the same way this entry isolated dephasing generally, and check whether the single T1-based exponential formula still holds at its extreme. If it does not extend the current model's accuracy further, task #21 (extending the D-screen protocol to Sherbrooke and Cairo) is the next candidate.

Research Log — Index Addendum (Entry 028)

| Entry     | Title                                                      | Date         | Status   |
|-----------|------------------------------------------------------------|--------------|----------|
| Entry 028 | Fixing Two Decoherence Bugs (Double-Charge, Wrong Channel) | Aug 13, 2026 | Complete |

*`emulator_v4.dephasing()` now uses  $T_1$  (relaxation) instead of  $T_2$  (dephasing) -- pure  $T_2$  dephasing is invisible to a computational-basis  $\{00,11\}$ / $\{000,111\}$  measurement. `exact_dwell_cost()` now charges decoherence to a route's single carrier qubit per SWAP leg, matching v4's own BFS-model design intent.*

ENTRY 029 — A Provably Correct Fix That Made the Model Worse — August 13, 2026 — Status: COMPLETE (fix not shipped)

## Entry 029 — When Being More Right Makes the Model More Wrong

*Chasing the Remaining bell\_near / bell\_far Gap Finds a Real Formula Bug, and a Reason Not to Fix It Yet*

### 1. The Remaining Gap After Entry 028

bell\_near\_77\_82 (10.81-point gap) and bell\_far\_0\_126 (11.21-point gap) were still the two worst circuits after Entry 028's fixes. Both routes dwell on qubit 80,  $T_1 = 5.0 \mu\text{s}$  -- the same extreme-decoherence qubit Entry 023 originally flagged. Isolating gate error, decoherence, and readout separately for the single worst leg (the (80,79) SWAP in bell\_near's route) against Aer found gate error and readout both matched closely, but the emulator's decoherence-only prediction for that one leg (70.58%) was far below Aer's isolated result (93.15%).

### 2. A Real, Derivable Bug -- Verified

Every decoherence call site since Entry 023 has computed a hazard probability  $\gamma = 1 - e^{-t/T_1}$  and used  $(1 - \gamma)$  directly as the success multiplier -- treating amplitude damping as an all-or-nothing failure. Working through the Kraus operators for one qubit of a Bell pair under amplitude damping shows this is wrong: the decay-triggered branch (probability  $\gamma/2$ ) always leaks outside {00,11}, but the no-decay branch (probability  $1 - \gamma/2$ ) ALWAYS still lands inside it. The correct success factor is  $1 - \gamma/2$ , not  $1 - \gamma$ . Verified directly: an isolated single-qubit amplitude-damping Aer test at four  $T_1$  values (5, 10, 20, 50  $\mu\text{s}$ ) matched  $1 - \gamma/2$  to within shot noise at every single point.

### 3. Applying It Made Things Worse

The fix was implemented and re-validated against the full Entry 022 Kyiv reference set. bell\_mid\_77\_100 and bell\_far\_0\_126 both improved. bell\_near\_77\_82 got dramatically worse -- its gap nearly tripled, from 10.81 to 24.08 points -- because relaxing the decoherence penalty pushed an already-too-optimistic prediction even further from the real answer. On the full 5-circuit set, mean gap moved from 5.13 to 6.61 points and worst-case from 11.21 to 24.08. A four-way sweep (single-carrier vs. both SWAP endpoints charged, times  $\gamma$  vs.  $\gamma/2$ ) confirmed Entry 028's exact combination -- single carrier, full  $\gamma$  -- is the best-performing option of the four on this reference set, despite  $\gamma/2$  being the more physically correct formula in isolation.

| combination                                        | mean gap | worst gap |
|----------------------------------------------------|----------|-----------|
| single carrier, full $\gamma$ (Entry 028, shipped) | 5.14     | 11.08     |
| single carrier, $\gamma/2$ (this entry's attempt)  | 6.66     | 24.25     |
| both endpoints, full $\gamma$                      | 6.45     | 19.33     |
| both endpoints, $\gamma/2$                         | 5.62     | 13.23     |

### Entry 029 — Conclusion

The correct single-qubit formula does not survive contact with the full model. A naive check confirms why: multiplying Aer's own individually-isolated gate-error, decoherence, and readout success numbers together for bell\_near's route gives roughly 72% -- but the real combined-noise result is 49%. Even a perfect decomposition of each channel in isolation, recombined by simple multiplication, misses a 23-point interaction effect. Entry 028's technically-wrong  $\gamma$  formula was, on this route, accidentally absorbing some of that missing interaction term. Fixing the formula in isolation removed a real bug and simultaneously removed the accidental compensation for a different, larger, unmodeled one -- leaving the combined prediction worse. The  $\gamma/2$  fix is NOT shipped. The codebase is unchanged from Entry 028.

### 4. What This Actually Means

The project's multiplicative model -- gate error, decoherence, and readout as independent probabilities multiplied together -- has a structural limit on routes that dwell on extreme- $T_1$  qubits. The channels are not independent there: whatever damage decoherence does to the state appears to make it MORE vulnerable to the other error sources than treating the channels separately predicts. Closing this gap for real would require modeling that interaction directly, not

patching the decoherence formula alone -- a materially bigger task than this entry's scope.

5. Next Step

Given the interaction effect is specific to extreme-T1 routes and the current model (Entry 028's state) already gives a 5.13-point mean gap across the reference set -- a large improvement over where the project stood two entries ago -- task #21 (extending the D-screen protocol to Sherbrooke and Cairo) is the better use of the next entry. The channel-independence question is left open, documented here rather than chased further, and can be revisited if a future chip's reference set turns up the same pattern.

Research Log — Index Addendum (Entry 029)

| Entry     | Title                                                   | Date         | Status                 |
|-----------|---------------------------------------------------------|--------------|------------------------|
| Entry 029 | A Correct Formula Fix That Regressed the Combined Model | Aug 13, 2026 | Complete (not shipped) |

*t1\_survival()* ( $1 - \gamma/2$ ) is derived and verified correct in isolation but NOT applied in `emulator_v4.py` or `exact_dwell_routing.py` -- it regresses the Entry 022 Kyiv reference set. Codebase left at Entry 028's state. Documented as an open channel-independence question for future chips.

ENTRY 030 Extending Validation to Sherbrooke — August 13, 2026 — Status: COMPLETE

## Entry 030 — A Second Chip Confirms One Fix and Surfaces a New Gap

*Task #21: Extending the Kyiv-Validated Model to Sherbrooke, and Why Cairo Is Excluded*

### 1. Scope Decision: Sherbrooke Yes, Cairo Not Yet

Sherbrooke (127 qubits) shares Kyiv's uniform ecr-only basis, so it can get the same Aer/MPS ground-truth validation used since Entry 022. Cairo (27 qubits) genuinely mixes cx and ecr per edge -- forcing an ecr-only basis for a Cairo circuit throws a disconnected-coupling-map error, confirming this is a real hardware property, not an artifact of an old validation harness. Entry 022 already flagged Cairo as unsuitable for circuit-level validation for this reason. Cairo is excluded from this entry's full-circuit test; extending to it would need a validation approach that respects per-edge native gates rather than a single forced basis, left for a future entry.

### 2. Sherbrooke Reference Set and Results

Five circuits built the same way as the Entry 022 Kyiv set, using real Sherbrooke qubit pairs: an adjacent pair, a route deliberately chosen through qubit 57 ( $T_2 = 2.6 \mu\text{s}$ , the chip's worst by far, but  $T_1 = 301.9 \mu\text{s}$  -- healthy), a 12-hop and a 26-hop pair, and a local GHZ. Ground truth from Aer/MPS against Sherbrooke's real noise model, 4 seeds  $\times$  4096 shots.

| circuit                  | Aer ref | v4.1 predicted | gap          |
|--------------------------|---------|----------------|--------------|
| bell_adjacent_0_1        | 95.53%  | 95.53%         | 0.00         |
| bell_near_57_61          | 95.08%  | 79.10%         | 15.98        |
| bell_mid_0_58 (12 hops)  | 49.40%  | 66.97%         | 17.57        |
| bell_far_0_126 (26 hops) | 49.51%  | 64.41%         | 14.90        |
| ghz_local_0_1_14         | 91.48%  | 90.43%         | 1.05         |
| MEAN / WORST             |         |                | 9.90 / 17.57 |

Mean gap 9.90 points, worst 17.57 -- both real improvements over v3 (13.23 mean) and v4-BFS (13.84 mean) on this same set, but noticeably worse than the 5.13-point mean the same v4.1 model achieves on Kyiv. The model generalizes -- imperfectly.

### 3. Confirmation: T1, Not T2, Is What Matters

bell\_near\_57\_61's route was chosen deliberately to cross qubit 57, the chip's single worst  $T_2$  ( $2.6 \mu\text{s}$ ) by a wide margin -- yet its real measured success (95.08%) was barely below the clean adjacent circuit's (95.53%). Under the OLD (pre-Entry-028)  $T_2$ -based decoherence formula this route would have predicted catastrophic loss. Under the current  $T_1$ -based formula it doesn't, because qubit 57's  $T_1$  ( $301.9 \mu\text{s}$ ) is healthy -- and the real hardware agrees almost exactly. This is independent confirmation, on a chip never used to derive or tune the fix, that Entry 028's  $T_2$ -to- $T_1$  correction was the right one.

### Entry 030 — Conclusion

The v4.1 model built and tuned entirely on Kyiv transfers to Sherbrooke without modification and beats the older v3/v4-BFS models there too -- and the  $T_1$ -vs- $T_2$  physics from Entry 028 holds up on a qubit the fix was never calibrated against. But the mean gap nearly doubles ( $5.13 \rightarrow 9.90$ ), so "generalizes" should not be read as "transfers cleanly." The two long routes (mid, far) are both over-predicted by 15-18 points, and the near-57 route is under-predicted by 16 -- different circuits failing in different directions, which rules out one single missing constant fixing all of them.

### 4. A New Lead: An Outlier Edge That Doesn't Behave Like a Depolarizing Channel

Isolating gate error alone for bell\_near\_57\_61's route in Aer found the edge (57,58) -- reported gate\_error 11.68%, roughly  $15\times$  the chip's median (0.78%) and the largest of only 4 edges on the whole chip above 5% -- produces far

less real damage (98.71% isolated survival) than modeling it as a depolarizing channel of that magnitude predicts (83.5%, via the same closed form that matched Kyiv's edges to within about a point). Either this specific calibration entry doesn't correspond to a simple depolarizing process, or extreme outlier edges behave differently from the typical edges the closed-form baseline was derived and verified against. Not resolved in this entry -- flagged as the likely largest single contributor to Sherbrooke's residual gap.

5. Next Step — Entry 031

Isolate edge (57,58) the way Entry 025 isolated a single Kyiv edge: sweep gate count on that one edge alone under a pure depolarizing channel at p=11.68% and check whether the closed-form baseline still matches at this much larger error magnitude, or whether it was only verified in a regime (Kyiv's edges topped out near 4.3%) that doesn't extend to genuine outliers. If the baseline breaks down at high p specifically, that would explain both the near-57 gap here and motivate re-checking whether any Kyiv edges near the top of its range carry the same undetected issue.

Research Log — Index Addendum (Entry 030)

| Entry     | Title                                                                            | Date         | Status   |
|-----------|----------------------------------------------------------------------------------|--------------|----------|
| Entry 030 | Extending Validation to Sherbrooke; T1 Fix Confirmed, New Outlier-Edge Gap Found | Aug 13, 2026 | Complete |

quantumbridge\_data/coherence\_sherbrooke.json and coherence\_cairo.json exported. Cairo full-circuit validation deferred (mixed cx/ecr basis, Entry 022). Sherbrooke reference set committed as validate\_v4\_sherbrooke.py, results saved to quantumbridge\_data/entry030\_sherbrooke\_validation.json.

ENTRY 031 — The Model Doesn't Know When Entanglement Actually Happens — August 13, 2026 — Status: COMPLETE

## Entry 031 — Route-Then-Entangle: A Structural Assumption Error

*The Outlier Edge Was a Red Herring; the Real Bug Is About WHEN a SWAP Chain Actually Starts Carrying Entangled Information*

### 1. Ruled Out: The Closed Form Doesn't Break at High p

Entry 030 flagged Sherbrooke edge (57,58) -- gate\_error 11.68%, ~15x the chip median -- as a possible case where the closed-form baseline (verified on Kyiv up to ~4.3% error) simply doesn't hold at larger magnitudes. Isolating that exact edge alone, sweeping gate count the same way Entry 025 did, found the closed form matches Aer to within 0.13 points on average even at p=11.68% -- essentially exact. The formula itself is not the problem, at any error magnitude tested so far.

### 2. The Real Bug

Re-isolating the FULL bell\_near\_57\_61 route (4 real edges, matching the exact transpiled circuit used for the Aer reference) reproduced Entry 030's gap directly: 98.71% real gate-error-only survival against 83.5% predicted. Since the per-edge formula is now proven correct, the bug had to be in how results from different edges combine. A minimal toy circuit isolated it: build a Bell pair by applying H to one qubit, then routing BOTH halves toward each other via SWAPs, and creating the actual entanglement with a SINGLE cx/ecr gate only once they're adjacent -- exactly what the real transpiler does for these circuits, confirmed by inspecting the transpiled gate sequence directly (only one non-SWAP two-qubit gate appears, at the very end).

Every model since Entry 023 assumed the opposite: that entanglement exists from the start and every SWAP afterward is moving an already-entangled payload, equally vulnerable to depolarizing damage at every hop. That assumption is wrong for exactly the circuits this project measures. Before the final entangling gate, the two qubits being routed toward each other are carrying ordinary, UNENTANGLED product-state information -- and a depolarizing event on a product state does not destroy a correlation that doesn't exist yet. It does less damage to the final {00,11} outcome than the same event would do to an already-entangled pair.

| test                                             | Aer    | closed-form (old model) | diff   |
|--------------------------------------------------|--------|-------------------------|--------|
| toy: entangle-first-then-route (wrong structure) | 50.32% | 93.62%                  | -43.30 |
| toy: route-first-then-entangle (real structure)  | 99.29% | 93.62%                  | +5.67  |
| real bell_near_57_61, gate-error-only            | 98.71% | 83.52%                  | +15.19 |

*The first row is a deliberately broken toy (re-entangling an already-entangled pair) included only to show how sensitive this structure is -- getting the entanglement-timing wrong doesn't cause a small error, it inverts the whole prediction. The second row, with the structure corrected to match what Qiskit's transpiler actually produces, shows the SAME direction and rough scale of under-prediction as the real circuit in row three. That consistency is what makes this the real explanation rather than a coincidence.*

### Entry 031 — Conclusion

The persistent over-prediction of damage on long, multi-hop routes (bell\_mid and bell\_far on both Kyiv and Sherbrooke, and now bell\_near\_57\_61) traces to one structural assumption: exact\_dwell\_cost and route\_cost both treat every SWAP identically, as if it always carries an already-entangled qubit. Real transpiled circuits for a single Bell/GHZ preparation typically route the uninvolved qubits into position FIRST and entangle them LAST, meaning most of a long route's SWAP legs are moving product-state information that is measurably less vulnerable to depolarizing noise than the model assumes. This is not a tuning problem or an edge-magnitude problem -- it is a category error in what physical process the model thinks is happening during most of a circuit's real gate sequence.

### 3. Why This Wasn't Caught Earlier

Entry 025's isolation test, which the closed form was derived and verified against, deliberately used "cancelling CX pairs" applied repeatedly to an circuit that is ALREADY entangled from the first gate -- a clean, controlled test of the



compounding formula, but one that never exercises the pre-entanglement regime a real multi-hop route spends most of its gates in. The formula was proven correct in exactly the case it was tested in, and that case turns out not to be representative of the majority of gates in a long real route.

4. Next Step — Entry 032

This entry identifies the mechanism but does not yet fix it -- deriving the correct closed form for a depolarizing event on a pre-entanglement product-state SWAP (as opposed to a post-entanglement one) needs its own Kraus-operator derivation, analogous to Entry 025's and Entry 029's, before it can be verified and shipped. The practical implication for `exact_dwll_cost`: gate error needs to be split into two categories per route -- SWAPs before the pending pair's entangling gate, and SWAPs/gates after it -- rather than one uniform survival product across the whole route.

Research Log — Index Addendum (Entry 031)

| Entry     | Title                                                                     | Date         | Status                                   |
|-----------|---------------------------------------------------------------------------|--------------|------------------------------------------|
| Entry 031 | Route-Then-Entangle: A Structural Assumption Error in Gate-Error Modeling | Aug 13, 2026 | Complete (root cause found, fix pending) |

*entry031\_high\_p\_isolation.py confirms the closed-form baseline holds at p=11.68% (rules out edge-magnitude as the cause). entry031\_route\_then\_entangle.py isolates and confirms the real cause. Neither emulator\_v4.py nor exact\_dwll\_routing.py is fixed yet -- left for Entry 032.*

ENTRY 032 Deriving and Shipping the Route-Then-Entangle Fix — August 13, 2026 — Status: COMPLETE

# Entry 032 — The Control Qubit's Routing Errors Don't Matter

*A Kraus-Operator Derivation, Verified by an Error Sweep That Shouldn't Move the Answer -- and Doesn't*

## 1. The Derivation

Entry 031 found that most SWAP legs on a real route move UNENTANGLED product-state qubits, not an already-entangled payload. This entry works out exactly what that means. Call the qubit that receives the initial H the CONTROL (it carries |+> before any routing) and the qubit that starts at |0> the TARGET. Tracking the four combinations of "control corrupted / clean" × "target corrupted / clean" through the final entangling gate via its Kraus operators: if the control gets forced to a maximally mixed state before the final gate, it is now a CLASSICAL random bit (diagonal in the computational basis) -- combining a classical random control with a clean target through a CX/ECR still produces a definite, correlated pair (00 or 11), so it costs nothing. If the TARGET gets corrupted instead, the final gate combines a genuine coherent superposition with a classically randomized partner, producing a 50/50 mixture of a correct and an incorrect Bell state -- landing exactly at the floor. The two roles are not interchangeable.

Closed form for a single-pair circuit:  $\text{success} = \text{target\_survive} \times (1 - \text{final\_raw}/2) + (1 - \text{target\_survive}) \times 0.5$ , where  $\text{target\_survive}$  is the product of survival across only the target's routing legs and  $\text{final\_raw}$  is the entangling gate's own error. In the no-routing limit ( $\text{target\_survive} = 1$ ) this reduces exactly to Entry 025's single-gate formula,  $0.5 + 0.5(1-p)$ .

## 2. Verified: Zero Sensitivity to the Control's Error

Sweeping the control-side edge's depolarizing error from 1% to 60% in an Aer simulation of the exact `bell_near_57_61` route structure left the measured success probability completely unchanged at every point (99.286%, to three decimal places) -- while this entry's closed form (evaluated without depending on that edge at all) matched it to within 0.11 points throughout. This is about as clean a confirmation as this kind of derivation gets.

## 3. Scope and Implementation

Implemented in `exact_dwll_cost()` for single-pair (plain Bell) circuits only -- 4 of the 5 circuits in every reference set used this session. Multi-pair (GHZ-style) circuits keep the old floor-based model unchanged, since later entangling gates there act on qubits already entangled from earlier gates, and the derivation above assumed a fresh product-state input -- extending it correctly would need its own derivation, not attempted here.

## 4. Results

| circuit (chip)                              | Aer ref | gap before   | gap after    |
|---------------------------------------------|---------|--------------|--------------|
| <code>bell_adjacent_77_78</code> (Kyiv)     | 96.07%  | 0.09         | 0.15         |
| <code>bell_near_77_82</code> (Kyiv)         | 48.74%  | 10.81        | 11.65        |
| <code>bell_mid_77_100</code> (Kyiv)         | 83.69%  | 3.39         | 2.31         |
| <code>bell_far_0_126</code> (Kyiv)          | 69.90%  | 11.21        | 6.07         |
| <code>ghz_local_77_78_79</code> (Kyiv)      | 88.53%  | 0.10         | 0.10         |
| KYIV MEAN / WORST                           |         | 5.13 / 11.21 | 4.06 / 11.65 |
| <code>bell_adjacent_0_1</code> (Sherbrooke) | 95.53%  | 0.00         | 0.00         |
| <code>bell_near_57_61</code> (Sherbrooke)   | 95.08%  | 15.98        | 1.65         |
| <code>bell_mid_0_58</code> (Sherbrooke)     | 49.40%  | 17.57        | 20.19        |
| <code>bell_far_0_126</code> (Sherbrooke)    | 49.51%  | 14.90        | 20.39        |
| <code>ghz_local_0_1_14</code> (Sherbrooke)  | 91.48%  | 1.05         | 1.05         |
| SHERBROOKE MEAN / WORST                     |         | 9.90 / 17.57 | 8.66 / 20.39 |

Entry 032 — Conclusion

Mean gap improved on both chips (Kyiv 5.13 → 4.06, Sherbrooke 9.90 → 8.66), and bell\_near\_57\_61 -- the specific circuit this whole investigation started from -- dropped from a 15.98-point gap to 1.65, close to shot noise. That is a genuine, derived, twice-verified fix, and it ships as of this entry. It is not a clean sweep, though: bell\_mid and bell\_far got WORSE on Sherbrooke (17.57 → 20.19 and 14.90 → 20.39) and bell\_near got marginally worse on Kyiv (10.81 → 11.65). Removing gate-error charges from the control's path made the model more optimistic everywhere that charge used to apply -- correctly so, per the derivation -- but on the very longest routes, the real measured success is low enough that removing ANY pessimistic term moves the prediction further from reality. That is not evidence the fix is wrong; the control-sweep test proves it isn't. It is evidence that bell\_mid and bell\_far have a separate, still-unidentified source of loss big enough that it was previously masked by an unrelated bug's over-charging.

5. Next Step — Entry 033

Isolate bell\_far\_0\_126 and bell\_mid\_0\_58/bell\_mid\_77\_100 the same way this entry isolated bell\_near\_57\_61: ablate gate error, decoherence, and readout separately against Aer for these SPECIFIC long routes now that the route-then-entangle and control/target asymmetry are both accounted for, to find what's actually left unmodeled. Candidates worth checking first: whether the TARGET's own long chain has a similar asymmetry still unaccounted for once it spans many hops, or whether something entirely new (crosstalk between simultaneously-active spectator qubits on a 26-hop route, for instance) only becomes visible at this route length.

Research Log — Index Addendum (Entry 032)

| Entry     | Title                                                                   | Date         | Status   |
|-----------|-------------------------------------------------------------------------|--------------|----------|
| Entry 032 | Route-Then-Entangle Fix Shipped: Control-Side Gate Error Doesn't Matter | Aug 13, 2026 | Complete |

*exact\_dwelling\_routing.py's exact\_dwelling\_cost() now splits single-pair gate-error handling into a target-survival closed form, verified via a control-edge sensitivity sweep. GHZ/multi-pair circuits unchanged. bell\_mid/bell\_far remain open (Entry 033).*

ENTRY 033 Confirming the Channel-Independence Ceiling — August 13, 2026 — Status: COMPLETE

## Entry 033 — Every Formula Is Right, and the Route Still Loses 20 Points

*bell\_far\_0\_126's Regression Traces to the Same Structural Limit Entry 029 Found, Now Confirmed on an Unrelated Route*

### 1. Isolating bell\_far\_0\_126 on Sherbrooke

Entry 032 improved the reference set overall but made bell\_far\_0\_126 and bell\_mid\_0\_58 worse (gaps up to 20.39 points). Rather than guess, each channel was isolated against Aer the same way Entries 028 and 029 did for bell\_mid\_77\_100 and bell\_near\_57\_61 -- this route crosses 25 real SWAP legs (76 total two-qubit gate applications) toward qubit 126, the longest in the whole project.

| channel                            | emulator predicts | Aer isolated | gap      |
|------------------------------------|-------------------|--------------|----------|
| gate error (Entry 032 formula)     | 84.51%            | 83.28%       | 1.23 pts |
| decoherence (Entry 028 T1 formula) | 86.44%            | 89.72%       | 3.28 pts |
| readout error                      | 95.91%            | 95.70%       | 0.21 pts |

*Every single formula checks out well individually -- the largest gap is 3.3 points, on decoherence. None of Entries 028's, 029's, or 032's fixes are wrong.*

### 2. Where the Gap Actually Comes From

Multiplying the three Aer-ISOLATED results together (as if the channels were independent) gives  $0.8328 \times 0.8972 \times 0.9570 = 71.5\%$ . The REAL combined-noise Aer result for this circuit is 49.51%. A 22-point gap appears purely from treating three genuinely well-measured, individually-correct channels as if they don't interact -- with nothing left to blame on any of this project's formulas, since each one was just independently verified against its own isolated ground truth.

### Entry 033 — Conclusion

This is not a new problem. It is Entry 029's finding -- "even a perfect decomposition of each channel in isolation, recombined by simple multiplication, misses a real interaction effect" -- showing up again on a completely different route, on a different chip, driven by a different cause (25 accumulated SWAP legs here, versus one catastrophically low-T1 qubit on Kyiv's bell\_near\_77\_82). Two independent circuits now show the same signature: individually correct channels, multiplicatively combined, systematically overshoot real fidelity once a route accumulates enough total exposure -- whether that exposure comes from one extreme qubit or from many ordinary ones. This confirms it as a general property of the model's multiplicative-composition assumption, not a coincidence specific to qubit 80 on Kyiv.

### 3. Why This Isn't Chased Further Right Now

Fixing this for real requires modeling how gate error and decoherence correlate along a route -- effectively simulating correlated noise rather than composing independently-measured channel probabilities, which is close to what Aer itself already does at the circuit level. That is a materially different, larger undertaking than any fix shipped this session, and outside a closed form the rest of this project's formulas (Entries 025, 028, 032) were built around. The current model's real, verified achievement stands: mean gap on Kyiv's reference set fell from 13.28 (Entry 026) to 4.06 points (Entry 032), and on Sherbrooke from 9.90 to 8.66, with every individual physical channel -- gate error, decoherence, readout -- now independently verified correct. The two longest routes on each chip are a known, understood, documented ceiling on that approach, not an unexplained failure.

### 4. Where This Leaves the Project

The physics-debugging arc that ran from Entry 022 through this entry has taken v4.1 from a validation harness that silenced 53% of its own noise model to a closed-form emulator with no fitted, chip-specific coefficients, verified on two chips it wasn't tuned on. The remaining gap is well-characterized rather than mysterious. That is a reasonable, honest stopping point for this arc -- the next entry moves to building the batch-circuit dataset for the ML phase, using this

validated model as the benchmark any learned model will need to beat.

Research Log — Index Addendum (Entry 033)

| Entry     | Title                                                         | Date         | Status   |
|-----------|---------------------------------------------------------------|--------------|----------|
| Entry 033 | Confirming the Channel-Independence Ceiling on a Second Route | Aug 13, 2026 | Complete |

No code changes. Confirms Entry 029's channel-independence finding generalizes beyond Kyiv's qubit 80 to any route with enough cumulative exposure. Closes the Entries 022-033 physics-debugging arc; next entry begins the ML batch-dataset phase.

ENTRY 034 The Batch-Circuit Dataset — August 16, 2026 — Status: COMPLETE

## Entry 034 — From 10 Hand-Picked Circuits to 116 Sampled Ones

*Batch Methodology Surfaces a New Failure Pattern the Reference Sets Never Showed*

### 1. Why Batch Instead of One-Off

Every finding from Entry 022 through Entry 033 rested on 5 hand-picked circuits per chip (adjacent / near / mid / far / one local GHZ) -- the right tool for chasing an individual bug, but a 10-point sample for any claim about how the model behaves in general. `entry034_batch_dataset.py` instead samples qubit pairs stratified across six BFS hop-distance bins (1, 2-3, 4-7, 8-15, 16-25, 26+), builds a plain Bell circuit for each, runs it through the same Aer/MPS ground-truth pipeline used since Entry 022 (4 seeds × 4096 shots, `optimization_level=3`), and records the v4.1 (Entry 032) prediction alongside it -- plus a smaller batch of 3-qubit GHZ circuits for structural variety. Every result is checkpointed to JSON as it completes, so the batch can run in resumable chunks.

| chip       | bell circuits | ghz circuits | total | mean gap | median gap | worst gap |
|------------|---------------|--------------|-------|----------|------------|-----------|
| Kyiv       | 45            | 10           | 55    | 5.38 pts | 2.18 pts   | 31.39 pts |
| Sherbrooke | 51            | 10           | 61    | 7.48 pts | 2.14 pts   | 40.89 pts |

*116 circuits total, roughly 12x the combined size of every reference set built so far this project.*

### 2. The Expected Pattern: Gap Grows With Route Length

Bucketing by hop distance confirms what the hand-picked circuits suggested but never proved at scale: adjacent pairs (1 hop) predict almost perfectly (median gap 0.1-0.2 points on both chips), and the median gap climbs steadily to 4-14 points by the 16-25 hop bin. Correlation between gap and hop distance across all 96 Bell circuits is +0.20 -- real but modest, because hop distance alone does not explain the worst outliers (below).

### 3. The New Finding: Short Routes That Collapse to the Floor

Sorting by gap regardless of distance surfaces something the 5-circuit reference sets never could: 7 circuits, 5 of them on Sherbrooke, whose REAL Aer result sits within 2 points of the 50% floor -- the outcome of complete entanglement failure -- despite short 3-9 hop routes the model predicts should succeed 60-91% of the time.

| chip       | pair    | hops | Aer (real) | v4.1 predicts | gap      |
|------------|---------|------|------------|---------------|----------|
| Sherbrooke | 3 → 6   | 3    | 50.0%      | 88.5%         | 38.5 pts |
| Sherbrooke | 2 → 6   | 4    | 50.2%      | 91.1%         | 40.9 pts |
| Sherbrooke | 11 → 22 | 9    | 48.9%      | 77.2%         | 28.3 pts |
| Sherbrooke | 28 → 5  | 7    | 50.3%      | 84.8%         | 34.5 pts |
| Kyiv       | 95 → 72 | 8    | 50.0%      | 60.9%         | 10.9 pts |

Physical qubit 6 (readout error 25.7%, roughly 4-8x every neighboring qubit's readout error on Sherbrooke) and qubit 5 ( $T_2 = 78.9\mu\text{s}$  against a  $172.7\mu\text{s}$   $T_1$  -- the largest  $T_1/T_2$  split on any route in this batch) both recur across these routes. Neither value alone is extreme enough in the current model to explain a full collapse to the floor -- the model's readout and decoherence terms would only cost these circuits 10-20 points, not 35-40.

### Entry 034 — Conclusion

The batch approach does exactly what it was proposed to do: instead of finding one bug at a time by hand-picking a circuit, running many circuits and aggregating surfaced a failure mode -- short routes that still crash to the 50% floor -- with 7 independent examples on the first pass, concentrated on two specific physical qubits. That is a concrete, high-confidence lead for the next entry, discovered by frequency rather than by guessing which circuit to isolate next.

4. What This Dataset Is Also For

Beyond bug-hunting, every one of the 116 records is saved with its chip, qubit pair, BFS hop distance, real transpiled edge/gate counts, worst raw edge error and worst T1 along the BFS path, the Aer ground truth, the v4.1 prediction, and the gap -- circuit-and-calibration features paired with a measured label. This is the first dataset in the project large enough to be a plausible starting point for the ML/GNN phase discussed since Entry 033, and it will keep growing the same way as more circuits get added in resumable batches.

5. Next Step

Entry 035: isolate the qubit-5/qubit-6 (Sherbrooke) floor-collapse outliers against Aer the same way Entries 028, 029, and 033 isolated earlier anomalies -- gate error, decoherence, and readout separately -- to find what specifically drives a short route all the way to 50%, since none of the three known channels individually accounts for a drop this size.

Research Log — Index Addendum (Entry 034)

| Entry     | Title                                                       | Date         | Status   |
|-----------|-------------------------------------------------------------|--------------|----------|
| Entry 034 | The Batch-Circuit Dataset (116 circuits, Kyiv + Sherbrooke) | Aug 16, 2026 | Complete |

Script: entry034\_batch\_dataset.py (resumable, checkpointed). Data: quantumbridge\_data/entry034\_batch\_dataset.json (116 records), quantumbridge\_data/entry034\_batch\_analysis.json (aggregated bins and outlier list). Surfaces a new floor-collapse pattern on 7 short routes for Entry 035 to isolate. Also the first dataset scaled for the ML/GNN phase.

ENTRY 035   Replication: Is It the Qubits, or the Chip? — August 16, 2026 — Status: COMPLETE

# Entry 035 — A Second, Independent Batch Rules Out "Bad Qubits"

Two Experiments, Zero Shared Outlier Qubits, Same Pattern — This Is a Chip-Level Effect, Not a Qubit-Level One

## 1. The Question This Entry Answers

Entry 034's batch found 7 short routes (3-9 hops) that collapse completely to the 50% floor, clustered on Sherbrooke qubits 5 and 6. Before spending an entry chasing what makes those two qubits special, the proposed test was: run a second, independently-sampled batch with a different random seed and see whether the SAME qubits show up again (a qubit-specific defect worth isolating on its own) or whether a similar pattern reappears on DIFFERENT qubits (a general, chip-level effect worth chasing as one shared problem). entry035\_replication\_batch.py re-runs Entry 034's exact methodology with seed=99 instead of seed=7, drawing a fresh, non-overlapping set of qubit pairs.

## 2. Result: The Pattern Recurs, the Qubits Don't

|                                   | circuits | floor-collapse outliers | outlier qubits                      |
|-----------------------------------|----------|-------------------------|-------------------------------------|
| Experiment 1 (Entry 034, seed=7)  | 116      | 7                       | 97,114,95,72,3,6,2,5,28,86,91,11,22 |
| Experiment 2 (Entry 035, seed=99) | 122      | 5                       | 83,103,82,84,73,80,40,56,15,9       |

Zero overlap. Not one physical qubit that produced a floor-collapse outlier in experiment 1 appears in experiment 2's outlier list, or vice versa -- despite both experiments sampling the same two chips with the same stratified hop-distance methodology. Overall gap statistics also matched closely between the two independent draws (mean 6.48 vs 6.10 points, gap-vs-distance correlation 0.20 vs 0.26), confirming this isn't two unrelated experiments -- it's the same underlying phenomenon sampled twice.

## 3. What Actually Distinguishes the Outliers: the Chip, Not the Qubit

Pooling both experiments and normalizing by how many short-route ( $\leq 10$  hop) circuits were sampled on each chip:

| chip       | short-route circuits | floor-collapse | rate  |
|------------|----------------------|----------------|-------|
| Kyiv       | 65                   | 2              | 3.1%  |
| Sherbrooke | 69                   | 10             | 14.5% |

Sherbrooke collapses to the floor on short routes roughly 4.7x more often than Kyiv, consistently across both independent samples. Both chips share the same uniform ecr-only basis and the same v4.1 model -- the difference has to come from something about Sherbrooke's actual topology or calibration surface that this project hasn't identified yet, not from a handful of bad qubits.

## Entry 035 — Conclusion

The replication test gives a clear, decisive answer: floor-collapse is a general, recurring phenomenon rather than a defect isolated to qubits 5 and 6. It shows up on a completely different set of physical qubits in an independent sample, at a comparable rate, heavily skewed toward Sherbrooke over Kyiv. That rules out a qubit-specific patch as the right fix -- the next entry should isolate a FRESH instance of the collapse (not qubits 5/6, since those are now known not to be special) using the same gate-error / decoherence / readout separation Entries 028, 029, and 033 used, with the explicit goal of finding what property of the ROUTE or the CHIP -- not the qubit -- predicts it.

## 4. Next Step

Entry 036: pick 2-3 fresh floor-collapse instances from the combined pool of both experiments' outliers, isolate gate error, decoherence, and readout separately against Aer for each (as in Entries 028/029/033), and check whether the collapse traces to one specific channel consistently across instances despite the different qubits involved -- that would identify what the qubits have in common functionally, even without sharing physical identity.



Research Log — Index Addendum (Entry 035)

| Entry     | Title                                                             | Date         | Status   |
|-----------|-------------------------------------------------------------------|--------------|----------|
| Entry 035 | Replication Batch — Floor-Collapse Is Chip-Level, Not Qubit-Level | Aug 16, 2026 | Complete |

Script: `entry035_replication_batch.py` (122 circuits, `seed=99`, reuses `entry034_batch_dataset.py`'s methodology). Data: `quantumbridge_data/entry035_replication_dataset.json`, `quantumbridge_data/entry035_comparison_summary.json`. Zero qubit overlap with Entry 034's outliers; Sherbrooke's short-route floor-collapse rate (14.5%) is 4.7x Kyiv's (3.1%). Rules out a qubit-specific fix; next entry isolates channels on fresh instances.

ENTRY 036 Floor-Collapse Is the Independence Breakdown, Amplified — August 16, 2026 — Status: COMPLETE

## Entry 036 — Every Channel Looks Fine in Isolation, Combined They Erase the Circuit

*Three Fresh Instances, None Involving Qubits 5/6, Same Signature as Entries 029 and 033 -- Just Far Stronger and on Much Shorter Routes*

### 1. Method

Three floor-collapse instances were picked from the combined Entry 034/035 outlier pool, deliberately avoiding qubits 5 and 6 (Entry 035 already showed they aren't special) and choosing cases where v4.1 predicted well above the floor -- genuine misses, not places the model already (correctly) predicted near-50%. Each was isolated against Aer into gate-error-only, decoherence-only, and readout-only noise models, the same ablation used in Entries 028, 029, and 033.

### 2. Result: Every Channel Checks Out, the Combination Doesn't

| route            | edges | gate err. | decoherence | readout | naive product | REAL combined | v4.1 predicts |
|------------------|-------|-----------|-------------|---------|---------------|---------------|---------------|
| Sherbrooke 3→6   | 3     | 97.7%     | 97.7%       | 91.9%   | 87.8%         | 50.0%         | 88.5%         |
| Sherbrooke 40→56 | 5     | 93.3%     | 84.6%       | 90.8%   | 71.6%         | 50.7%         | 81.9%         |
| Kyiv 95→72       | 8     | 90.4%     | 71.5%       | 99.4%   | 64.2%         | 50.0%         | 60.9%         |

In every instance, each individual channel isolated against Aer looks completely ordinary -- 72-98% survival, nothing alarming. Multiplying even Aer's own perfectly isolated per-channel results together lands at 64-88%, matching v4.1's prediction closely (as expected, since v4.1 is built on the same multiplicative-composition assumption). But the REAL combined-noise result on all three routes sits within 1 point of the 50% floor -- total entanglement failure, with nothing in any single channel foreshadowing it.

### 3. This Is Entries 029/033's Finding -- Just Much Stronger, on Much Shorter Routes

Entry 029 found an 11-point channel-independence gap on Kyiv's worst-T1 route. Entry 033 found a 22-point gap on Sherbrooke's longest route (25 SWAP legs). This entry's gaps are 37.8, 21.0, and 14.2 points -- comparable to or LARGER than Entry 033's, but on routes of just 3, 5, and 8 edges. Entry 033 concluded the breakdown was driven by "accumulated exposure," implicitly assuming more hops means more correlation. A 3-edge route producing the single largest independence gap measured anywhere in this project rules that assumption out: route length is not the variable that predicts how badly the channels interact. Something about which specific edges combine -- not how many -- drives the correlation strength.

### Entry 036 — Conclusion

Floor-collapse is not a new bug and not a qubit defect -- it is the channel-independence breakdown documented in Entries 029 and 033, occurring far more severely than previously seen, and on routes short enough that route length alone cannot be the predictor. This raises the stakes on the limitation Entry 033 flagged but set aside: it isn't just costing a few points on the two longest routes per chip, it is erasing 14.5% of Sherbrooke's short routes completely. The multiplicative-composition framework this project's closed forms (Entries 025, 028, 032) are built on has a real, now well-evidenced ceiling, and no closed-form patch discovered so far reaches past it.

### 4. Where This Leaves the Project

Fixing this properly requires modeling correlated noise across a route rather than composing independently-measured channel probabilities -- effectively a step toward simulating the density matrix along the path instead of multiplying survival probabilities, which Entry 033 already identified as the real fix and set aside as out of scope for a closed-form model. That assessment stands. What changes here is urgency and framing: this is exactly the kind of structured, high-value target a learned model (GNN or otherwise) could plausibly pick up where the closed-form approach hits its ceiling -- the batch dataset from Entries 034/035 already contains 12 labeled floor-collapse examples for it to learn

from directly.

5. Next Step

Move into the ML/GNN phase using the Entry 034/035 batch dataset (238 circuits combined) as the training/validation set, with v4.1's predictions as the closed-form benchmark any learned model needs to beat -- particularly on the floor-collapse cases this entry shows the closed-form approach structurally cannot reach.

Research Log — Index Addendum (Entry 036)

| Entry     | Title                                                   | Date         | Status   |
|-----------|---------------------------------------------------------|--------------|----------|
| Entry 036 | Floor-Collapse Is the Independence Breakdown, Amplified | Aug 16, 2026 | Complete |

Script: entry036\_isolate\_floor\_collapse.py. Data: quantumbridge\_data/entry036\_isolate\_floor\_collapse.json. Confirms floor-collapse is Entries 029/033's channel-independence breakdown, stronger and on shorter routes than previously seen -- rules out route length as the predictor. No code changes; closes the floor-collapse investigation opened in Entry 034. Next: ML/GNN phase.

ENTRY 037 The First ML Baseline — August 16, 2026 — Status: COMPLETE

## Entry 037 — Phase 1 Closes, Phase 2 Opens: Can a Learned Model Beat the Closed Form?

*A Tabular Baseline First, Not a GNN -- 238 Circuits Isn't Enough Data for One Yet*

### 1. Why a Tabular Model, Not a GNN, First

The original plan (Entry 033's close) was a GNN over the chip's connectivity graph. 238 labeled circuits is not enough to train a graph neural network without either overfitting or leaning so hard on regularization that any comparison to v4.1 is meaningless -- GNNs typically want thousands of labeled graphs, not hundreds. This entry instead trains gradient-boosted trees and a random forest on the same scalar features v4.1's formulas already use (hop distance, real edge/gate counts from the actual transpiled route, worst edge error and worst T1 along it, plus v4.1's own prediction as an input feature), evaluated with 5-fold cross-validation. If a model can't beat v4.1 with today's dataset, a bigger architecture won't either -- this is the honest floor to clear before scaling up.

### 2. Result: Roughly Tied Overall, Modest Gains Where It Matters

| model                     | CV MAE (all 238) | CV R <sup>2</sup> | MAE on the 12 floor-collapse cases |
|---------------------------|------------------|-------------------|------------------------------------|
| <b>v4.1 (closed form)</b> | 6.29 pts         | 0.605             | 20.85 pts                          |
| <b>Gradient boosting</b>  | 6.48 pts         | 0.594             | 16.21 pts                          |
| <b>Random forest</b>      | 6.39 pts         | 0.617             | 18.81 pts                          |

Across the whole dataset, both learned models land within noise of v4.1's own accuracy -- not a surprise, since v4.1's prediction is itself one of the input features and dominates the model's learned importance (87.3%). On the 12 floor-collapse cases specifically -- exactly where Entry 036 showed the closed form has a structural ceiling -- gradient boosting cuts the error from 20.85 to 16.21 points, about 22% relative. Real, but far from solved: the model has seen roughly 2-3 floor-collapse examples per cross-validation fold, nowhere near enough to learn the pattern Entry 036 couldn't even isolate to a single channel.

### 3. What the Model Actually Learned

Feature importances confirm the honest read: v4\_1\_prediction (87.3%) dominates, with n\_2q\_gates\_total, worst\_edge\_error, worst\_t1\_us, n\_real\_edges\_used, and bfs\_hop\_distance splitting most of the remaining 12.7% roughly evenly. The model is best described as learning a small correction on top of v4.1 rather than an independent prediction -- which chip and circuit kind (Bell vs GHZ) contributed almost nothing (under 1% combined), consistent with Entry 035's finding that the floor-collapse pattern isn't tied to chip identity in any way these coarse features capture.

### Entry 037 — Conclusion

The first ML baseline does what a first baseline should: it roughly matches the physics model everywhere the physics model already works, and shows a real but modest edge exactly where the physics model is known to structurally fail. That is evidence the learning approach is pointed at the right problem, not yet evidence it has solved it. 238 circuits, 12 of them floor-collapse cases, is too thin a sample for a tree model -- let alone a GNN -- to do much better than this.

### 4. Next Step

Grow the batch dataset (Entries 034/035's scripts are already resumable and checkpointed) toward the low thousands of circuits, specifically oversampling short routes on Sherbrooke where floor-collapse concentrates, then revisit both the tabular baseline and a proper GNN over the chip connectivity graph with enough floor-collapse examples per fold to actually learn the pattern Entry 036 could isolate but not explain.

| Entry     | Title                                                            | Date         | Status   |
|-----------|------------------------------------------------------------------|--------------|----------|
| Entry 037 | The First ML Baseline (gradient boosting vs. v4.1, 238 circuits) | Aug 16, 2026 | Complete |

Script: entry037\_ml\_baseline.py. Data: quantumbridge\_data/entry037\_ml\_baseline.json. Opens Phase 2 (ML) of the project. Ties v4.1 overall, ~22% relative MAE reduction on floor-collapse cases. Next: grow the dataset before scaling to a GNN.

ENTRY 038 Growing the Dataset, Sharpening the Estimate — August 16, 2026 — Status: COMPLETE

## Entry 038 — 238 Circuits to 558, and the Sherbrooke/Kyiv Ratio Comes Back Down to Earth

*Oversampling Short Routes Triples the Floor-Collapse Sample and Corrects an Overconfident Early Estimate*

### 1. Why Short Routes Specifically

Entry 036 showed floor-collapse doesn't need a long route -- the worst channel-independence gap measured in the whole project came from a 3-edge route. Growing the dataset efficiently therefore means concentrating new circuits in the 1-15 hop range, which is also the cheapest region to simulate (no long SWAP chains).

entry038\_grow\_shortroute.py reuses Entry 034's exact methodology with a third, independent seed, restricted to four short/mid bins, generating 320 new Bell circuits (160 per chip) on top of Entries 034/035's 238.

### 2. The Dataset, and the Floor-Collapse Rate, Both Grew

|                             | before (Entries 034/035) | after (+ Entry 038) |
|-----------------------------|--------------------------|---------------------|
| total circuits              | 238                      | 558                 |
| floor-collapse cases        | 12                       | 37                  |
| Kyiv short-route rate       | 3.1%                     | 6.1%                |
| Sherbrooke short-route rate | 14.5%                    | 12.5%               |
| Sherbrooke / Kyiv ratio     | 4.7x                     | 2.05x               |

The direction of Entry 035's finding holds -- Sherbrooke still collapses on short routes more than twice as often as Kyiv -- but the 4.7x ratio reported on 134 short-route samples was itself noisy. With 396 short-route samples the estimate settles to roughly 2x. This is worth stating plainly: the early number wasn't wrong in direction, but a batch of ~130 samples was too small to trust its exact magnitude, which is exactly the kind of correction a bigger dataset is supposed to produce.

### 3. Re-Running Entry 037's ML Baseline on the Larger Sample

| model                  | MAE, all circuits | MAE, floor-collapse only | n floor-collapse |
|------------------------|-------------------|--------------------------|------------------|
| v4.1 (closed form)     | 4.60 pts          | 18.20 pts                | 37               |
| Gradient boosting (CV) | 4.57 pts          | 14.79 pts                | 37               |
| Random forest (CV)     | 4.36 pts          | 15.34 pts                | 37               |

With 37 floor-collapse examples instead of 12, every cross-validation fold now contains floor-collapse cases (5 of 5, versus an uneven 2-3 before) -- a materially more trustworthy estimate. The relative gain holds: gradient boosting cuts floor-collapse MAE by about 19%, random forest by about 16%, consistent with Entry 037's ~22% finding on the smaller sample. Overall MAE for every model, v4.1 included, dropped as well (6.3 to ~4.5 points) -- this batch's short-route bias makes the whole dataset easier on average, not because anything got fixed.

### Entry 038 — Conclusion

Growing the dataset did what it was supposed to: it didn't just add volume, it corrected an overconfident early ratio (4.7x to 2.05x) and confirmed the ML baseline's floor-collapse advantage is real and consistent, not an artifact of a 12-example sample. The floor-collapse phenomenon itself remains exactly as unexplained as Entry 036 left it -- this entry is about data quality and confidence, not a new mechanism.

### 4. Next Step

Continue growing toward the low thousands using the same resumable scripts (Entries 034, 035, 038 all checkpoint to JSON and can resume in chunks), then retrain and re-evaluate the ML baseline again -- watching whether the floor-collapse MAE gap keeps narrowing or plateaus, which would itself be informative about whether a tabular model

can ever fully close it or whether it genuinely needs correlated-noise modeling as Entry 036 suggested.

Research Log — Index Addendum (Entry 038)

| Entry     | Title                                                                 | Date         | Status   |
|-----------|-----------------------------------------------------------------------|--------------|----------|
| Entry 038 | Growing the Dataset to 558 Circuits, Refining the Floor-Collapse Rate | Aug 16, 2026 | Complete |

Script: entry038\_grow\_shortroute.py. Data: quantumbridge\_data/entry038\_shortroute\_dataset.json (320 new circuits), quantumbridge\_data/entry038\_combined\_dataset.json (558 total, merged), quantumbridge\_data/entry038\_growth\_summary.json. Sherbrooke/Kyiv floor-collapse ratio revised from 4.7x to a more reliable 2.05x; ML baseline's floor-collapse advantage confirmed with full 5-fold coverage.

ENTRY 039 790 Circuits: the Estimates Are Converging — August 16, 2026 — Status: COMPLETE

## Entry 039 — A Fourth Independent Batch, and the Numbers Stop Moving

*Balanced Across All Six Hop-Distance Bins Again, Restoring the Long-Route Coverage Entry 038 Traded Away*

### 1. Why Balance This Time

Entry 038 deliberately skewed toward short/mid routes to grow the floor-collapse sample cheaply, which worked, but left the dataset thin on long routes again. entry039\_grow\_balanced.py returns to all six hop-distance bins from Entries 034/035 (with a fourth independent seed) plus a GHZ batch, adding 232 more circuits: 232 new (117 Kyiv, 115 Sherbrooke), on top of the 558 from Entries 034/035/038.

### 2. The Floor-Collapse Ratio Holds Steady

|                             | after Entry 038 (n=396 short-route) | after Entry 039 (n=535 short-route) |
|-----------------------------|-------------------------------------|-------------------------------------|
| total circuits              | 558                                 | 790                                 |
| floor-collapse cases        | 37                                  | 47                                  |
| Kyiv short-route rate       | 6.1%                                | 6.0%                                |
| Sherbrooke short-route rate | 12.5%                               | 11.6%                               |
| Sherbrooke / Kyiv ratio     | 2.05x                               | 1.93x                               |

Two independent growth rounds in a row now land within a few tenths of a point of each other -- 2.05x, then 1.93x. That's the signature of an estimate that has actually converged rather than one still drifting with more data, in contrast to the 4.7x-to-2.05x jump between the original small sample and the first growth round. Sherbrooke floor-collapsing on short routes roughly twice as often as Kyiv looks like a stable, real property of the two chips.

### 3. The ML Baseline's Advantage Also Holds

|                                      | after Entry 038 (37 fc cases) | after Entry 039 (47 fc cases) |
|--------------------------------------|-------------------------------|-------------------------------|
| v4.1 floor-collapse MAE              | 18.20 pts                     | 17.59 pts                     |
| Gradient boosting floor-collapse MAE | 14.79 pts (-19%)              | 14.07 pts (-20%)              |
| Random forest floor-collapse MAE     | 15.34 pts (-16%)              | 15.70 pts (-11%)              |

Gradient boosting's relative advantage over v4.1 on floor-collapse cases holds essentially flat (-19% to -20%) across two growth rounds. Random forest's edge narrowed slightly. Neither model is closing the gap further as data grows within this range -- consistent with Entry 036's read that the missing ingredient is a correlated-noise feature these models don't have access to, not simply more of the same kind of data.

### Entry 039 — Conclusion

Two consecutive independent growth rounds (558 to 790 circuits) now agree closely on both headline numbers: Sherbrooke's short-route floor-collapse rate is roughly 2x Kyiv's, and a tabular ML model's edge over v4.1 on those cases plateaus around 15-20% relative MAE reduction. That plateau is itself informative -- it suggests the current feature set has been mostly saturated, and further gains will need either a genuinely new feature (something that captures cross-channel correlation directly) or the GNN architecture originally planned, once there is enough data to support one.

### 4. Where This Leaves the Project

790 circuits is a real, useful dataset -- roughly 79x the original 5-circuit reference sets this project started validating against in Entry 022 -- but still short of what a GNN needs, and the plateau in section 3 suggests raw volume alone won't move the floor-collapse number much further without a new feature or architecture. This is a reasonable point to pause pure data growth and let the next entry decide between two paths: keep scaling the existing feature set toward



a GNN, or first engineer a feature that captures the specific correlated-noise signature Entry 036 found.

Research Log — Index Addendum (Entry 039)

| Entry     | Title                                                        | Date         | Status   |
|-----------|--------------------------------------------------------------|--------------|----------|
| Entry 039 | Fourth Growth Batch (790 circuits total); Estimates Converge | Aug 16, 2026 | Complete |

Script: entry039\_grow\_balanced.py. Data: quantumbridge\_data/entry039\_balanced\_dataset.json (232 new), quantumbridge\_data/entry039\_combined\_dataset.json (790 total, merged), quantumbridge\_data/entry039\_growth\_summary.json. Floor-collapse ratio and ML advantage both stabilized across two growth rounds -- a plateau, not further movement.

ENTRY 040 1,376 Circuits: the Plateau Was Premature — August 16, 2026 — Status: COMPLETE

## Entry 040 — A Fifth and Sixth Growth Batch Reopen the Question Entry 039 Thought Was Closed

*790 to 1,376 Circuits, and the ML Model Improved Again -- More Data Was Still the Right Lever*

### 1. Method

Two more independent batches: entry040\_grow\_short2.py (seed=777, short/mid bins only, 480 new Bell circuits, 240 per chip) and entry040b\_grow\_far.py (seed=888, far bins + GHZ, 106 more circuits) -- 586 new circuits total, taking the combined dataset from 790 to 1,376. One checkpoint file was corrupted mid-write when a run was interrupted (a truncated JSON string from a kill mid-save); it was recovered by trimming back to the last complete record boundary, losing only the single in-flight circuit rather than the run.

### 2. The Floor-Collapse Ratio Keeps Narrowing

|                             | 790 circuits (Entry 039) | 1,376 circuits (Entry 040) |
|-----------------------------|--------------------------|----------------------------|
| floor-collapse cases        | 47                       | 70                         |
| Kyiv short-route rate       | 6.0%                     | 5.7%                       |
| Sherbrooke short-route rate | 11.6%                    | 9.3%                       |
| Sherbrooke / Kyiv ratio     | 1.93x                    | 1.63x                      |

The ratio has now moved 4.7x → 2.05x → 1.93x → 1.63x across four growth rounds -- still trending down, not yet flat. Sherbrooke still collapses more often than Kyiv, but each larger sample makes the gap look smaller than the last one suggested. This hasn't converged as cleanly as Entry 039 read it.

### 3. Correcting Entry 039: the ML Plateau Was Premature

|                                      | 790 circuits (Entry 039) | 1,376 circuits (Entry 040) |
|--------------------------------------|--------------------------|----------------------------|
| Gradient boosting CV R <sup>2</sup>  | 0.725                    | 0.785                      |
| Gradient boosting CV MAE (all)       | 4.52 pts                 | 3.92 pts                   |
| v4.1 floor-collapse MAE              | 17.59 pts                | 17.01 pts                  |
| Gradient boosting floor-collapse MAE | 14.07 pts                | 13.78 pts                  |

Entry 039 read two similar-looking growth rounds as a plateau and concluded more volume wouldn't help further. This round contradicts that: CV R<sup>2</sup> rose meaningfully (0.725 to 0.785) and overall MAE improved by over half a point -- real gains, not noise. The lesson isn't that Entry 039 reasoned badly, it's that two data points looking flat isn't enough to call a plateau, especially this early. That's a useful correction to carry into how future entries interpret small numbers of growth rounds.

### Entry 040 — Conclusion

More data is still the highest-leverage thing this project can do before touching model architecture. Both headline numbers moved in the same direction as every prior round -- ratio narrowing, ML advantage strengthening -- and the previous entry's read that they'd stabilized turned out to be one round too early. 1,376 circuits is real progress toward the low thousands a GNN would need, and the trend curve, not just the current values, is now informative: nothing here says growth has stopped paying off yet.

### 4. Next Step

Continue growing the dataset using the same resumable, checkpointed approach (Entries 034, 035, 038, 039, 040 all use the same methodology and can run in further independent batches) until gains genuinely flatten across two OR MORE consecutive rounds -- not one -- before committing to that as the point to stop and move to a GNN architecture.

Research Log — Index Addendum (Entry 040)

| Entry     | Title                                                          | Date         | Status   |
|-----------|----------------------------------------------------------------|--------------|----------|
| Entry 040 | Fifth/Sixth Growth Batches (1,376 circuits); Plateau Retracted | Aug 16, 2026 | Complete |

Scripts: entry040\_grow\_short2.py, entry040b\_grow\_far.py. Data: quantumbridge\_data/entry040\_short\_dataset.json, quantumbridge\_data/entry040\_far\_dataset.json, quantumbridge\_data/entry040\_combined\_dataset.json (1,376 total), quantumbridge\_data/entry040\_growth\_summary.json. Retracts Entry 039's plateau conclusion -- growth is still improving both the floor-collapse ratio estimate and the ML baseline.

## ENTRY 041 The First GNN — August 16, 2026 — Status: COMPLETE

# Entry 041 — A Graph Neural Network Cuts the Floor-Collapse Error by 63%

*Real Route Structure, Not Hand-Picked Summaries, Is What the Tabular Model Was Missing*

## 1. Architecture

Each circuit's real physical route (Entry 031/032's route-then-entangle structure) is represented as a small graph: nodes are the physical qubits the transpiler actually touched (2-37 per circuit), node features are that qubit's own T1/T2/readout; edges are the real 2-qubit gates issued (SWAPs and the final entangling gate), edge features are that edge's gate error and duration. Two rounds of message passing (16-dim hidden state, mean aggregation) let information several hops away reach a node's embedding, then a masked mean-pool over nodes feeds a small MLP alongside chip/kind/hop-distance and v4.1's own prediction (included as an input feature, same as Entry 037's tabular baseline, so the comparison is fair). Built in JAX + Optax (no PyTorch available in this environment); graphs are padded to 40 nodes/edges with masks so the whole batch vmaps. entry041\_build\_graphs.py re-derives the exact graph for all 1,376 Entry 034-040 circuits from their stored (chip, qubit pairs) via route\_with\_explicit\_swaps -- no new simulation needed, the labels already exist.

## 2. Result: A Real, Large Improvement

| model                             | CV MAE (all 1,376) | CV R <sup>2</sup> | MAE, floor-collapse only |
|-----------------------------------|--------------------|-------------------|--------------------------|
| v4.1 (closed form)                | 4.50 pts           | n/a               | 17.01 pts                |
| Gradient boosting (Entry 037/040) | 3.92 pts           | 0.785             | 13.78 pts                |
| GNN (this entry)                  | 2.64 pts           | 0.868             | 6.22 pts                 |

The GNN beats the tabular model everywhere: 33% lower overall MAE (3.92 to 2.64), and on the 70 floor-collapse cases specifically, a 63% reduction from v4.1's 17.01 points down to 6.22 -- more than triple the tabular model's ~20% relative gain. Spot-checking individual floor-collapse predictions confirms this isn't an aggregate artifact: on 13 of 15 sampled cases the GNN lands within 5 points of the true ~50% floor, while v4.1 predicts 80-91% on several of the same circuits.

## 3. Why the Graph Structure Helps

Entry 036 showed floor-collapse comes from correlated noise across a route that no per-channel closed form captures, and that route length alone doesn't predict it -- a 3-edge route produced the single largest independence-breakdown gap measured in the project. The tabular model's features (worst edge error, worst T1, hop count) are scalar summaries that throw away exactly the arrangement information -- WHICH qubits sit next to which -- that message passing preserves. The GNN's large jump specifically on floor-collapse cases, rather than a uniform improvement everywhere, is consistent with it picking up on a structural signature in the route graph that no single scalar feature could encode.

## Entry 041 — Conclusion

The GNN was worth building now, at 1,376 circuits, rather than waiting for the low thousands originally planned: it already beats both the closed form and the tabular baseline by a wide margin, especially on exactly the failure mode (floor-collapse) neither could explain. That answers the diagnostic question this entry was built to answer -- the project was NOT dataset-starved for a first useful GNN. The floor-collapse gap the closed-form model treated as a hard ceiling (Entry 036) is not actually a hard ceiling for a model that sees the real route graph.

## 4. Honest Caveats

This is a first prototype: 16-dim hidden states, 2 message-passing rounds, 180 epochs, no hyperparameter search, and v4.1's own prediction is still an input feature rather than the GNN reasoning entirely from scratch. Five-fold CV on 1,376 examples with 70 floor-collapse cases (roughly 14 per fold) is a real but still modest sample for the exact subset that matters most -- the 6.22-point floor-collapse MAE should be read as promising, not final. The natural next check is

whether it holds up as the dataset continues to grow.

5. Next Step

Grow the dataset further with the GNN now in place as the metric to track (not just the tabular baseline), watching specifically whether floor-collapse MAE keeps falling -- and consider a small ablation removing v4.1's prediction as an input feature, to see how much of the GNN's edge is genuinely new structural learning versus a residual correction on top of the closed form.

Research Log — Index Addendum (Entry 041)

| Entry     | Title                                                       | Date         | Status   |
|-----------|-------------------------------------------------------------|--------------|----------|
| Entry 041 | The First GNN (JAX/Optax, message passing over real routes) | Aug 16, 2026 | Complete |

Scripts: entry041\_build\_graphs.py, entry041\_train\_gnn.py. Data: quantumbridge\_data/entry041\_graph\_dataset.json (1,376 graphs), quantumbridge\_data/entry041\_gnn\_folds.json, quantumbridge\_data/entry041\_gnn\_results.json. GNN beats v4.1 (63% lower floor-collapse MAE) and the tabular baseline (33% lower overall MAE) -- the biggest single-entry improvement since Entry 032.

ENTRY 042 2,198 Circuits, the GNN Keeps Improving — August 20, 2026 — Status: COMPLETE

# Entry 042 — More Data Helps the GNN Too, and a Parallelization Detour That Wasn't Worth It

1,376 to 2,198 Circuits; GNN Floor-Collapse MAE Falls Again, 6.22 to 5.31 Points

## 1. Growing the Dataset, and a Parallelization Attempt

Tried parallelizing circuit generation across the sandbox's 4 CPU cores via multiprocessing.Pool, keeping identical precision (4 seeds x 4096 shots, same methodology as every prior entry). It worked initially but gave only a 1.3-1.7x speedup, not 4x -- most of the per-circuit cost turned out to be Python/Qiskit overhead (circuit construction, transpile scheduling) rather than raw linear algebra, which doesn't parallelize as cleanly. It then began reliably hanging after several hundred circuits for reasons that didn't reproduce in isolated tests -- reverted to the proven single-process approach from Entries 034-040. The sandbox's own throughput also degraded sharply mid-session (from roughly 0.4 to 0.06 circuits/second) for reasons outside the model or scripts, most likely shared infrastructure load. Net result: 822 new circuits added this entry, bringing the combined dataset to 2,198 -- real progress, just slower than hoped.

## 2. Retraining the GNN

|                         | 1,376 circuits (Entry 041) | 2,198 circuits (Entry 042) |
|-------------------------|----------------------------|----------------------------|
| floor-collapse cases    | 70                         | 112                        |
| GNN CV R <sup>2</sup>   | 0.868                      | 0.881                      |
| GNN CV MAE (all)        | 2.64 pts                   | 2.04 pts                   |
| GNN floor-collapse MAE  | 6.22 pts                   | 5.31 pts                   |
| v4.1 floor-collapse MAE | 17.01 pts                  | 14.30 pts                  |

The GNN improved on every metric with more data, same as the tabular model did in Entries 038-040 -- confirming the growth strategy pays off for the graph architecture too, not just the simpler baseline. The GNN's margin over v4.1 on floor-collapse cases remains enormous (5.31 vs 14.30 points, a 63% relative reduction, essentially unchanged from Entry 041's finding).

## Entry 042 — Conclusion

The GNN is confirmed as the best model built in this project across every dataset size tested so far, and it is not yet showing diminishing returns from more data the way the closed-form model structurally can't improve at all. The path to 5,000 and eventually 10,000 circuits remains the right one; it will just take more sessions than originally hoped, since raw circuit-generation throughput -- not modeling -- is now the actual bottleneck.

## 3. Next Step

Continue growth toward 5,000 in future sessions using the proven sequential approach (entry042\_grow\_sequential.py), retraining the GNN periodically to keep tracking whether floor-collapse MAE keeps falling or starts to plateau for real this time. In parallel, build an interactive visualization of the trained GNN's message-passing process on an example circuit, showing how information flows across the real route graph toward a prediction.

## Research Log — Index Addendum (Entry 042)

| Entry     | Title                                                 | Date         | Status   |
|-----------|-------------------------------------------------------|--------------|----------|
| Entry 042 | Dataset to 2,198 Circuits; GNN Retrained and Improved | Aug 20, 2026 | Complete |

*Scripts: entry042\_grow\_parallel.py (partial, superseded), entry042\_grow\_sequential.py (active), entry042\_train\_gnn.py. Data: quantumbridge\_data/entry042\_parallel\_dataset.json, quantumbridge\_data/entry042\_combined\_dataset.json (2,198 total), quantumbridge\_data/entry042\_gnn\_results.json, quantumbridge\_data/entry042\_growth\_summary.json.*

ENTRY 043   A Small Increment Is Too Noisy to Read — August 20, 2026 — Status: COMPLETE

# Entry 043 — Checking Whether the GNN Has Plateaued, Before Grinding Toward 5,000

2,198 to 2,288 Circuits (+4%) Moved the Numbers the Wrong Way -- but That's Not the Same as a Real Plateau

## 1. Why This Check, Now

Growing the dataset toward 5,000 at full precision (Entry 042's methodology, no shot-count shortcuts) turned out to be much slower than expected -- roughly 10-20 circuits per batch round even after a free optimization (combining the 4 separate 4096-shot Aer calls into one 16,384-shot call, ~18% faster, statistically identical). Rather than keep grinding for the dozens of hours that would take, this entry asks the cheaper question first: is the GNN's improvement curve from Entries 041/042 still climbing, or did it already start to flatten with the 2,198 circuits already in hand.

## 2. The Result Is Genuinely Ambiguous

| dataset size       | GNN CV MAE | GNN CV R <sup>2</sup> | floor-collapse MAE | floor-collapse n |
|--------------------|------------|-----------------------|--------------------|------------------|
| 1,376 (Entry 041)  | 2.64 pts   | 0.868                 | 6.22 pts           | 70               |
| 2,198 (Entry 042)  | 2.04 pts   | 0.881                 | 5.31 pts           | 112              |
| 2,288 (this entry) | 2.17 pts   | 0.869                 | 6.07 pts           | 117              |

The 1,376-to-2,198 jump (+60% more data) showed a clear, unambiguous improvement across every metric. The 2,198-to-2,288 jump (+4% more data) moved every metric in the WRONG direction -- but a 4% increment is far too small to distinguish a real plateau from ordinary training noise: each fold uses a different random 80/20 split and a freshly-initialized network, and with only ~23 floor-collapse cases per test fold, a handful of hard examples landing in one fold versus another can swing the reported MAE by more than this entire increment moved it.

## Entry 043 — Conclusion

This entry does not answer whether the GNN has plateaued -- it answers that the question can't be answered from an increment this small, honestly reported rather than over-interpreted either direction. What it DOES confirm: the GNN remains dramatically ahead of v4.1 at every checkpoint (floor-collapse MAE 6.07 vs 14.67 points here, a 59% reduction, consistent with Entries 041 and 042), and R<sup>2</sup> has stayed in the same healthy 0.87-0.88 band across three retrainings. 2,288 circuits is a solid, validated working dataset -- not proven to be enough for a truly bigger GNN, but not shown to be insufficient either.

## 3. Next Step

Continue growing the dataset opportunistically in future sessions using the proven sequential, full-precision approach -- but without treating each small increment as a verdict on the model's ceiling. A real read on whether more of this kind of data still helps needs either a much bigger jump (the way 1,376 to 2,198 was) or averaging several training seeds per checkpoint to separate signal from noise, which Entry 043 did not have the compute budget to do this session.

## Research Log — Index Addendum (Entry 043)

| Entry     | Title                                                  | Date         | Status   |
|-----------|--------------------------------------------------------|--------------|----------|
| Entry 043 | Checking the GNN's Improvement Curve at 2,288 Circuits | Aug 20, 2026 | Complete |

Scripts: entry043\_train\_gnn.py. Data: quantumbridge\_data/entry043\_combined\_dataset.json (2,288), quantumbridge\_data/entry043\_gnn\_folds.json, quantumbridge\_data/entry043\_gnn\_results.json, quantumbridge\_data/entry043\_growth\_summary.json. Also includes a free, precision-preserving speedup to the Aer ground-truth pipeline in entry042\_grow\_parallel.py (one 16,384-shot call instead of four 4,096-shot calls).